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Robust and sparse support vector machine via hybrid truncated loss for supervised classification

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ABSTRACT

Choosing a suitable loss function for support vector machines (SVMs) is difficult. Convex losses can be affected by outliers, while bounded non-convex losses are harder to optimize. We propose a family of hybrid truncated losses that are sparse and bounded, and use the member with $\delta = 2/3$ to build the L_{ht}-SVM model. We introduce P-stationarity and use it to give first-order necessary and sufficient conditions for local optimality. We then solve the model by an alternating direction method of multipliers with a working-set strategy to lower the computational cost. If the generated sequence converges, its limit is P-stationary and therefore a local minimizer. We also extend L_{ht}-SVM to multi-view learning. The resulting MvL_{ht}-SVM uses structural information and adaptive view weights to combine shared and complementary information from different views. Experiments on synthetic, UCI, LIBSVM, and STL-10 datasets compare our models with five single-view and six multi-view methods. The proposed models rank among the best methods on average and use relatively few support vectors. They also remain competitive under label noise, feature noise, outliers, class imbalance, and changes to test samples.

1. Introduction

Support vector machines (SVMs) are widely used in machine learning (Cortes & Vapnik, 1995; Maggioni & Spinelli, 2025), computer vision (Chen et al., 2025b; Senokosov et al., 2024), natural language processing (Alkhodhairy & Saleem, 2025; Wu et al., 2023), and biomedical analysis (Taha, 2025). Given a binary classification dataset $\{(\mathbf{x}_i, y_i)\}_{i=1}^m$, where $\mathbf{x}_i \in \mathbb{R}^n$ and $y_i \in \{-1, +1\}$, the soft-margin SVM solves

$$\min_{\mathbf{w},b} \ \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{m} \ell\big(1 - y_i(\mathbf{w}^\top \mathbf{x}_i + b)\big), \tag{1}$$

where $C > 0$ is the penalty parameter and $\ell(\cdot)$ is the loss function. The choice of loss strongly affects the sparsity, robustness, and computational efficiency of the resulting classifier (Akhtar et al., 2025; Wang et al., 2025).

Among convex losses, the hinge loss (see Fig. 1(a)) (Cortes & Vapnik, 1995) can produce sparse solutions, but its penalty is unbounded for large margin violations, which makes it sensitive to outliers (Brahmi, 2024; Wang et al., 2023). The pinball loss (see Fig. 1(b)) (Huang et al., 2014) extends the hinge loss to quantile settings, but it is still unbounded (Huang et al., 2017). The least-squares loss (see Fig. 1(c)) (Suykens & Vandewalle, 1999) gives a smooth quadratic objective, but it penalises all samples, even those that are already classified correctly, and therefore loses sparsity (Liu et al., 2022; Wang & Hu, 2005). In short, these convex losses do not provide both sparsity and bounded penalty at the same time.

Bounded non-convex losses provide an alternative. The ramp loss (see Fig. 1(d)) (Collobert et al., 2006) clips the hinge loss at a finite threshold and thus improves both sparsity and robustness to outliers. The $L_{0/1}$ loss (see Fig. 1(e)) (Wang et al., 2022) assigns a unit penalty to each misclassified sample and can be viewed as a natural counting-type loss. Other bounded losses, such as the sigmoid loss (Mason et al., 1999), rescaled hinge loss (Xu et al., 2017), truncated pinball loss (Shen et al., 2017), capped squared loss (Wang et al., 2024a), and V-shaped ravine loss (Liu et al., 2026), have also been proposed to balance robustness and smoothness. The main difficulty is that non-convex losses are harder to optimise and may have many local minimisers (Akhtar et al., 2025). For example, optimisation under the $L_{0/1}$ loss is NP-hard in general.

A practical way to handle bounded non-convex losses was introduced in the $L_{0/1}$-SVM of Wang et al. (2022), where an alternating direction method of multipliers (ADMM) was combined with a working-set strategy and the P-stationary point (PSP) was used to characterise the non-convex subproblem. This ADMM-plus-working-set framework was later extended to several other bounded-loss models, including the ramp

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fraction loss (Wang & Shao, 2024), the capped squared loss (Wang et al., 2024a), a sparse and robust variant (Wang et al., 2025), and the multi-view $SL_{0/1}$-SVM (Chen et al., 2025a).

Different from these studies, the present work starts from the loss itself: the proposed linear–quadratic–constant loss admits an exact proximal mapping, from which the P-stationary conditions and the active-sample characterization are derived, and the working-set rule of the solver is built directly on this characterization.

In many real applications, data come from multiple sources or can be described by different feature extractors. For example, an image can be represented by colour histograms, texture descriptors, and deep features, while a web page can be described by both text content and hyperlink topology. Multi-view learning (MVL) uses these complementary representations jointly and often performs better than single-view methods in both theory and practice (Berahmand et al., 2025; Farquhar et al., 2005; Jia et al., 2026).

Most MVL methods are built on two principles: consistency and complementarity. Consistency encourages agreement across views, whereas complementarity makes use of the information that is unique to each view (Berahmand et al., 2025). Effective multi-view classification models should satisfy both principles, which are often called the 2C principles.

Many studies have combined SVM with MVL. SVM-2K (Farquhar et al., 2005) enforces consistency between two views through a KCCA-based constraint, but it does not explicitly use complementary information. MvTSVM (Xie & Sun, 2015) speeds up training by solving two smaller quadratic programmes, but it also ignores complementarity. MvSVM-2C (Xie & Sun, 2020) addresses this issue by introducing adaptive view weights so that both 2C principles are considered. Later models extended this line of work. For example, MPWTSVM (Xu & Wang, 2022) incorporates privileged information, MvLSSVC-2C (Zhang et al., 2025) uses hierarchical agglomerative clustering to enhance inter-view interaction, the deep multi-view LSSVM in Jia et al. (2026) adds stacking layers, and Mv-vSVM (He et al., 2026) reduces the burden of parameter selection in SVM-2K.

However, most existing multi-view SVM models still rely on convex losses and may therefore degrade under outlier contamination. Some recent methods have introduced bounded non-convex losses for better robustness, such as the LINEX loss in MVASY-BX (Tang et al., 2023), the wave loss in Wave-MvSVM (Quadir et al., 2025), and the $L_{0/1}$ loss in Mv$SL_{0/1}$-SVM (Chen et al., 2025a). To the best of our knowledge, the hybrid truncated loss has not yet been studied in a multi-view SVM framework.

Although deep networks are dominant on large-scale perception tasks, where end-to-end representation learning from abundant labeled data is essential, many practical classification problems still involve tabular or small-to-medium data with fixed feature representations. In this regime, a linear classifier determined by a few support vectors is interpretable at the sample level, a bounded loss limits the influence of any single corrupted observation, and training requires neither large-scale annotation nor pre-training. The models studied in this paper are designed for this regime and are complementary to deep architectures; the bounded loss can also be used as a classification component on top of learned representations.

Motivated by these observations, we introduce a family of hybrid truncated losses in which an interior junction $\delta \in (0, 1)$ controls the transition from linear penalization to smooth saturation. The resulting L_{ht}-SVM retains a zero-loss region for sparsity and a constant upper branch for bounded influence, while the intermediate quadratic segment provides a continuously differentiable connection to the cap. We derive its proximal mapping and P-stationary conditions, develop a working-set ADMM solver, and extend the formulation to multi-view classification through structural regularization and adaptive view weighting. The main model uses $\delta = 2/3$, whose empirical sensitivity is examined in Section 6.1.7.

This paper makes the following contributions.

- We construct a hybrid truncated loss family that combines a zero-loss region, linear margin penalization, smooth quadratic saturation, and a bounded cap, and analyse the role of its junction parameter δ.
- We derive the regular subdifferential and exact proximal mapping of the $\delta = 2/3$ specialization and establish the associated P-stationary optimality conditions for L_{ht}-SVM.
- Based on established ADMM and working-set techniques, we design a solver whose working sets are determined by the support-vector characterization of the L_{ht} loss, and extend the resulting classifier to multi-view learning with structural covariance regularization and adaptive view weighting.
- We evaluate predictive performance, sparsity, fitting time, noise robustness, and junction sensitivity on synthetic, UCI, LIBSVM, and image datasets.

Since adaptive view weighting and covariance regularization are established multi-view mechanisms, the novelty of the multi-view model lies in embedding the L_{ht} loss, its proximal updates, and the sparse working sets into this framework. Section 5.3 details the cross-view structural term, and Appendix D analyses a naturally multi-view dataset through ablation, view corruption, learned weights, and support-set overlap.

The rest of this paper is organised as follows. Section 2 reviews related loss functions and multi-view SVM models. Section 3 introduces the L_{ht} loss and presents the optimality theory for L_{ht}-SVM. Section 4 describes the ADMM solver with working-set acceleration. Section 5 presents the multi-view extension. Section 6 reports the experimental results. Section 7 concludes the paper.

2. Related work

We begin by fixing notation and then survey the SVM loss functions, optimisation algorithms, and multi-view SVM models that are most relevant to the present work. Let $S^{(V)} = \{(\mathbf{x}_i^{(1)}, \ldots, \mathbf{x}_i^{(V)}, y_i)\}_{i=1}^m$ denote a V-view dataset with feature spaces $\mathbf{x}_i^{(v)} \in \mathcal{X}_v$ ($v \in [V]$) and label space $\mathcal{Y} = \{-1, +1\}$. Table 1 collects the principal symbols used throughout the paper.

Table 1
List of main notations.

Symbol	Denotation	Scope
m	Number of training samples	$\mathbb{N}$
n	Total dimension of the sample	$\mathbb{N}$
n_v	Dimension in vth view	$\mathbb{N}$
V	Number of the views	$\mathbb{N}$
$\mathbf{w}$	Weight vector of hyperplane	$\mathbb{R}^n$
$\mathbf{w}^{(v)}$	Weight vector in vth view	$\mathbb{R}^{n_v}$
b	Bias of the hyperplane	$\mathbb{R}$
$b^{(v)}$	Bias in vth view	$\mathbb{R}$
$\theta^{(v)}$	Weight of vth view	$[0, 1]$
$\mathbf{X}^{(v)}$	Data matrix of vth view	$\mathbb{R}^{m\times n_v}$
$\mathbf{y}$	$(y_1, y_2, \ldots, y_m)$	$\{-1, 1\}^m$
$\mathbf{Y}$	$\mathrm{diag}(y_1, \ldots, y_m)$	$\mathbb{R}^{m\times m}$
$\mathbf{1}$	$(1, \ldots, 1)^\top$	$\mathbb{R}^m$
N	$[y_1\mathbf{x}_1 \cdots y_m\mathbf{x}_m]^\top$	$\mathbb{R}^{m\times n}$
$M^{(v)}$	$[y_1\mathbf{x}_1^{(v)} \cdots y_m\mathbf{x}_m^{(v)}]^\top$	$\mathbb{R}^{m\times n_v}$
$\mathbf{t}$	$(t_1, \ldots, t_m)^\top$	$\mathbb{R}^m$
$t_i^{(v)}$	$1 - y_i f^{(v)}(\mathbf{x}_i^{(v)})$	$\mathbb{R}$
$\|\cdot\|_0$	Number of non-zero elements	
$\mathbf{q}$	Auxiliary variable	$\mathbb{R}^m$
λ	Lagrangian multiplier	$\mathbb{R}^m$
$\boldsymbol{\xi}, \boldsymbol{\eta}$	Slack variables	$\mathbb{R}_+^m$
C, C_1, C_2, D, ϵ	Nonnegative parameters	$\mathbb{R}_+$
α, η, β	Nonnegative parameters	$\mathbb{R}_+$
ξ	ADMM penalty parameter	$\mathbb{R}_+$
$\Sigma^{(v)}$	Structural covariance matrix	$\mathbb{R}^{n_v\times n_v}$
l	Outer loop counter	$\mathbb{N}$
k	Inner loop counter	$\mathbb{N}$

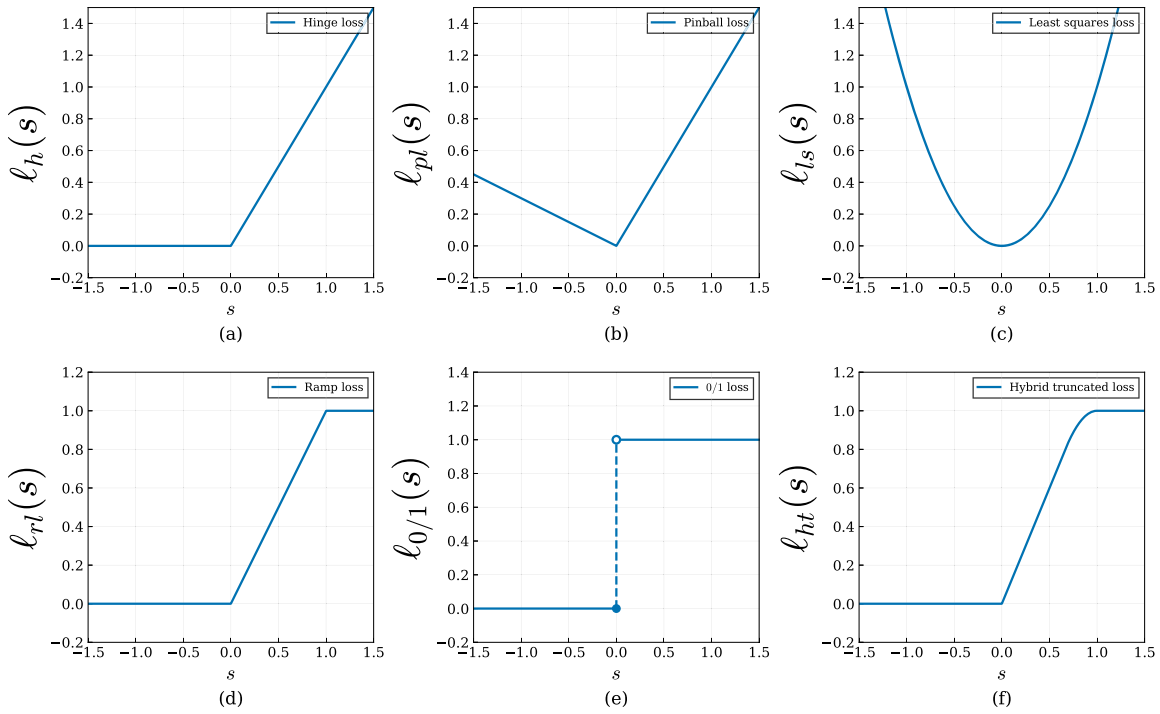


Fig. 1. The figures of six loss functions. (a) Hinge loss. (b) Pinball loss. (c) Least squares loss. (d) Ramp loss. (e) $L_{0/1}$ loss. (f) Hybrid truncated loss.

2.1. Loss functions and optimization algorithms for SVM

A wide variety of loss functions have been investigated for SVM. We group them into convex and non-convex families and briefly discuss representative members.

- **Hinge loss.** Introduced in the original SVM (Cortes & Vapnik, 1995), the hinge loss (see Fig. 1(a)) is defined as

$$\ell_h(s) := \begin{cases} 0, & s < 0, \\ s, & s \geq 0. \end{cases}$$

  Samples satisfying $s < 0$ incur zero penalty, yielding sparse solutions. On the positive side, however, the loss increases linearly without bound, making the model susceptible to large-margin outliers (Brahmi, 2024; Wang et al., 2023).
- **Pinball loss.** A generalisation of the hinge loss with a tunable slope on the negative side (Huang et al., 2014) (see Fig. 1(b)):

$$\ell_{pl}(s) := \begin{cases} \imath s, & s < 0, \\ s, & s \geq 0, \end{cases}$$

  where $\imath \in [0, 1]$. It is non-smooth at $s = 0$ and imposes a linear penalty on every sample, leading to denser solutions and limited noise tolerance (Huang et al., 2017).
- **Least squares loss.** The quadratic penalty (Liu et al., 2022; Suykens & Vandewalle, 1999) (see Fig. 1(c)) reads

$$\ell_{ls}(s) := s^2, \quad s \in \mathbb{R}.$$

  Its differentiability simplifies training, but the unbounded quadratic growth amplifies the influence of outliers and eliminates sparsity (Wang & Hu, 2005).

Convex losses benefit from well-established optimisation theory, yet their unboundedness limits robustness under contaminated data. Non-convex alternatives address this shortcoming at the expense of a harder optimisation landscape.

- **Ramp loss.** The ramp loss (Wang & Shao, 2024) (see Fig. 1(d)) truncates the hinge loss at a finite ceiling $\overline{\chi} \geq 1$:

$$\ell_{rl}(s) := \begin{cases} \overline{\chi}, & s \geq \overline{\chi}, \\ s, & s \in (0, \overline{\chi}), \\ 0, & s \leq 0. \end{cases}$$

  The output is confined to $[0, \overline{\chi}]$, securing both sparsity and bounded influence on outliers, although non-smoothness at $s = 0$ and $s = \overline{\chi}$ must be handled carefully.
- $L_{0/1}$ **loss.** Proposed by Wang et al. (2022), the $L_{0/1}$ loss (see Fig. 1(e)) is the purest counting loss:

$$\ell_{0/1}(s) := \begin{cases} 1, & s > 0, \\ 0, & s \leq 0. \end{cases}$$

  Each misclassified point contributes a unit cost irrespective of the margin violation magnitude, which maximally bounds the outlier influence. The resulting problem, however, is NP-hard, and practical solvers can only guarantee convergence to local optimisers (Akhtar et al., 2025).

In summary, bounded non-convex losses can limit the influence of extreme margin violations, but their practical robustness depends on the data and comes with a more demanding optimization problem.

Optimization algorithms. Solving SVM models with non-convex losses requires specialised algorithms. Full-sample gradient-based methods, such as the concave-convex procedure (CCCP) (Wang et al., 2021), semi-smooth Newton schemes (Yin & Li, 2019), and the conjugate gradient method (Suykens & Vandewalle, 1999), are straightforward but incur high per-iteration cost because they process the entire training set. Sub-sample strategies, including sequential minimal optimisation (SMO) (Huang et al., 2014), stochastic gradient descent (SGD) (Allen-Zhu, 2017), and coordinate descent (Liu et al., 2021), alleviate this burden yet may exhibit slow convergence on non-convex objectives.

A different strategy is ADMM, which splits the original problem into tractable subproblems solved in alternation (Dai et al., 2023; Feng & Xu,

2022). ADMM has been applied to SVM through, e.g., hinge-ADMM (Ye et al., 2011), elastic net ADMM (Wang et al., 2024b), and ramp-ADMM (Wang et al., 2024a). A common bottleneck of these approaches is the need to invert a matrix involving all training samples at every iteration. The $L_{0/1}$-SVM (Wang et al., 2022) addresses this bottleneck by coupling ADMM with a working-set rule that restricts the linear-system update to an active subset and characterizes convergent limit points through P-stationarity.

2.2. MVL models

2.2.1. SVM-2K

SVM-2K (Farquhar et al., 2005) builds upon the hinge loss and enforces consistency between two views via KCCA theory. For a two-view dataset $S^{(2)} = \{(\mathbf{x}_i^{(1)}, \mathbf{x}_i^{(2)}, y_i)\}_{i=1}^m$, the optimisation problem reads:

$$\begin{aligned}\min_{\mathbf{w}^{(v)},\boldsymbol{\xi}^{(v)},\boldsymbol{\eta}} \ & \frac{1}{2}\left\|\mathbf{w}^{(1)}\right\|_2^2 + \frac{1}{2}\left\|\mathbf{w}^{(2)}\right\|_2^2 \\ & + C_1\mathbf{1}^\top\boldsymbol{\xi}^{(1)} + C_2\mathbf{1}^\top\boldsymbol{\xi}^{(2)} + D\mathbf{1}^\top\boldsymbol{\eta} \\ \text{s.t.} \quad & \left|\mathbf{X}^{(1)}\mathbf{w}^{(1)} - \mathbf{X}^{(2)}\mathbf{w}^{(2)}\right| \le \boldsymbol{\eta} + \epsilon\mathbf{1}, \\ & \mathbf{Y}\mathbf{X}^{(v)}\mathbf{w}^{(v)} \ge \mathbf{1} - \boldsymbol{\xi}^{(v)}, \\ & \boldsymbol{\xi}^{(v)} \ge \mathbf{0},\ \boldsymbol{\eta} \ge \mathbf{0}, \quad v = 1, 2. \end{aligned} \tag{2}$$

Here $\mathbf{Y} := \mathrm{diag}(y_1, \ldots, y_m) \in \mathbb{R}^{m\times m}$ is the label matrix, $\mathbf{1} := (1, \ldots, 1)^\top \in \mathbb{R}^m$, $C_1, C_2, D > 0$ are penalty parameters, and $\epsilon \ge 0$ is a tolerance constant. Because the hinge loss is unbounded, SVM-2K lacks robustness to outliers. Additionally, it only imposes a consensus constraint between the two views and does not capture complementary information.

2.2.2. MvSVM-2C

MvSVM-2C (Xie & Sun, 2020) extends SVM-2K to an arbitrary number of views while satisfying both the 2C principles. It retains the hinge loss but introduces a learnable weight vector $\boldsymbol{\theta}$ that adaptively balances view-specific contributions. The formulation is:

$$\begin{aligned}\min_{\mathbf{w}^{(v)},\boldsymbol{\xi}^{(v)},\boldsymbol{\eta}^{(j)},\boldsymbol{\theta}} \ & \frac{1}{2}\sum_{v=1}^{V}\theta^{(v)}\left\|\mathbf{w}^{(v)}\right\|^2 + \frac{\beta}{2}\|\boldsymbol{\theta}\|^2 \\ & + D\sum_{v=1}^{V}\sum_{j=1}^{V}\mathbf{1}^\top\boldsymbol{\eta}^{(j)} + C\sum_{v=1}^{V}\mathbf{1}^\top\boldsymbol{\xi}^{(v)} \\ \text{s.t.} \quad & \left|\mathbf{X}^{(i)}\mathbf{w}^{(i)} - \mathbf{X}^{(j)}\mathbf{w}^{(j)}\right| \le \boldsymbol{\eta}^{(j)} + \epsilon\mathbf{1}, \\ & \mathbf{Y}\mathbf{X}^{(v)}\mathbf{w}^{(v)} \ge \mathbf{1} - \boldsymbol{\xi}^{(v)}, \\ & \boldsymbol{\xi}^{(v)} \ge \mathbf{0},\ \boldsymbol{\eta}^{(j)} \ge \mathbf{0},\ \theta^{(v)} > 0,\ \sum_{v=1}^{V}\theta^{(v)} = 1, \\ & i, j, v \in [V],\ 1 \le i \le j \le V. \end{aligned} \tag{3}$$

Here V is the total number of views, $C, D > 0$ are penalty parameters, $\epsilon \ge 0$ is a tolerance constant, and $\beta > 0$ regularises the weight distribution $\boldsymbol{\theta}$. The $\binom{V}{2}$ pairwise constraints encode consensus, while the adaptive weights exploit complementary information across views. A key limitation of MvSVM-2C is its reliance on the unbounded hinge loss, which restricts robustness under outlier or label-noise corruption.

3. Optimality conditions of L_{ht}-SVM

3.1. Hybrid-truncated loss function

For $\delta \in (0, 1)$, consider the hybrid-truncated loss family

$$\ell_{\mathrm{ht},\delta}(s) := \begin{cases} 0, & s \le 0, \\ \dfrac{2s}{1+\delta}, & s \in (0, \delta], \\ 1 - \dfrac{(1-s)^2}{1-\delta^2}, & s \in (\delta, 1), \\ 1, & s \ge 1. \end{cases} \tag{4}$$

Here $s = 1 - yf(\mathbf{x})$ is the margin violation. Thus, $s \le 0$ identifies samples satisfying the unit-margin condition, whereas $s \ge 1$ corresponds to a non-positive classification margin. The outer points 0 and 1 are normalization choices. The point 0 aligns the zero-to-positive transition with the hinge, ramp, and $L_{0/1}$ losses. The point 1 aligns the cap with the normalized ramp loss in Fig. 1(d) and the unit value of the $L_{0/1}$ loss in Fig. 1(e), making their shapes directly comparable. The ramp threshold is not intrinsically fixed at 1; a different upper point could be introduced by rescaling the margin variable. Here the outer points are fixed so that δ alone controls the interior junction.

To obtain the two interior pieces, write the linear branch as $a_\delta s$ and the quadratic branch as $1 - b_\delta(1-s)^2$. Continuity and first-order matching at $s = \delta$ require

$$a_\delta\delta = 1 - b_\delta(1-\delta)^2, \qquad a_\delta = 2b_\delta(1-\delta),$$

and hence

$$a_\delta = \frac{2}{1+\delta}, \qquad b_\delta = \frac{1}{1-\delta^2}.$$

Consequently, the derivative on $(\delta, 1)$ decreases from $2/(1+\delta)$ to zero, making $\ell_{\mathrm{ht},\delta}$ continuously differentiable at both $s = \delta$ and $s = 1$. This smooth saturation distinguishes it from the abrupt upper truncation of the ramp loss and common capped formulations, while the graded penalty on $(0, 1)$ retains more margin information than the $L_{0/1}$ loss.

The theoretical development below uses $\ell_{\mathrm{ht}} = \ell_{\mathrm{ht},2/3}$, namely

$$\ell_{\mathrm{ht}}(s) := \begin{cases} 1, & s \ge 1, \\ (-9s^2 + 18s - 4)/5, & s \in (2/3, 1), \\ 6s/5, & s \in (0, 2/3], \\ 0, & s \le 0. \end{cases} \tag{5}$$

The value $2/3$ is chosen as a reference junction because the loss preserves a ramp-like linear increase over the first two thirds of $(0, 1)$ but replaces the abrupt upper truncation by smooth quadratic saturation. It is not claimed to be theoretically optimal. The subsequent proximal and P-stationary analysis specializes the family at $\delta = 2/3$, which accounts for $2/3$ and its derived thresholds in the lemmas and proofs. Its empirical sensitivity is examined in Section 6.1.7. For a general junction $\delta \in (0, 1)$, the proximal operator of $\ell_{\mathrm{ht},\delta}$ keeps the three-regime form of Lemma 2, with the slope $6/5$ replaced by $2/(1+\delta)$, the breakpoints $5/18$ and $25/18$ replaced by $(1-\delta^2)/2$ and $(1+\delta)^2/2$, the endpoints $2/3 + 6\nu C/5$ and $5/6 + 3\nu C/5$ replaced by $\delta + 2\nu C/(1+\delta)$ and $(1+\delta)/2 + \nu C/(1+\delta)$, and the threshold $\sqrt{2\nu C}$ unchanged. The proofs of Lemma 2 and Theorems 3–5 use only this piecewise structure and hold for every δ after these substitutions. The working set (30) follows the same rule, and Theorem 6 does not involve δ. The other junctions in Section 6.1.7 are computed with the same solver and the corresponding constants.

Substituting $\ell(\cdot)$ in (1) with $\ell_{\mathrm{ht}}(\cdot)$ yields a new SVM formulation:

$$\min_{\mathbf{w}\in\mathbb{R}^n, b\in\mathbb{R}} \ \frac{1}{2}\|\mathbf{w}\|^2 + C\sum_{i=1}^{m}\ell_{\mathrm{ht}}\big(1 - y_i(\langle\mathbf{w}, \mathbf{x}_i\rangle + b)\big). \tag{6}$$

Introducing an auxiliary variable $\mathbf{q} \in \mathbb{R}^m$, the above problem can be equivalently reformulated as:

$$\min_{\mathbf{w}\in\mathbb{R}^n, b\in\mathbb{R}, \mathbf{q}\in\mathbb{R}^m} \ \frac{1}{2}\|\mathbf{w}\|^2 + CL_{\mathrm{ht}}(\mathbf{q}) \quad \text{s.t.} \quad \mathbf{q} + N\mathbf{w} + b\mathbf{y} = \mathbf{1}, \tag{7}$$

where $L_{\mathrm{ht}}(\mathbf{q}) := \sum_{i=1}^m \ell_{\mathrm{ht}}(q_i)$, $\mathbf{1} := (1, \ldots, 1)^\top \in \mathbb{R}^m$, $\mathbf{y} := (y_1, \ldots, y_m)^\top \in \mathbb{R}^m$, and $N := [y_1\mathbf{x}_1 \cdots y_m\mathbf{x}_m]^\top \in \mathbb{R}^{m\times n}$.

Eqs. (6) and (7) are collectively referred to as the L_{ht}-SVM. The subsequent subsections derive the regular subdifferential and proximal operator of ℓ_{ht} and establish the optimality theory.

3.2. L_{ht} regular subdifferential

In this subsection, we present the regular subdifferential of the hybrid-truncated loss function.

Lemma 1. *For $\mathbf{q} = (q_1, q_2, \dots, q_m)^\top \in \mathbb{R}^m$, the regular subdifferential of L_{ht} is given by*

$$\partial L_{\mathrm{ht}}(\mathbf{q}) := (\partial \ell_{\mathrm{ht}}(q_1), \dots, \partial \ell_{\mathrm{ht}}(q_m))^\top \in \mathbb{R}^m, \tag{8}$$

$$\partial \ell_{\mathrm{ht}}(q_i) := \begin{cases} 0, & q_i \geq 1, \\ (18 - 18q_i)/5, & q_i \in (2/3, 1), \\ 6/5, & q_i \in (0, 2/3], \\ [0, 6/5], & q_i = 0, \\ 0, & q_i < 0, \end{cases} \qquad i \in [m]. \tag{9}$$

Proof. The proof follows directly from the definition of regular subdifferential and is omitted here. □

3.3. ℓ_{ht} proximal operator

The proximal step connects the loss geometry to the algorithm. For a scalar input z, it balances fidelity to z against the bounded penalty $\nu C \ell_{\mathrm{ht}}(s)$. Since the loss has zero, linear, quadratic, and constant pieces, only a small number of piecewise stationary points and boundary points can be minimizers, and the product νC determines which candidates survive: small values retain all four loss regions, intermediate values remove the concave quadratic candidate, and sufficiently large values reduce the update to a hard choice between 0 and z. This yields the three regimes in Lemma 2.

Definition 1. Given $z \in \mathbb{R}$, $\nu, C > 0$, the proximal operator of ℓ_{ht} is defined by

$$\mathrm{prox}_{\nu C \ell_{\mathrm{ht}}}(z) = \arg\min_{s \in \mathbb{R}} \nu C \ell_{\mathrm{ht}}(s) + \frac{1}{2}(s - z)^2. \tag{10}$$

The following lemma provides the closed-form expression of the ℓ_{ht} proximal operator.

Lemma 2. *For $z \in \mathbb{R}$, $\nu, C > 0$, we have the following results.*
(i) When $\nu C \in (0, 5/18)$, the ℓ_{ht} proximal operator has the following exact formula:

$$\mathrm{prox}_{\nu C \ell_{\mathrm{ht}}}(z) = \begin{cases} z, & z > 1, \\ \frac{5z - 18\nu C}{5 - 18\nu C}, & z \in (\frac{2}{3} + 6\nu C/5, 1], \\ z - 6\nu C/5, & z \in (6\nu C/5, 2/3 + 6\nu C/5], \\ 0, & z \in [0, 6\nu C/5], \\ z, & z < 0. \end{cases} \tag{11}$$

(ii) When $\nu C \in [5/18, 25/18)$, the ℓ_{ht} proximal operator has the following exact formula:

$$\mathrm{prox}_{\nu C \ell_{\mathrm{ht}}}(z) = \begin{cases} z, & z > 5/6 + 3\nu C/5, \\ z - 6\nu C/5, & z \in (6\nu C/5, 5/6 + 3\nu C/5], \\ 0, & z \in [0, 6\nu C/5], \\ z, & z < 0. \end{cases} \tag{12}$$

(iii) For $\nu C \geq 25/18$, the ℓ_{ht} proximal operator has the following exact formula:

$$\mathrm{prox}_{\nu C \ell_{\mathrm{ht}}}(z) = \begin{cases} z, & z > \sqrt{2\nu C}, \\ z \text{ or } 0, & z = \sqrt{2\nu C}, \\ 0, & z \in [0, \sqrt{2\nu C}), \\ z, & z < 0. \end{cases} \tag{13}$$

Proof. Let $\kappa := \nu C$. By (10), the scalar objective is

$$\varphi(s) := \begin{cases} \frac{(s-z)^2}{2}, & s \leq 0, \\ \frac{6\kappa s}{5} + \frac{(s-z)^2}{2}, & s \in (0, \frac{2}{3}], \\ \frac{\kappa(-9s^2 + 18s - 4)}{5} + \frac{(s-z)^2}{2}, & s \in (\frac{2}{3}, 1], \\ \kappa + \frac{(s-z)^2}{2}, & s > 1. \end{cases} \tag{14}$$

The candidate minimizers from the four pieces and the junction at $s = 0$ are

$$s_1 = z, \quad s_2 = z - \frac{6\kappa}{5}, \quad s_3 = \frac{5z - 18\kappa}{5 - 18\kappa},$$
$$s_4 = z, \quad s_5 = 0.$$

subject to the interval in which each candidate is feasible.

Case (i): $0 < \kappa < 5/18$. Here the quadratic coefficient of the third piece is positive, so s_3 remains a candidate. Direct comparison of the feasible candidate values gives: $s_4 = z$ for $z > 1$; $s_3 = (5z - 18\kappa)/(5 - 18\kappa)$ for $z \in (2/3 + 6\kappa/5, 1]$; $s_2 = z - 6\kappa/5$ for $z \in (6\kappa/5, 2/3 + 6\kappa/5]$; $s_5 = 0$ for $z \in [0, 6\kappa/5]$; and $s_1 = z$ for $z < 0$. These intervals and candidates give (11).

Case (ii): $5/18 \leq \kappa < 25/18$. The third piece is no longer strictly convex and cannot contribute an interior minimizer. Comparing s_1, s_2, s_4, and s_5 yields $s_4 = z$ for $z > 5/6 + 3\kappa/5$; $s_2 = z - 6\kappa/5$ for $z \in (6\kappa/5, 5/6 + 3\kappa/5]$; $s_5 = 0$ for $z \in [0, 6\kappa/5]$; and $s_1 = z$ for $z < 0$. This is (12).

Case (iii): $\kappa \geq 25/18$. Only the unchanged candidate z and the zero candidate can be globally optimal on the nonnegative half-line. Their objective values are $\varphi(z) = \kappa$ and $\varphi(0) = z^2/2$ whenever the unchanged candidate lies in the capped region. Hence z is selected when $z > \sqrt{2\kappa}$, both z and 0 minimize the objective at equality, and 0 is selected for $z \in [0, \sqrt{2\kappa})$. For $z < 0$, the zero-loss branch gives the unique minimizer $s = z$. This proves (13). □

Definition 2. Given $\nu, C > 0$, the proximal operator of L_{ht} at $\mathbf{z} = (z_1, z_2, \dots, z_m)^\top \in \mathbb{R}^m$ is defined by

$$\mathrm{prox}_{\nu C L_{\mathrm{ht}}}(\mathbf{z}) := \begin{pmatrix} \mathrm{prox}_{\nu C \ell_{\mathrm{ht}}}(z_1) \\ \vdots \\ \mathrm{prox}_{\nu C \ell_{\mathrm{ht}}}(z_m) \end{pmatrix} \in \mathbb{R}^m, \tag{15}$$

where $\mathrm{prox}_{\nu C \ell_{\mathrm{ht}}}(z_i)$ is given in Lemma 2.

In this paper, we will utilize Eqs. (8)–(9) and (15) to establish the optimality theory of L_{ht}-SVM. Additionally, Eq. (15) will be used to develop an efficient algorithm for solving L_{ht}-SVM.

3.4. First-order local optimality conditions

We next establish first-order conditions for local optimality of (7). These conditions motivate the algorithmic stopping rule developed in Section 4. We begin with the definition of a proximal stationary (P-stationary) point induced by the L_{ht} proximal operator.

Definition 3. The $(\mathbf{w}^*; b^*; \mathbf{q}^*)$ is called a P-stationary point of (7) if there is $\nu, C > 0$ and $\boldsymbol{\psi}^* \in \mathbb{R}^m$ such that

$$\begin{cases} \mathbf{w}^* + N^\top \boldsymbol{\psi}^* = \mathbf{0}, \\ \langle \mathbf{y}, \boldsymbol{\psi}^* \rangle = 0, \\ \mathbf{q}^* + N\mathbf{w}^* + b^* \mathbf{y} = \mathbf{1}, \\ \mathrm{prox}_{\nu C L_{\mathrm{ht}}}(\mathbf{q}^* - \nu \boldsymbol{\psi}^*) \ni \mathbf{q}^*. \end{cases} \tag{16}$$

The four relations in (16) have distinct roles. The first balances the regularization term against the sample multipliers; the second is the stationarity condition for the unregularized intercept; the third restores the definition of the margin-residual variable $\mathbf{q}$; and the fourth requires every residual to be a fixed point of the loss-specific proximal mapping. The last relation is what converts the piecewise loss into an active-sample characterization: residuals on a flat branch have zero multiplier and do not determine the separating hyperplane, whereas residuals selected by the non-flat branches can remain active. In this sense, P-stationarity serves both as a local optimality concept and as a computable stopping target for the algorithm.

Given $C > 0$ and $\mathbf{q}^* \in \mathbb{R}^m$, denote

$$\mathcal{A}^* := \{k \in [m] : q_k^* > 1\}, \quad \mathcal{B}^* := \{k \in [m] : q_k^* \in (\tfrac{2}{3}, 1]\},$$
$$\mathcal{C}^* := \{k \in [m] : q_k^* \in (0, \tfrac{2}{3}]\}, \quad \mathcal{D}^* := \{k \in [m] : q_k^* = 0\},$$

$$\mathcal{E}^* := \{k \in [m] : q_k^* < 0\}, \quad \mathcal{P}^* := \{k \in [m] : q_k^* \in (0, \tfrac{5}{6})\},$$
$$\mathcal{Q}^* := \{k \in [m] : q_k^* = \tfrac{5}{6}\}, \quad \mathcal{R}^* := \{k \in [m] : q_k^* > \tfrac{5}{6}\},$$

$$v_1^* := \begin{cases} \frac{5}{18C}, & \mathcal{B}^* \neq \emptyset, \\ +\infty, & \mathcal{B}^* = \emptyset, \end{cases} \tag{17}$$

$$v_2^* := \begin{cases} \min_{k\in\mathcal{R}^*} \frac{5(6q_k^*-5)}{18C}, & \mathcal{R}^* \neq \emptyset, \\ +\infty, & \mathcal{R}^* = \emptyset, \end{cases} \tag{18}$$

$$v_3^* := \begin{cases} \min_{k\in\mathcal{P}^*} \frac{5(5-6q_k^*)}{18C}, & \mathcal{P}^* \neq \emptyset, \\ +\infty, & \mathcal{P}^* = \emptyset, \end{cases} \tag{19}$$

$$v_4^* := \begin{cases} \frac{25}{18C}, & \mathcal{D}^* \neq \emptyset, \\ +\infty, & \mathcal{D}^* = \emptyset, \end{cases} \tag{20}$$

$$v_5^* := \begin{cases} +\infty, & \mathbf{q}^* \leq 0, \\ \min\{\frac{(q_k^*)^2}{2C} : q_k^* > 0\}, & \text{otherwise.} \end{cases} \tag{21}$$

Using the above notation, we give necessary and sufficient conditions for local optimality of (7).

Theorem 1 (First-order necessary condition). *Let $(\mathbf{w}^*; b^*; \mathbf{q}^*)$ be a local minimizer of (7). Then it is a P-stationary point with $0 < \nu \leq \nu^* := \min\{v_1^*, v_2^*, v_3^*, v_4^*, v_5^*\}$.*

Proof outline. Local optimality first gives the multiplier system consisting of the first three equations in (16) and $\mathbf{0} \in \boldsymbol{\psi}^* + C\partial L_{\text{ht}}(\mathbf{q}^*)$. Lemma 1 specifies ψ_i^* on each loss interval. The bounds $\nu \leq v_j^*$ place $q_i^* - \nu\psi_i^*$ in the corresponding interval of Lemma 2, so the coordinate-wise proximal fixed-point condition follows. Appendix A gives the complete case analysis. □

Theorem 2 (First-order sufficient condition). *Suppose that $(\mathbf{w}^*; b^*; \mathbf{q}^*)$ with $\boldsymbol{\psi}^*$ is a P-stationary point of (7). Then it is a local minimizer of (7).*

Proof outline. For a feasible point near the P-stationary point, convexity of $\|\mathbf{w}\|^2$ and the first three relations in (16) reduce the local objective difference to coordinate-wise loss terms involving $\boldsymbol{\psi}^*$. The proximal fixed-point relation partitions the coordinates into the same three νC regimes as Lemma 2; on each active interval the resulting loss–multiplier contribution is nonnegative, while the two flat branches are locally constant. Summing these contributions proves local minimality. The complete inequalities are given in Appendix B. □

Theorems 1 and 2 show that, under the stated bounds on the proximal parameter ν, P-stationary points coincide with local minimizers of (7). Thus, an exact P-stationary point, including the limit in Theorem 6, is not a saddle point. Since the classical Hessian test is not applicable at the loss junctions or on its flat branches, finite-iteration escape from other nonsmooth critical configurations is not established. Initialization sensitivity is instead examined empirically in Section 6.1.5. In the next section, the P-stationary characterization is used to define support vectors and a computable stopping condition.

4. ADMM with a working set

We now turn the optimality theory into an implementable solver. The support-vector characterization below identifies the samples that can affect $\mathbf{w}$ at a P-stationary point; the algorithm exploits this structure by restricting the linear-system update to a working set of such samples, while the residual computations still involve all samples.

4.1. L_{ht} support vectors

The separating hyperplane of L_{ht}-SVM is fully determined by its support vectors; all remaining training samples can be discarded. Reducing the support-vector count can therefore improve prediction sparsity and lower inference-time storage. Using the optimality conditions derived in Section 3, we formally define the L_{ht} support vectors below.

Theorem 3 (L_{ht} support vectors for $\nu C \in (0, 5/18)$). *For $\nu C \in (0, 5/18)$, let $(\mathbf{w}^*; b^*; \mathbf{q}^*)$ with $\boldsymbol{\psi}^*$ be a P-stationary point of (7). Then, the $\mathbf{w}^*$ satisfies*

$$\mathbf{w}^* = -\sum_{i\in\mathbb{G}^*} \psi_i^* y_i \mathbf{x}_i \text{ and } \psi_i^* = 0, i \in \overline{\mathbb{G}}^*, \tag{22}$$

$$\mathbb{G}_1^* := \{i \in [m] : p_i^* < 0\}, \quad \mathbb{G}_2^* := \{i \in [m] : p_i^* \in [0, \widetilde{\beta}]\},$$
$$\mathbb{G}_3^* := \{i \in [m] : p_i^* \in (\widetilde{\beta}, \widehat{\beta})\},$$
$$\mathbb{G}_4^* := \{i \in [m] : p_i^* \in [\widehat{\beta}, 1)\}, \quad \mathbb{G}_5^* := \{i \in [m] : p_i^* \geq 1\},$$

where $\mathbf{p}^* := \mathbf{q}^* - \nu\boldsymbol{\psi}^*$, $\widetilde{\beta} := 6\nu C/5$, $\widehat{\beta} := 2/3 + 6\nu C/5$, $\mathbb{G}^* := \mathbb{G}_2^* \cup \mathbb{G}_3^* \cup \mathbb{G}_4^*$ and $\overline{\mathbb{G}}^* := [m]\backslash\mathbb{G}^*$. The training vectors $\{\mathbf{x}_i \in \mathbb{R}^n : i \in \mathbb{G}^*\}$ are called the L_{ht} support vectors and meet

$$\begin{cases} y_i(\langle \mathbf{w}^*, \mathbf{x}_i\rangle + b^*) = 1, & i \in \mathbb{G}_2^*, \\ y_i(\langle \mathbf{w}^*, \mathbf{x}_i\rangle + b^*) \in (1/3, 1), & i \in \mathbb{G}_3^*, \\ y_i(\langle \mathbf{w}^*, \mathbf{x}_i\rangle + b^*) \in (0, 1/3], & i \in \mathbb{G}_4^*. \end{cases} \tag{23}$$

Proof. The proof is presented in Appendix C. □

Hence, we can deduce that the L_{ht}-SVM has sparsity since the well-classified training vectors with $y_i(\langle \mathbf{w}^*, \mathbf{x}_i\rangle + b^*) > 1$ are not support vectors and the L_{ht}-SVM enjoys robustness to outliers since the misclassified training vectors with $y_i(\langle \mathbf{w}^*, \mathbf{x}_i\rangle + b^*) \leq 0$ fall into the flat region of the loss and are not support vectors.

Theorem 4. *[L_{ht} support vectors for $\nu C \in [5/18, 25/18)$] For $\nu C \in [5/18, 25/18)$, let $(\mathbf{w}^*; b^*; \mathbf{q}^*)$ with $\boldsymbol{\psi}^*$ be a P-stationary point of (7). Then, the $\mathbf{w}^*$ satisfies*

$$\mathbf{w}^* = -\sum_{i\in\mathbb{I}^*} \psi_i^* y_i \mathbf{x}_i \text{ and } \psi_i^* = 0, i \in \overline{\mathbb{I}}^*, \tag{24}$$

$$\mathbb{I}_1^* := \{i \in [m] : p_i^* < 0\}, \quad \mathbb{I}_2^* := \{i \in [m] : p_i^* \in [0, \widetilde{\beta}]\},$$
$$\mathbb{I}_3^* := \{i \in [m] : p_i^* \in (\widetilde{\beta}, \widehat{\beta}) \text{ or } p_i^* = \widehat{\beta}, \psi_i^* \neq 0\},$$
$$\mathbb{I}_4^* := \{i \in [m] : p_i^* > \widehat{\beta} \text{ or } p_i^* = \widehat{\beta}, \psi_i^* = 0\},$$

where $\mathbf{p}^* := \mathbf{q}^* - \nu\boldsymbol{\psi}^*$, $\widetilde{\beta} := 6\nu C/5$, $\widehat{\beta} := 5/6 + 3\nu C/5$, $\mathbb{I}^* := \mathbb{I}_2^* \cup \mathbb{I}_3^*$ and $\overline{\mathbb{I}}^* := [m]\backslash\mathbb{I}^*$. The training vectors $\{\mathbf{x}_i \in \mathbb{R}^n : i \in \mathbb{I}^*\}$ are called the L_{ht} support vectors and meet

$$\begin{cases} y_i(\langle \mathbf{w}^*, \mathbf{x}_i\rangle + b^*) = 1, & i \in \mathbb{I}_2^*, \\ y_i(\langle \mathbf{w}^*, \mathbf{x}_i\rangle + b^*) \in [1/6 + 3\nu C/5, 1), & i \in \mathbb{I}_3^*. \end{cases} \tag{25}$$

The proof is similar to that of Theorem 3 and thus is omitted. We can deduce that the L_{ht}-SVM has sparsity since the well-classified training vectors with $y_i(\langle \mathbf{w}^*, \mathbf{x}_i\rangle + b^*) > 1$ are not support vectors and the L_{ht}-SVM enjoys robustness to outliers since the training vectors with $y_i(\langle \mathbf{w}^*, \mathbf{x}_i\rangle + b^*) < 1/6 + 3\nu C/5$, including all misclassified ones, are not support vectors.

Theorem 5 (L_{ht} support vectors for $\nu C \geq 25/18$). *For $\nu C \geq 25/18$, suppose that $(\mathbf{w}^*; b^*; \mathbf{q}^*)$ with $\boldsymbol{\psi}^*$ is a P-stationary point of (7). Then, the $\mathbf{w}^*$ satisfies*

$$\mathbf{w}^* = -\sum_{i\in\mathbb{K}^*} \psi_i^* y_i \mathbf{x}_i \text{ and } \psi_i^* = 0, i \in \overline{\mathbb{K}}^*, \tag{26}$$

$$\mathbb{K}_1^* := \{i \in [m] : p_i^* < 0\}, \quad \mathbb{K}_2^* := \{i \in [m] : p_i^* \in [0, \sqrt{2\nu C})\},$$
$$\mathbb{K}_3^* := \{i \in [m] : p_i^* = \sqrt{2\nu C}, \psi_i^* \neq 0\},$$
$$\mathbb{K}_4^* := \{i \in [m] : p_i^* > \sqrt{2\nu C} \text{ or } p_i^* = \sqrt{2\nu C}, \psi_i^* = 0\},$$

where $\mathbf{p}^* := \mathbf{q}^* - \nu\boldsymbol{\psi}^*$, $\mathbb{K}^* := \mathbb{K}_2^* \cup \mathbb{K}_3^*$ and $\overline{\mathbb{K}}^* := [m]\backslash\mathbb{K}^*$. The training vectors $\{\mathbf{x}_i \in \mathbb{R}^n : i \in \mathbb{K}^*\}$ are called the L_{ht} support vectors and meet

$$y_i(\langle \mathbf{w}^*, \mathbf{x}_i\rangle + b^*) = 1, \quad i \in \mathbb{K}^*. \tag{27}$$

The proof is similar to that of Theorem 3 and thus is omitted. We can deduce that the L_{ht}-SVM has sparsity since the well-classified training vectors with $y_i(\langle \mathbf{w}^*, \mathbf{x}_i\rangle + b^*) > 1$ are not support vectors and the L_{ht}-SVM enjoys robustness to outliers since the training vectors with $y_i(\langle \mathbf{w}^*, \mathbf{x}_i\rangle + b^*) < 1$, including all misclassified ones, are not support vectors. Based on Theorems 3–5, we deduce that the L_{ht} support vectors are a small portion of the training set and the L_{ht}-SVM possesses sparsity and robustness to outliers, which plays an important role in constructing a new working set in the next part. Motivated by this, we will establish an ADMM with working set for dealing with L_{ht}-SVM (7).

4.2. L_{ht}-ADMM framework

To handle the equality constraint in (7), we introduce the augmented Lagrangian function

$$\mathcal{L}_\xi(\mathbf{w}; b; \mathbf{q}; \boldsymbol{\psi}) = \frac{1}{2}\|\mathbf{w}\|^2 + CL_{\mathrm{ht}}(\mathbf{q}) + \langle \boldsymbol{\psi}, \mathbf{q} - \mathbf{1} + N\mathbf{w} + b\mathbf{y}\rangle + \frac{\xi}{2}\|\mathbf{q} - \mathbf{1} + N\mathbf{w} + b\mathbf{y}\|^2, \tag{28}$$

where $\boldsymbol{\psi} \in \mathbb{R}^m$ represents the Lagrangian multiplier and $\xi > 0$ is the penalty parameter. Denote $\nu := 1/\xi$. Starting from an initial point $(\mathbf{w}^0; b^0; \mathbf{q}^0; \boldsymbol{\psi}^0)$, the ADMM iteration proceeds as

$$\begin{aligned}
\mathbf{q}^{k+1} &= \arg\min_{\mathbf{q}\in\mathbb{R}^m} \mathcal{L}_\xi(\mathbf{w}^k, b^k, \mathbf{q}, \boldsymbol{\psi}^k),\\
\mathbf{w}^{k+1} &= \arg\min_{\mathbf{w}\in\mathbb{R}^n} \frac{\xi}{2}\|\mathbf{w} - \mathbf{w}^k\|^2_{Z_k} + \mathcal{L}_\xi(\mathbf{w}, b^k, \mathbf{q}^{k+1}, \boldsymbol{\psi}^k),\\
b^{k+1} &= \arg\min_{b\in\mathbb{R}} \mathcal{L}_\xi(\mathbf{w}^{k+1}, b, \mathbf{q}^{k+1}, \boldsymbol{\psi}^k),\\
\boldsymbol{\psi}^{k+1} &= \boldsymbol{\psi}^k + \tau\xi(\mathbf{q}^{k+1} - \mathbf{1} + N\mathbf{w}^{k+1} + b^{k+1}\mathbf{y}).
\end{aligned} \tag{29}$$

where $Z_k \in \mathbb{R}^{n\times n}$ is a symmetric matrix and $\tau \in (0, (1+\sqrt{5})/2)$ denotes the dual step-size parameter. Here $\|\mathbf{w} - \mathbf{w}^k\|^2_{Z_k} := \langle \mathbf{w} - \mathbf{w}^k, Z_k(\mathbf{w} - \mathbf{w}^k)\rangle$.

For the construction of Z_k, we first establish a working set F_k based on the L_{ht} support vectors. Let $\mathbf{p}^k := \mathbf{1} - N\mathbf{w}^k - b^k\mathbf{y} - \nu\boldsymbol{\psi}^k$ and define

$$\begin{aligned}
&T_k^1 := \{i \in [m] : p_i^k \in [0, \widetilde{\beta}]\}, \quad T_k^2 := \{i \in [m] : p_i^k \in (\widetilde{\beta}, \widehat{\beta})\},\\
&J_k^1 := \{i \in [m] : p_i^k \in (\widetilde{\beta}, \widehat{\beta}) \text{ or } p_i^k = \widehat{\beta}, \psi_i^k \neq 0\},\\
&I_k := \{i \in [m] : p_i^k \in [0, \sqrt{2\nu C}) \text{ or } p_i^k = \sqrt{2\nu C}, \psi_i^k \neq 0\},
\end{aligned}$$

where $\widetilde{\beta} := 6\nu C/5$ and

$$\widehat{\beta} := \begin{cases} 2/3 + 6\nu C/5, & \nu C \in (0, 5/18),\\ 5/6 + 3\nu C/5, & \nu C \in [5/18, 25/18),\\ \sqrt{2\nu C}, & \nu C \geq 25/18. \end{cases}$$

Based on these index sets, the working set F_k at iteration k is constructed as

$$F_k := \begin{cases} T_k^1 \cup T_k^2 \cup T_k^3, & \nu C \in (0, \frac{5}{18}),\\ J_k^1 \cup J_k^2, & \nu C \in [\frac{5}{18}, \frac{25}{18}),\\ I_k, & \nu C \geq \frac{25}{18}, \end{cases} \tag{30}$$

with $T_k^3 := \{i \in [m] : p_i^k \in [\widehat{\beta}, 1)\}$ and $J_k^2 := \{i \in [m] : p_i^k \in [0, \widetilde{\beta}]\}$. Denote $\overline{F}_k := [m]\backslash F_k$ as the complement of F_k. The matrix Z_k is then chosen as

$$Z_k = -N_{\overline{F}_k}^\top N_{\overline{F}_k}, \tag{31}$$

where $N_{\overline{F}_k} \in \mathbb{R}^{|\overline{F}_k|\times n}$ is the sub-matrix formed by rows of N corresponding to indices in $\overline{F}_k$, and $|\overline{F}_k|$ represents its cardinality. This selection of Z_k and F_k reduces the linear-system dimension by excluding current non-support-vector samples from that update. We now derive the closed-form solution for each sub-problem in (29).

(i) Updating $\mathbf{q}^{k+1}$. The $\mathbf{q}$-subproblem in (29) can be reformulated as

$$\begin{aligned}
\mathbf{q}^{k+1} &= \arg\min_{\mathbf{q}\in\mathbb{R}^m} CL_{\mathrm{ht}}(\mathbf{q}) + \langle \boldsymbol{\psi}^k, \mathbf{q}\rangle + \frac{\xi}{2}\|\mathbf{q} - \mathbf{1} + N\mathbf{w}^k + b^k\mathbf{y}\|^2\\
&= \arg\min_{\mathbf{q}\in\mathbb{R}^m} CL_{\mathrm{ht}}(\mathbf{q}) + \frac{\xi}{2}\|\mathbf{q} - \mathbf{p}^k\|^2 = \mathrm{prox}_{\nu CL_{\mathrm{ht}}}(\mathbf{p}^k),
\end{aligned} \tag{32}$$

where $\mathbf{p}^k = \mathbf{1} - N\mathbf{w}^k - b^k\mathbf{y} - \nu\boldsymbol{\psi}^k$ as defined above. Combining this with (11)–(13) yields the closed-form solution via Lemma 2.

(ii) Updating $\mathbf{w}^{k+1}$. The $\mathbf{w}$-subproblem takes the form

$$\begin{aligned}
\mathbf{w}^{k+1} &= \arg\min_{\mathbf{w}\in\mathbb{R}^n} \frac{1}{2}\|\mathbf{w}\|^2 + \frac{\xi}{2}\|\mathbf{w} - \mathbf{w}^k\|^2_{Z_k}\\
&\quad + \langle \boldsymbol{\psi}^k, N\mathbf{w}\rangle + \frac{\xi}{2}\|\mathbf{q}^{k+1} - \mathbf{1} + N\mathbf{w} + b^k\mathbf{y}\|^2.
\end{aligned} \tag{33}$$

Since this is a convex quadratic problem with differentiable objective, the optimality condition leads to the equation

$$\mathbf{0} = \mathbf{w} - \xi N_{\overline{F}_k}^\top N_{\overline{F}_k}(\mathbf{w} - \mathbf{w}^k) + N^\top\boldsymbol{\psi}^k + \xi N^\top(\mathbf{q}^{k+1} - \mathbf{1} + N\mathbf{w} + b^k\mathbf{y}). \tag{34}$$

This can be equivalently rewritten as the linear system

$$(I + \xi N_{F_k}^\top N_{F_k})\mathbf{w} = \xi N_{F_k}^\top \boldsymbol{\chi}_{F_k}^k, \tag{35}$$

where $\boldsymbol{\chi}^k := -(\mathbf{q}^{k+1} + b^k\mathbf{y} - \mathbf{1} + \boldsymbol{\psi}^k/\xi)$. The derivation of (35) from (34) relies on the identity

$$N_{F_k}^\top N_{F_k} = N^\top N - N_{\overline{F}_k}^\top N_{\overline{F}_k}.$$

Notably, samples indexed by $\overline{F}_k$ do not appear in (35), meaning that the working set strategy allows us to skip computations involving $\{\mathbf{x}_j, j \in \overline{F}_k\}$. This lowers the linear-system cost when $|F_k|$ is small relative to m, although the full residual calculation described in Section 4.4 remains. The linear system (35) is solved differently depending on the relationship between n and $|F_k|$. When $n \leq |F_k|$, we directly solve (35) via

$$\mathbf{w}^{k+1} = (I + \xi N_{F_k}^\top N_{F_k})^{-1}\xi N_{F_k}^\top \boldsymbol{\chi}_{F_k}^k. \tag{36}$$

When $n > |F_k|$, we apply the Sherman–Morrison–Woodbury formula to obtain

$$(I + \xi N_{F_k}^\top N_{F_k})^{-1} = I - \xi N_{F_k}^\top (I + \xi N_{F_k} N_{F_k}^\top)^{-1} N_{F_k},$$

which gives the update

$$\mathbf{w}^{k+1} = \xi N_{F_k}^\top (I + \xi N_{F_k} N_{F_k}^\top)^{-1}\boldsymbol{\chi}_{F_k}^k. \tag{37}$$

(iii) Updating b^{k+1}. For the b-subproblem, we minimize

$$b^{k+1} = \arg\min_{b\in\mathbb{R}} \langle \boldsymbol{\psi}^k, b\mathbf{y}\rangle + \frac{\xi}{2}\|\mathbf{q}^{k+1} - \mathbf{1} + N\mathbf{w}^{k+1} + b\mathbf{y}\|^2.$$

This univariate quadratic minimization admits the closed-form solution

$$b^{k+1} = \langle \mathbf{y}, \mathbf{r}^k\rangle/\|\mathbf{y}\|^2 = \langle \mathbf{y}, \mathbf{r}^k\rangle/m, \tag{38}$$

where $\mathbf{r}^k := -N\mathbf{w}^{k+1} + \mathbf{1} - \mathbf{q}^{k+1} - \boldsymbol{\psi}^k/\xi$.

(iv) Updating $\boldsymbol{\psi}^{k+1}$. The multiplier update from (29) becomes

$$\boldsymbol{\psi}_{F_k}^{k+1} = \boldsymbol{\psi}_{F_k}^k + \tau\xi\Lambda_{F_k}^{k+1}, \quad \boldsymbol{\psi}_{\overline{F}_k}^{k+1} = \mathbf{0}, \tag{39}$$

where $\boldsymbol{\Lambda}^{k+1} := \mathbf{q}^{k+1} - \mathbf{1} + N\mathbf{w}^{k+1} + b^{k+1}\mathbf{y}$. The assignment $\boldsymbol{\psi}_{\overline{F}_k}^{k+1} = \mathbf{0}$ follows from Theorems 3–5.

The four updates play distinct roles. The $\mathbf{q}$-update applies the exact proximal mapping and decides which margin residuals are suppressed, graded, or capped; the $\mathbf{w}$-update fits the hyperplane on the current working set; the scalar b-update recenters the decision function; and the multiplier update reduces violation of the residual-defining equality. This sequence links the loss geometry in Section 3.3 directly to the stopping conditions in (16).

The complete procedure is summarized in Algorithm 1.

The working set is recomputed from $\mathbf{p}^k$ and $\boldsymbol{\psi}^k$ at every iteration; exclusion is therefore not permanent. A sample outside F_k can re-enter F_{k+1} when its updated residual crosses a proximal threshold. Conversely, membership at an early iteration does not certify that the sample will remain a support vector at the limit. This distinction is important because the finite-iterate working set is an algorithmic device, whereas

Algorithm 1 L_{ht}-ADMM for handling problem (7).

Input: Data $\{(\mathbf{x}_i, y_i)\}_{i=1}^m$, parameters $C, \xi > 0$, maximum iterations K.
Initialize $(\mathbf{w}^0; b^0; \mathbf{q}^0; \boldsymbol{\psi}^0)$. Set $\nu := 1/\xi$, $k := 0$.
while $k \le K$ and the stop condition does not hold **do**
  Compute F_k as in (30).
  Compute $\mathbf{q}^{k+1}$ by (32).
  Compute $\mathbf{w}^{k+1}$ by (36) if $n \le |F_k|$ and (37) otherwise.
  Compute b^{k+1} by (38).
  Calculate $\boldsymbol{\psi}^{k+1}$ by (39).
  Set $k = k + 1$.
end while
return the solution $(\mathbf{w}^k, b^k)$ to (7).

Table 2
Details of the 15 real datasets.

Datasets	Source	m	m_t	n
australian	UCI	690	–	14
breast-cancer	UCI	683	–	9
mushroom	UCI	8124	–	22
wine-red	UCI	1599	–	11
wine-white	UCI	4898	–	11
a1a	LIBSVM*	1605	30,956	123
a2a	LIBSVM*	2265	30,296	123
a4a	LIBSVM*	4781	27,780	123
a6a	LIBSVM*	11,220	21,341	123
a7a	LIBSVM*	16,100	16,461	123
a8a	LIBSVM*	22,696	9865	123
a9a	LIBSVM*	32,561	16,281	123
w2a	LIBSVM*	3470	46,279	300
w5a	LIBSVM*	9888	39,861	300
w6a	LIBSVM*	17,188	32,561	300

m: number of training samples; m_t: number of testing samples; n: number of features.
* LIBSVM datasets use the predefined train/test split.

Theorems 3–5 characterize a P-stationary point. Empirical re-entry frequencies, together with a comparison against the full update that keeps all samples in every $\mathbf{w}$-subproblem, are reported in Table 12.

Following Theorem 2, the P-stationary point serves as the termination criterion for L_{ht}-ADMM. Specifically, the algorithm terminates when the current iterate $(\mathbf{w}^k; b^k; \mathbf{q}^k; \boldsymbol{\psi}^k)$ approximately satisfies (16), i.e.,

$$\max\{\epsilon_1^k, \epsilon_2^k, \epsilon_3^k, \epsilon_4^k\} < \epsilon,$$

where $\epsilon > 0$ is the prescribed tolerance, and

$$\epsilon_1^k := \frac{\|\mathbf{w}^k + N_{F_k}^\top \boldsymbol{\psi}_{F_k}^k\|}{1 + \|\mathbf{w}^k\|}, \quad \epsilon_2^k := \frac{|\langle \mathbf{y}_{F_k}, \boldsymbol{\psi}_{F_k}^k \rangle|}{1 + |F_k|},$$
$$\epsilon_3^k := \frac{\|\mathbf{q}^k - \mathbf{1} + N\mathbf{w}^k + b^k\mathbf{y}\|}{\sqrt{m}},$$
$$\epsilon_4^k := \frac{\|\mathbf{q}^k - \mathrm{prox}_{\nu C L_{\mathrm{ht}}}(\mathbf{p}^k)\|}{1 + \|\mathbf{q}^k\|}.$$

Here ϵ_1^k and ϵ_2^k monitor stationarity with respect to the hyperplane and intercept, ϵ_3^k is the normalized primal-feasibility residual, and ϵ_4^k measures the proximal fixed-point error. Requiring their maximum to be small prevents termination based only on a stable objective value while one of the defining P-stationary relations is still inaccurate.

4.3. Limit-point stationarity of a convergent sequence

We characterize the limit of the algorithm under the explicit assumption that the generated sequence converges.

Theorem 6 (Limit-point stationarity). *If the sequence $\{(\mathbf{w}^k; b^k; \mathbf{q}^k; \boldsymbol{\psi}^k)\}$ generated by L_{ht}-ADMM converges to $(\mathbf{w}^*; b^*; \mathbf{q}^*; \boldsymbol{\psi}^*)$, then $(\mathbf{w}^*; b^*; \mathbf{q}^*)$ is a P-stationary point of (7) and consequently a local minimizer of (7).*

Proof. Since $F_k \subseteq [m]$ has only finitely many configurations, there is an infinite index set Ψ on which $F_i \equiv F$. Write $\Phi^k = (\mathbf{w}^k, b^k, \mathbf{q}^k, \boldsymbol{\psi}^k)$. The assumed convergence of $\{\Phi^k\}$ implies that both $\{\Phi^i\}_{i\in\Psi}$ and $\{\Phi^{i+1}\}_{i\in\Psi}$ converge to Φ^*. Taking the limit of (39) along Ψ gives

$$\boldsymbol{\psi}_F^* = \boldsymbol{\psi}_F^* + \tau\xi\boldsymbol{\Lambda}_F^*, \qquad \boldsymbol{\psi}_{\bar{F}}^* = \mathbf{0}, \tag{40}$$

where $\boldsymbol{\Lambda}^{k+1} = \mathbf{q}^{k+1} - \mathbf{1} + N\mathbf{w}^{k+1} + b^{k+1}\mathbf{y}$. Thus $\boldsymbol{\Lambda}_F^* = \mathbf{0}$. For $i \in \bar{F}$, the working-set construction places p_i^* on a proximal identity branch, and $\psi_i^* = 0$; hence $q_i^* = p_i^* = 1 - (N\mathbf{w}^*)_i - b^* y_i$. Therefore $\boldsymbol{\Lambda}_{\bar{F}}^* = \mathbf{0}$ as well, proving

$$\mathbf{q}^* + N\mathbf{w}^* + b^*\mathbf{y} = \mathbf{1}. \tag{41}$$

Taking the limit in the $\mathbf{q}$-update yields

$$\mathbf{q}^* = \mathrm{prox}_{\nu C L_{\mathrm{ht}}}(\mathbf{p}^*), \qquad \mathbf{p}^* = \mathbf{1} - N\mathbf{w}^* - b^*\mathbf{y} - \nu\boldsymbol{\psi}^*.$$

Using (41), this becomes

$$\mathbf{q}^* \in \mathrm{prox}_{\nu C L_{\mathrm{ht}}}(\mathbf{q}^* - \nu\boldsymbol{\psi}^*). \tag{42}$$

Next, the limit of (35) is

$$(I + \xi N_F^\top N_F)\mathbf{w}^* = \xi N_F^\top \boldsymbol{\chi}_F^*,$$

where $\boldsymbol{\chi}^* = \mathbf{1} - \mathbf{q}^* - b^*\mathbf{y} - \boldsymbol{\psi}^*/\xi$. Eq. (41) gives $\boldsymbol{\chi}_F^* = N_F\mathbf{w}^* - \boldsymbol{\psi}_F^*/\xi$. Substitution and (40) then imply

$$\mathbf{w}^* = -N_F^\top \boldsymbol{\psi}_F^* = -N^\top \boldsymbol{\psi}^*.$$

Finally, taking the limit in (38) gives $b^* = \langle \mathbf{y}, \mathbf{r}^* \rangle / m$, where $\mathbf{r}^* = \mathbf{1} - N\mathbf{w}^* - \mathbf{q}^* - \boldsymbol{\psi}^*/\xi$. Again using (41), $\mathbf{r}^* = b^*\mathbf{y} - \boldsymbol{\psi}^*/\xi$, and hence

$$b^* = b^* - \frac{\langle \mathbf{y}, \boldsymbol{\psi}^* \rangle}{m\xi},$$

so $\langle \mathbf{y}, \boldsymbol{\psi}^* \rangle = 0$. Together with (41) and (42), these are exactly the four conditions in (16). Theorem 2 then gives local minimality. □

Theorem 6 assumes that the generated sequence converges, an assumption also adopted in the analysis of $L_{0/1}$-ADMM (Wang et al., 2022). Since (7) is non-convex and non-smooth, establishing convergence of the whole sequence, together with its rate, is left for future work.

4.4. Computational complexity analysis

We now examine the per-iteration cost of L_{ht}-ADMM. Constructing the working set F_k requires $\mathcal{O}(m)$ operations. For the $\mathbf{q}$-update, the dominant cost is the matrix-vector product $N\mathbf{w}^k$, requiring $\mathcal{O}(mn)$ operations. The $\mathbf{w}$-update is dominated by computing $N\mathbf{w}^{k+1}$, costing $\mathcal{O}(mn)$. The b-update reuses the product $N\mathbf{w}^{k+1}$, adding $\mathcal{O}(m)$ cost. The linear system solve for $\mathbf{w}^{k+1}$ (matrix inversion) costs $\mathcal{O}(\min\{n^2, |F_k|^2\}\max\{n, |F_k|\})$. The Gram matrix $N_{F_k}N_{F_k}^\top$ requires $\mathcal{O}(n|F_k|^2)$ operations, and its inverse $(I + \xi N_{F_k}N_{F_k}^\top)^{-1}$ costs $\mathcal{O}(|F_k|^\omega)$ where $\omega \in (2, 3)$ denotes the matrix multiplication exponent. Summing up, the per-iteration complexity of $\mathbf{w}^{k+1}$ is

$$\mathcal{O}(\min\{n^2, |F_k|^2\}\max\{n, |F_k|\}).$$

The overall per-iteration complexity is therefore

$$\mathcal{O}\big(mn + \min\{n^2, |F_k|^2\}\max\{n, |F_k|\}\big).$$

Thus, the working set reduces the linear-system component whenever $|F_k| \ll m$, while the $\mathcal{O}(mn)$ matrix–vector products required to evaluate the full set of margin residuals remain; the latter term dominates the per-iteration cost when n is large.

For dense storage, the data matrix and the primal/dual vectors require $\mathcal{O}(mn + m + n)$ memory. The active submatrix requires $\mathcal{O}(|F_k|n)$

Table 3
ACC (%) with standard deviations for the UCI datasets.

Dataset	Hinge-SVM	LS-SVM	Pin-SVM	Ramp-SVM	$L_{0/1}$-SVM	L_{ht}-SVM
australian	85.51±3.14	86.09±3.25	85.51±3.14	85.51±3.14	85.80±3.32	**86.09±2.35**
breast-cancer	96.92±0.97	96.05±1.50	96.49±1.25	96.39±1.03	97.07±1.22	**97.22±0.86**
mushroom	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
wine-red	73.98±1.34	74.55±1.58	74.05±1.44	74.16±1.93	**74.73±1.80**	74.61±1.54
wine-white	75.11±0.44	75.05±0.38	75.03±0.72	75.05±0.48	75.31±0.27	**75.38±0.53**
Avg. Acc.	86.30	86.35	86.22	86.22	86.58	**86.66**
Avg. Rank	4.10	3.70	4.70	4.40	2.30	**1.80**

Table 4
Number of support vectors (NSV) for the UCI datasets.

Dataset	Hinge-SVM	LS-SVM	Pin-SVM	Ramp-SVM	$L_{0/1}$-SVM	L_{ht}-SVM
australian	207	690	384	194	56	**44**
breast-cancer	259	683	546	175	**16**	25
mushroom	1008	8124	3555	824	395	**347**
wine-red	770	1599	1279	616	177	**153**
wine-white	2282	4898	3918	1697	**136**	188
Avg. NSV	905	3199	1936	701	156	**151**
Avg. Rank	4.00	6.00	5.00	3.00	1.60	**1.40**

Table 5
TIME (s) for the UCI datasets.

Dataset	Hinge-SVM	LS-SVM	Pin-SVM	Ramp-SVM	$L_{0/1}$-SVM	L_{ht}-SVM
australian	0.024	0.017	0.015	0.613	0.006	**0.005**
breast-cancer	0.047	0.029	0.198	7.871	0.009	**0.004**
mushroom	8.266	11.387	3.692	205.892	**0.570**	0.579
wine-red	0.729	1.063	0.607	5.845	0.126	**0.101**
wine-white	0.693	1.027	0.704	28.087	**0.139**	0.199
Avg. Time	1.952	2.705	1.043	49.662	**0.172**	0.176
Avg. Rank	4.00	4.40	3.60	6.00	1.60	**1.40**

Table 6
ACC (%) for the LIBSVM datasets.

Dataset	Hinge-SVM	LS-SVM	Pin-SVM	Ramp-SVM	$L_{0/1}$-SVM	L_{ht}-SVM
a1a	83.74	83.99	83.51	83.60	83.89	**84.08**
a2a	84.33	84.37	84.30	84.37	84.51	**84.68**
a4a	84.45	84.51	84.34	84.35	**84.63**	84.54
a6a	84.74	84.58	84.24	84.41	84.82	**84.93**
a7a	84.53	84.72	84.39	84.27	84.74	**84.81**
a8a	85.20	85.17	84.94	84.74	85.23	**85.40**
a9a	85.03	84.55	84.42	84.39	**85.05**	84.96
w2a	98.07	98.17	98.14	98.10	98.03	**98.21**
w5a	98.50	98.28	**98.51**	98.32	**98.51**	**98.51**
w6a	98.59	98.58	98.41	97.71	98.62	**98.67**
Avg. Acc.	88.72	88.69	88.52	88.43	88.80	**88.88**
Avg. Rank	3.70	3.55	4.90	5.15	2.30	**1.40**

Table 7
Number of support vectors (NSV) for the LIBSVM datasets.

Dataset	Hinge-SVM	LS-SVM	Pin-SVM	Ramp-SVM	$L_{0/1}$-SVM	L_{ht}-SVM
a1a	632	1605	1605	837	**74**	110
a2a	904	2265	2265	697	**96**	120
a4a	1762	4781	4781	855	227	**213**
a6a	4038	11,204	8388	1151	448	**397**
a7a	5824	16,052	13,626	4816	1217	**918**
a8a	8315	22,694	14,231	4112	1313	**625**
a9a	11,664	32,552	14,281	632	**411**	549
w2a	256	3470	1386	384	120	**115**
w5a	472	9869	2740	422	**169**	208
w6a	882	17,186	3401	723	380	**230**
Avg. NSV	3475	12,168	6670	1463	446	**348**
Avg. Rank	3.80	6.00	5.00	3.20	1.60	**1.40**

Table 8
TIME (s) for the LIBSVM datasets.

Dataset	Hinge-SVM	LS-SVM	Pin-SVM	Ramp-SVM	$L_{0/1}$-SVM	L_{ht}-SVM
a1a	2.170	0.440	0.200	12.820	0.190	**0.110**
a2a	5.780	1.320	**0.300**	89.610	0.410	0.380
a4a	1.190	2.120	3.920	388.210	1.600	**0.610**
a6a	12.240	14.980	7.130	499.660	4.470	**3.160**
a7a	27.210	27.170	7.280	839.180	5.610	**5.430**
a8a	44.950	75.840	12.350	1230.230	5.240	**4.780**
a9a	75.130	220.780	22.770	1713.840	**5.400**	7.440
w2a	2.340	1.590	4.330	218.150	**0.860**	0.960
w5a	14.320	41.610	6.680	388.950	**1.300**	1.430
w6a	31.660	40.070	9.100	911.920	7.080	**2.530**
Avg. Time	21.700	42.590	7.410	629.260	3.220	**2.680**
Avg. Rank	4.10	4.40	3.20	6.00	1.90	**1.40**

additional storage, while the matrix used by the selected linear-system branch requires $\mathcal{O}(n^2)$ or $\mathcal{O}(|F_k|^2)$. An upper bound on the peak storage is consequently

$$\mathcal{O}\big(mn + m + n + |F_k|n + \max\{n^2, |F_k|^2\}\big).$$

The working set can reduce temporary linear-algebra storage, but the $\mathcal{O}(mn)$ storage of a dense training matrix remains.

5. Application in multi-view learning

This section generalises L_{ht}-SVM to the multi-view setting. We first describe the multi-view task formulation, then construct the structural information matrices, and finally detail the optimisation procedure.

5.1. Multi-view learning task

Consider a multi-view training set $\mathcal{D} = \{(\mathbf{x}_i^{(1)}, \ldots, \mathbf{x}_i^{(V)}, y_i)\}_{i=1}^m$, where $\mathbf{x}_i^{(v)} \in \mathbb{R}^{n_v}$ is the feature representation of sample i in view v ($v \in [V] := \{1, \ldots, V\}$) and $y_i \in \{-1, +1\}$ is the label. Each view may encode distinct yet complementary discriminative cues.

For each view v, we define an affine classifier

$$f^{(v)}(\mathbf{x}^{(v)}) = \big(\mathbf{w}^{(v)}\big)^\top \mathbf{x}^{(v)} + b^{(v)}, \quad \mathbf{w}^{(v)} \in \mathbb{R}^{n_v},\ b^{(v)} \in \mathbb{R}.$$

A weight coefficient $\theta^{(v)} \geq 0$ is assigned to each view to reflect its contribution to the final decision. The view weights satisfy the simplex constraint

$$\theta^{(v)} \geq 0\ (v \in [V]), \quad \sum_{v=1}^{V} \theta^{(v)} = 1, \tag{43}$$

which ensures a convex combination across views. The prediction rule is

$$f(\mathbf{x}) = \operatorname{sign}\left(\sum_{v=1}^{V} \theta^{(v)} \left[\big(\mathbf{w}^{(v)}\big)^\top \mathbf{x}^{(v)} + b^{(v)}\right]\right). \tag{44}$$

The weight $\theta^{(v)}$ adaptively adjusts the importance of view v during training.

Table 9
ACC (%) with standard deviations on synthetic datasets ($n = 2$), where the SD represents the standard deviation.

m	Hinge-SVM	LS-SVM	Pin-SVM	Ramp-SVM	$L_{0/1}$-SVM	L_{ht}-SVM
5000	97.96±0.30	97.99±0.30	97.89±1.92	97.81±0.50	98.01±0.30	**98.04±0.30**
10000	98.01±0.17	98.05±0.16	98.01±0.39	98.05±0.33	**98.07±0.18**	**98.07±0.18**
15000	97.99±0.15	97.96±0.16	97.97±2.19	97.75±0.32	98.05±0.19	**98.06±0.16**
20000	98.01±0.13	98.01±0.12	97.95±3.20	97.77±0.31	97.99±0.15	**98.05±0.12**
25000	97.93±0.13	98.01±0.13	97.98±9.12	97.70±0.41	98.04±0.13	**98.07±0.13**
30000	98.06±0.11	98.06±0.11	97.96±8.01	97.92±0.31	98.06±0.11	**98.09±0.11**

Table 10
Number of support vectors (NSV) ± SD on synthetic datasets.

m	Hinge-SVM	LS-SVM	Pin-SVM	Ramp-SVM	$L_{0/1}$-SVM	L_{ht}-SVM
5000	432±12	5000±0	2480±136	172±13	20±19	**17±23**
10000	660±19	10000±0	4960±199	441±27	17±11	**12±9**
15000	1085±28	15000±0	7305±862	705±18	72±6	**69±6**
20000	1975±25	20000±0	9684±969	1054±12	140±41	**132±34**
25000	2029±33	25000±0	7909±823	1430±15	**137±64**	175±65
30000	2755±43	30000±0	10129±93	1972±25	218±85	**212±55**

5.2. MvL_{ht}-SVM model formulation

Extending the single-view L_{ht}-SVM of Section 3, we formulate the multi-view L_{ht}-SVM (MvL_{ht}-SVM). Fig. 2 depicts the overall framework. The resulting optimisation problem is

$$\begin{aligned}\min_{\substack{\mathbf{w}^{(v)}\in\mathbb{R}^{n_v},b^{(v)}\in\mathbb{R},\\ \theta^{(v)}\geq 0,\ v\in[V]}}\ & \frac{1}{2}\sum_{v=1}^{V}\theta^{(v)}\left\|\mathbf{w}^{(v)}\right\|^2+\frac{\alpha}{2}\|\boldsymbol{\theta}\|^2+\frac{\eta}{2}\sum_{v=1}^{V}\left(\mathbf{w}^{(v)}\right)^{\top}\Sigma^{(v)}\mathbf{w}^{(v)}\\ &+C\sum_{v=1}^{V}\sum_{i=1}^{m}\ell_{\mathrm{ht}}\left(1-y_i f^{(v)}\left(\mathbf{x}_i^{(v)}\right)\right)\\ \text{s.t.}\quad & \sum_{v=1}^{V}\theta^{(v)}=1,\end{aligned}\tag{45}$$

where $C>0$ is the penalty parameter for the L_{ht}-loss, $\alpha>0$ regularizes the view weights, $\eta>0$ controls the structural regularization strength, and $\Sigma^{(v)}\in\mathbb{R}^{n_v\times n_v}$ is a positive semi-definite covariance matrix capturing local structure in view v (see Section 5.3).

The objective in (45) consists of four terms: (i) $\frac{1}{2}\sum_v\theta^{(v)}\|\mathbf{w}^{(v)}\|^2$ for weighted margin regularization across views; (ii) $\frac{\alpha}{2}\|\boldsymbol{\theta}\|^2$ as the weight complexity penalty to avoid degenerate solutions; (iii) $C\sum_v\sum_i\ell_{\mathrm{ht}}(\cdot)$ for the L_{ht}-loss over all views, retaining sparsity and robustness; (iv) $\frac{\eta}{2}\sum_v(\mathbf{w}^{(v)})^{\top}\Sigma^{(v)}\mathbf{w}^{(v)}$ for structural information regularization.

Introducing auxiliary variables $\mathbf{t}^{(v)}\in\mathbb{R}^m$ for each view, problem (45) can be rewritten as

$$\begin{aligned}\min_{\substack{\mathbf{w}^{(v)},b^{(v)},\mathbf{t}^{(v)},\\ \theta^{(v)}\geq 0,\ v\in[V]}}\ & \frac{1}{2}\sum_{v=1}^{V}\theta^{(v)}\left\|\mathbf{w}^{(v)}\right\|^2+\frac{\alpha}{2}\|\boldsymbol{\theta}\|^2+\frac{\eta}{2}\sum_{v=1}^{V}\left(\mathbf{w}^{(v)}\right)^{\top}\Sigma^{(v)}\mathbf{w}^{(v)}\\ &+C\sum_{v=1}^{V}L_{\mathrm{ht}}\left(\mathbf{t}^{(v)}\right)\\ \text{s.t.}\quad & \mathbf{t}^{(v)}+M^{(v)}\mathbf{w}^{(v)}+b^{(v)}\mathbf{y}=\mathbf{1},\quad \sum_{v=1}^{V}\theta^{(v)}=1,\quad v\in[V],\end{aligned}\tag{46}$$

where $L_{\mathrm{ht}}(\mathbf{t}^{(v)}):=\sum_{i=1}^{m}\ell_{\mathrm{ht}}(t_i^{(v)})$ and $M^{(v)}:=[y_1\mathbf{x}_1^{(v)}\cdots y_m\mathbf{x}_m^{(v)}]^{\top}\in\mathbb{R}^{m\times n_v}$. The linear constraints separate the non-smooth loss from the regularization terms, which facilitates the ADMM-based solver in Section 5.4.

5.3. Structural information and construction of $\Sigma^{(v)}$

The quadratic term $(\mathbf{w}^{(v)})^{\top}\Sigma^{(v)}\mathbf{w}^{(v)}$ in (45) incorporates local geometric structure, drawing on the idea of manifold regularisation (Belkin et al., 2006) and its multi-view extensions (Liu et al., 2025).

We construct $\Sigma^{(v)}$ with hierarchical agglomerative clustering (HAC) (Ward, 1963). Within each view u, positive and negative samples are clustered separately; the number of clusters K_u is determined by the L-method. Denoting the index set of cluster j in view u by $C_j^{(u)}$, we form the within-cluster covariance of view v's features as

$$\Sigma_j^{(v,u)}=\frac{1}{|C_j^{(u)}|}\sum_{i\in C_j^{(u)}}\left(\mathbf{x}_i^{(v)}-\boldsymbol{\mu}_j^{(v,u)}\right)\left(\mathbf{x}_i^{(v)}-\boldsymbol{\mu}_j^{(v,u)}\right)^{\top},\tag{47}$$

where $\boldsymbol{\mu}_j^{(v,u)}=|C_j^{(u)}|^{-1}\sum_{i\in C_j^{(u)}}\mathbf{x}_i^{(v)}$ is the centroid.

The aggregate structural matrix for view v is obtained by summing over all views and clusters, weighted by the view coefficients:

$$\Sigma^{(v)}=\sum_{u=1}^{V}\theta^{(u)}\sum_{j=1}^{K_u}\Sigma_j^{(v,u)}.\tag{48}$$

Because every $\Sigma_j^{(v,u)}$ is positive semi-definite, so is $\Sigma^{(v)}$. The resulting regulariser discourages large projections along high-variance directions within clusters, steering the classifier towards locally smooth decision boundaries.

Eq. (48) also makes precise what is transferred across views. For $u=v$, $\Sigma_j^{(v,v)}$ preserves structure estimated and measured in the same view. For $u\neq v$, the sample partition is determined in view u, whereas covariance is measured in view v; these cross-view terms therefore regularize the classifier in view v using neighborhoods revealed by another representation. Removing the structural term ($\eta=0$) leaves only adaptive score aggregation, so comparison with that variant isolates the contribution beyond weighted aggregation.

The formulation does not impose an explicit pairwise equality constraint between view-specific decision functions; agreement is encouraged indirectly through the shared labels, the structural transfer, and the final fused decision.

5.4. Optimization algorithm

We solve (46) via a two-level alternating scheme. In the inner loop, the view weights $\boldsymbol{\theta}$ are held fixed and every view-specific classifier $(\mathbf{w}^{(v)},b^{(v)})$ is updated independently. In the outer loop, the classifiers are frozen and $\boldsymbol{\theta}$ is re-estimated. The two loops alternate until the change in $\boldsymbol{\theta}$ falls below a tolerance or the iteration budget $l_{\max}$ is reached.

After fixing $\boldsymbol{\theta}$, the sub-problem for each view becomes independent and shares the same structure as the single-view L_{ht}-SVM. For view v, let $(\mathbf{w}^{(v)*},b^{(v)*},\mathbf{t}^{(v)*})$ with $\boldsymbol{\lambda}^{(v)*}$ be a P-stationary point. Define $\mathbf{p}^{(v)*}:=\mathbf{t}^{(v)*}-\nu\boldsymbol{\lambda}^{(v)*}$ with $\nu:=1/\xi$. According to Theorems 3–5, the support vector index set for view v is

$$\mathcal{I}^{(v)*}:=\begin{cases}\mathcal{G}_2^{(v)*}\cup\mathcal{G}_3^{(v)*}\cup\mathcal{G}_4^{(v)*}, & \nu C\in(0,5/18),\\ \mathcal{I}_2^{(v)*}\cup\mathcal{I}_3^{(v)*}, & \nu C\in[5/18,25/18),\\ \mathcal{K}_2^{(v)*}\cup\mathcal{K}_3^{(v)*}, & \nu C\geq 25/18,\end{cases}\tag{49}$$

where the index sets $\mathcal{G}_j^{(v)*}$, $\mathcal{I}_j^{(v)*}$, $\mathcal{K}_j^{(v)*}$ are defined analogously to the single-view case (see Theorems 3–5) with view-specific variables. The classifier satisfies

$$\mathbf{w}^{(v)*}=-\left(\theta^{(v)}I+\eta\Sigma^{(v)}\right)^{-1}\sum_{i\in\mathcal{I}^{(v)*}}\lambda_i^{(v)*}y_i\mathbf{x}_i^{(v)}.\tag{50}$$

Table 11
TIME (s) ± SD on synthetic datasets.

m	Hinge-SVM	LS-SVM	Pin-SVM	Ramp-SVM	$L_{0/1}$-SVM	L_{ht}-SVM
5000	0.105±0.027	1.345±0.003	0.461±0.082	21.019±21.545	0.010±0.003	**0.002±0.000**
10000	0.317±0.054	12.151±0.024	1.385±0.113	229.475±188.209	0.017±0.004	**0.011±0.017**
15000	0.704±0.135	40.673±0.072	2.351±0.269	494.150±420.492	**0.015±0.009**	0.025±0.081
20000	2.021±0.327	94.595±0.311	4.295±0.385	1215.696±819.898	0.031±0.009	**0.010±0.001**
25000	2.219±0.321	181.278±0.370	4.845±0.415	1861.817±1316.984	0.032±0.005	**0.014±0.001**
30000	2.544±0.270	420.441±2.672	5.044±0.497	2886.044±1533.268	**0.012±0.003**	0.031±0.005

Table 12
Working-set dynamics on the synthetic datasets ($n = 2$) under frozen (C, ξ). "WS" denotes Algorithm 1; "Full" solves every $\mathbf{w}$-subproblem with all samples. "Re-entry" is the percentage of iterations in which at least one previously excluded sample re-enters the working set F_k, and $|F_K|/m$ is the final working-set fraction. The WS accuracies coincide with the L_{ht}-SVM entries in Table 9. TIME is the mean fitting time in seconds and Iter the mean number of ADMM iterations. Both variants use the same (C, ξ), initialization, tolerance, and iteration budget. Entries are written as WS / full.

m	ACC (%)		NSV (WS / Full)	TIME (s, WS / Full)	Iter (WS / Full)	Re-entry (%)	$\lvert F_K\rvert/m$ (%)
	WS	Full					
5000	98.04 ± 0.30	97.97 ± 0.18	17.0 / 19.9	0.002 / 0.011	27.3 / 20.6	92.3	13.6
10000	98.07 ± 0.18	97.99 ± 0.28	12.0 / 26.3	0.011 / 0.040	36.2 / 23.1	88.4	9.4
15000	98.06 ± 0.16	97.98 ± 0.14	69.0 / 337.0	0.025 / 0.138	48.7 / 18.4	93.3	4.2
20000	98.05 ± 0.12	97.96 ± 0.25	132.0 / 484.3	0.010 / 0.043	41.8 / 21.7	98.2	2.1
25000	98.07 ± 0.13	98.00 ± 0.10	175.0 / 1401.7	0.014 / 0.084	53.6 / 24.8	94.3	2.5
30000	98.09 ± 0.11	98.05 ± 0.07	212.0 / 652.5	0.031 / 0.109	59.4 / 29.2	99.2	1.3

Table 13
Optimization-stability summary on four UCI datasets. For each dataset, a baseline (C, ξ) pair is selected under the default configuration. The value of C remains fixed, and ξ remains at its baseline value except when penalty sensitivity is evaluated. The initialization columns report five-fold mean ACC ± standard deviation (%); the last two columns give the range of mean ACC (%) over the nine ADMM penalties $\xi \in \{2^{-4}, \ldots, 2^{4}\}$ and the three stopping tolerances $\{10^{-2}, 10^{-3}, 10^{-4}\}$.

	Initialization					
Dataset	Default	Zero	Warm start	Random	Range over ξ	Range over tol.
Australian	86.23 ± 3.16	44.49 ± 0.40	85.51 ± 3.51	83.86 ± 6.87	72.61–86.23	85.07–86.23
Breast-cancer	97.22 ± 0.86	97.22 ± 1.33	96.93 ± 0.96	97.22 ± 1.22	96.64–97.22	96.49–97.22
Wine-red	75.04 ± 2.42	75.04 ± 2.42	74.30 ± 1.44	75.04 ± 2.21	73.17–75.04	73.86–75.04
Wine-white	75.38 ± 0.53	75.38 ± 1.31	74.58 ± 0.66	75.38 ± 1.19	74.09–75.38	73.95–75.38

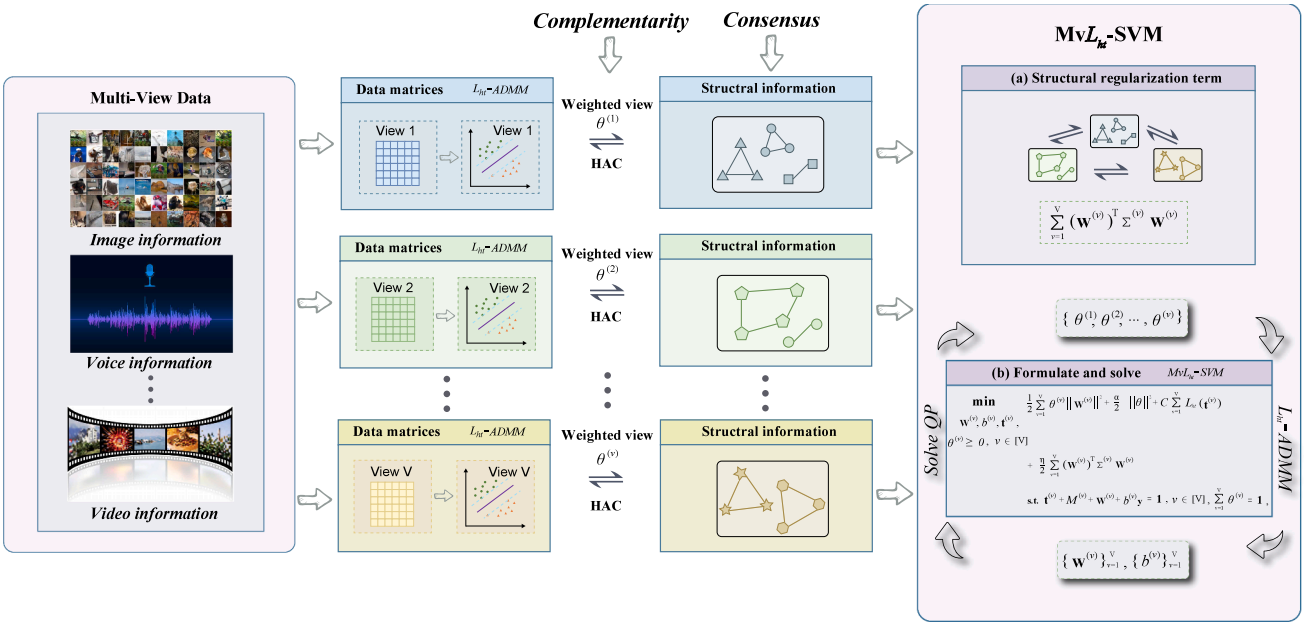


Fig. 2. Framework of MvL_{ht}-SVM. Multi-view data (e.g., image, voice, or video) are processed through view-specific L_{ht}-ADMM solvers to obtain $(\mathbf{w}^{(v)}, b^{(v)})$. HAC partitions constructed in one view are used to form structural covariance terms in the other views, while adaptive weights fuse the resulting decision scores. The optimization alternates between the view-weight subproblem and the view-specific classifier subproblems.

Table 14
Robustness comparison on the breast-cancer dataset under different label noise ratios.

ACC (%)						
r	Hinge-SVM	LS-SVM	Pin-SVM	Ramp-SVM	$L_{0/1}$-SVM	L_{ht}-SVM
0.00	96.92	96.05	96.49	96.39	97.07	**97.22**
0.02	95.17	94.43	95.02	95.46	95.46	**95.75**
0.04	92.98	92.54	92.83	93.42	**94.00**	93.71
0.06	91.66	90.78	91.22	91.51	92.09	**92.24**
0.08	89.16	88.72	89.31	89.45	**89.75**	**89.75**
0.10	87.26	87.26	87.26	87.85	88.00	**88.29**
NSV						
r	Hinge-SVM	LS-SVM	Pin-SVM	Ramp-SVM	$L_{0/1}$-SVM	L_{ht}-SVM
0.00	259	683	546	175	**16**	25
0.02	276	683	546	196	**17**	19
0.04	207	683	546	196	**17**	25
0.06	258	683	546	182	**17**	18
0.08	234	683	546	124	**17**	**17**
0.10	265	683	546	149	18	**16**
TIME (s)						
r	Hinge-SVM	LS-SVM	Pin-SVM	Ramp-SVM	$L_{0/1}$-SVM	L_{ht}-SVM
0.00	0.047	0.029	0.198	7.871	0.009	**0.004**
0.02	0.051	0.034	0.221	7.293	0.010	**0.009**
0.04	0.049	0.031	0.198	7.716	0.007	**0.006**
0.06	0.049	0.034	0.210	7.921	0.010	**0.006**
0.08	0.049	0.026	0.206	7.854	0.009	**0.005**
0.10	0.049	0.031	0.201	7.270	0.009	**0.008**

Table 15
Robustness comparison on the australian dataset under different label noise ratios.

ACC (%)						
r	Hinge-SVM	LS-SVM	Pin-SVM	Ramp-SVM	$L_{0/1}$-SVM	L_{ht}-SVM
0.00	85.51	**86.09**	85.51	85.51	85.80	**86.09**
0.02	84.78	**85.36**	84.78	84.78	84.35	84.93
0.04	83.33	83.62	83.33	83.33	83.62	**83.91**
0.06	82.03	82.09	82.03	82.03	81.30	**82.17**
0.08	80.00	80.15	80.00	80.00	79.57	**80.29**
0.10	78.55	78.64	78.55	78.55	77.68	**78.84**
NSV						
r	Hinge-SVM	LS-SVM	Pin-SVM	Ramp-SVM	$L_{0/1}$-SVM	L_{ht}-SVM
0.00	207	690	384	194	56	**44**
0.02	220	690	384	172	52	**49**
0.04	189	690	384	201	29	**18**
0.06	216	690	384	190	84	**36**
0.08	209	690	384	172	**21**	38
0.10	217	690	384	184	34	**23**
TIME (s)						
r	Hinge-SVM	LS-SVM	Pin-SVM	Ramp-SVM	$L_{0/1}$-SVM	L_{ht}-SVM
0.00	0.024	0.017	0.015	0.613	0.006	**0.005**
0.02	0.033	0.024	0.067	0.557	**0.009**	**0.009**
0.04	0.012	0.024	0.021	0.616	0.005	**0.004**
0.06	0.029	0.034	0.016	0.527	**0.007**	0.009
0.08	0.015	0.019	0.027	0.592	0.011	**0.006**
0.10	0.026	0.023	0.017	0.609	0.009	**0.005**

Compared with the single-view case (22), the regularization matrix changes from I to $\theta^{(v)}I+\eta\Sigma^{(v)}$, while the sparsity property is preserved.

5.4.1. Update classifiers via L_{ht}-ADMM-WS

For view v, the augmented Lagrangian function is defined as

$$\begin{aligned}&\mathcal{L}_{\xi}^{(v)}(\mathbf{w}^{(v)},b^{(v)},\mathbf{t}^{(v)},\boldsymbol{\lambda}^{(v)})\\&=\frac{\theta^{(v)}}{2}\|\mathbf{w}^{(v)}\|^2+CL_{\mathrm{ht}}(\mathbf{t}^{(v)})+\langle\boldsymbol{\lambda}^{(v)},\mathbf{t}^{(v)}-\mathbf{1}+M^{(v)}\mathbf{w}^{(v)}+b^{(v)}\mathbf{y}\rangle\\&\quad+\frac{\xi}{2}\|\mathbf{t}^{(v)}-\mathbf{1}+M^{(v)}\mathbf{w}^{(v)}+b^{(v)}\mathbf{y}\|^2+\frac{\eta}{2}(\mathbf{w}^{(v)})^{\top}\Sigma^{(v)}\mathbf{w}^{(v)},\end{aligned}\tag{51}$$

where $\boldsymbol{\lambda}^{(v)}\in\mathbb{R}^m$ is the Lagrangian multiplier and $\xi>0$ is the penalty parameter. Starting from an initial point, the ADMM iteration proceeds as

$$\begin{aligned}\mathbf{t}^{(v)^{k+1}}&=\arg\min_{\mathbf{t}^{(v)}}\mathcal{L}_{\xi}^{(v)}(\mathbf{w}^{(v)^k},b^{(v)^k},\mathbf{t}^{(v)},\boldsymbol{\lambda}^{(v)^k}),\\\mathbf{w}^{(v)^{k+1}}&=\arg\min_{\mathbf{w}^{(v)}\in\mathbb{R}^{n_v}}\mathcal{L}_{\xi}^{(v)}(\mathbf{w}^{(v)},b^{(v)^k},\mathbf{t}^{(v)^{k+1}},\boldsymbol{\lambda}^{(v)^k})\\&\quad+\frac{\xi}{2}\left\|\mathbf{w}^{(v)}-\mathbf{w}^{(v)^k}\right\|_{Z_k^{(v)}}^2,\\b^{(v)^{k+1}}&=\arg\min_{b^{(v)}\in\mathbb{R}}\mathcal{L}_{\xi}^{(v)}(\mathbf{w}^{(v)^{k+1}},b^{(v)},\mathbf{t}^{(v)^{k+1}},\boldsymbol{\lambda}^{(v)^k}),\\\boldsymbol{\lambda}^{(v)^{k+1}}&=\boldsymbol{\lambda}^{(v)^k}+\tau\xi\left(\mathbf{t}^{(v)^{k+1}}-\mathbf{1}+M^{(v)}\mathbf{w}^{(v)^{k+1}}+b^{(v)^{k+1}}\mathbf{y}\right).\end{aligned}\tag{52}$$

where $Z_k^{(v)}\in\mathbb{R}^{n_v\times n_v}$ is a symmetric matrix and $\tau\in(0,(1+\sqrt{5})/2)$ is the dual step-size.

Following the single-view case, we construct the working set $F_k^{(v)}$ for each view. Let $\mathbf{p}^{(v)^k}:=\mathbf{1}-M^{(v)}\mathbf{w}^{(v)^k}-b^{(v)^k}\mathbf{y}-\nu\boldsymbol{\lambda}^{(v)^k}$. The working set $F_k^{(v)}$ is constructed according to (30) with view-specific variables. Denote $\overline{F}_k^{(v)}:=[m]\backslash F_k^{(v)}$ and set

$$Z_k^{(v)}=-M_{\overline{F}_k^{(v)}}^{(v)\top}M_{\overline{F}_k^{(v)}}^{(v)}.\tag{53}$$

This choice excludes non-support-vector samples from computation, leading to significant speedup when $|F_k^{(v)}|\ll m$. Note that each view maintains its own working set, so a sample may be active in one view but not in another. This is intentional: the views have different feature geometries, and forcing identical active sets would suppress the complementary information that the views carry. Cross-view agreement is instead encouraged by the shared labels, the structural transfer in (48), and the fused decision rule; the resulting cross-view overlap of support and working sets, together with the disagreement between per-view predictions, is quantified in Appendix D.

(i) Updating $\mathbf{t}^{(v)^{k+1}}$. The $\mathbf{t}^{(v)}$-subproblem reduces to

$$\mathbf{t}^{(v)^{k+1}}=\arg\min_{\mathbf{t}^{(v)}}CL_{\mathrm{ht}}(\mathbf{t}^{(v)})+\frac{\xi}{2}\|\mathbf{t}^{(v)}-\mathbf{p}^{(v)^k}\|^2=\mathrm{prox}_{\nu CL_{\mathrm{ht}}}(\mathbf{p}^{(v)^k}),\tag{54}$$

where $\mathbf{p}^{(v)^k}=\mathbf{1}-M^{(v)}\mathbf{w}^{(v)^k}-b^{(v)^k}\mathbf{y}-\nu\boldsymbol{\lambda}^{(v)^k}$ as defined above. By Lemma 2, the closed-form solution is obtained component-wise.

(ii) Updating $\mathbf{w}^{(v)^{k+1}}$. Setting the gradient of the $\mathbf{w}^{(v)}$-subproblem to zero yields the linear system

$$\left(\theta^{(v)}I+\eta\Sigma^{(v)}+\xi M_{F_k^{(v)}}^{(v)\top}M_{F_k^{(v)}}^{(v)}\right)\mathbf{w}^{(v)}=\xi M_{F_k^{(v)}}^{(v)\top}\boldsymbol{\chi}_{F_k^{(v)}}^{(v)^k},\tag{55}$$

where $\boldsymbol{\chi}^{(v)^k}:=-(\mathbf{t}^{(v)^{k+1}}+b^{(v)^k}\mathbf{y}-\mathbf{1}+\boldsymbol{\lambda}^{(v)^k}/\xi)$. The derivation uses the identity

$$M_{F_k^{(v)}}^{(v)\top}M_{F_k^{(v)}}^{(v)}=M^{(v)\top}M^{(v)}-M_{\overline{F}_k^{(v)}}^{(v)\top}M_{\overline{F}_k^{(v)}}^{(v)}.$$

Depending on the relationship between n_v and $|F_k^{(v)}|$, we have two cases. When $n_v\le|F_k^{(v)}|$, we solve (55) directly:

$$\mathbf{w}^{(v)^{k+1}}=\left(\theta^{(v)}I+\eta\Sigma^{(v)}+\xi M_{F_k^{(v)}}^{(v)\top}M_{F_k^{(v)}}^{(v)}\right)^{-1}\xi M_{F_k^{(v)}}^{(v)\top}\boldsymbol{\chi}_{F_k^{(v)}}^{(v)^k}.\tag{56}$$

When $n_v>|F_k^{(v)}|$, applying the Sherman–Morrison–Woodbury formula gives

$$\begin{aligned}\mathbf{w}^{(v)^{k+1}}&=\xi(\theta^{(v)}I+\eta\Sigma^{(v)})^{-1}M_{F_k^{(v)}}^{(v)\top}\\&\quad\times\left(I+\xi M_{F_k^{(v)}}^{(v)}(\theta^{(v)}I+\eta\Sigma^{(v)})^{-1}M_{F_k^{(v)}}^{(v)\top}\right)^{-1}\boldsymbol{\chi}_{F_k^{(v)}}^{(v)^k}.\end{aligned}\tag{57}$$

Table 16
Additional robustness results on Australian. Each cell reports five-fold mean ACC ± std (%). The highest mean at each corruption level is shown in bold.

Corruption	Level	Hinge-SVM	LS-SVM	Pin-SVM	Ramp-SVM	$L_{0/1}$-SVM	L_{ht}-SVM
Boundary label	$r_b = 0.10$	85.94 ± 2.49	86.38 ± 2.95	86.09 ± 2.31	85.94 ± 2.49	86.52 ± 2.08	**87.25 ± 2.21**
	$r_b = 0.20$	86.23 ± 3.40	84.84 ± 4.13	85.55 ± 2.69	85.51 ± 3.52	86.09 ± 3.12	**86.81 ± 2.48**
	$r_b = 0.30$	81.45 ± 4.97	78.55 ± 1.92	77.83 ± 2.32	80.14 ± 4.39	82.46 ± 3.16	**83.55 ± 2.87**
Feature noise	$\gamma = 0.20$	86.09 ± 3.56	86.09 ± 3.02	85.65 ± 3.29	85.80 ± 3.26	86.38 ± 3.65	**87.10 ± 2.85**
	$\gamma = 0.40$	85.80 ± 3.26	84.78 ± 2.93	85.51 ± 3.14	85.51 ± 3.14	85.65 ± 3.29	**86.38 ± 2.67**
Sparse outliers	$r_o = 0.10$	86.81 ± 3.35	86.81 ± 2.73	86.67 ± 3.68	85.51 ± 3.14	86.81 ± 2.88	**86.96 ± 3.02**
	$r_o = 0.30$	85.51 ± 3.14	86.09 ± 2.44	85.51 ± 3.14	85.51 ± 3.14	85.51 ± 3.14	**86.23 ± 3.27**
Class imbalance	$\rho = 5$	82.03 ± 0.85	80.87 ± 1.69	81.01 ± 1.41	82.46 ± 3.16	83.04 ± 4.11	**84.20 ± 3.85**
	$\rho = 10$	75.36 ± 2.86	69.13 ± 1.98	64.20 ± 4.06	80.14 ± 5.55	80.72 ± 6.67	**82.17 ± 4.21**
Linear attack	$\epsilon = 0.05$	86.09 ± 3.56	86.09 ± 3.02	85.65 ± 3.29	85.80 ± 3.26	86.38 ± 2.40	**86.67 ± 2.49**
	$\epsilon = 0.10$	85.51 ± 3.14	84.78 ± 2.93	85.51 ± 3.14	85.51 ± 3.14	85.51 ± 3.14	**86.09 ± 3.35**

Table 17
Additional robustness results on breast-cancer. Each cell reports five-fold mean ACC ± std (%). The highest mean at each corruption level is shown in bold.

Corruption	Level	Hinge-SVM	LS-SVM	Pin-SVM	Ramp-SVM	$L_{0/1}$-SVM	L_{ht}-SVM
Boundary label	$r_b = 0.10$	97.07 ± 1.04	96.34 ± 1.38	96.49 ± 0.97	97.22 ± 1.17	97.65 ± 1.35	**97.81 ± 1.08**
	$r_b = 0.20$	96.49 ± 0.97	96.34 ± 1.46	96.49 ± 0.97	96.63 ± 0.99	96.63 ± 1.19	**97.12 ± 1.15**
	$r_b = 0.30$	96.63 ± 1.09	95.75 ± 1.25	96.63 ± 1.50	96.34 ± 1.03	96.49 ± 1.25	**96.80 ± 0.98**
Feature noise	$\gamma = 0.20$	96.92 ± 0.97	96.05 ± 1.50	96.63 ± 0.99	96.92 ± 0.97	97.07 ± 1.22	**97.51 ± 1.12**
	$\gamma = 0.40$	97.07 ± 0.80	95.90 ± 1.35	96.49 ± 1.17	96.63 ± 1.76	96.93 ± 1.16	**97.65 ± 0.92**
Sparse outliers	$r_o = 0.10$	96.49 ± 1.07	93.99 ± 2.24	96.63 ± 0.99	96.78 ± 0.74	96.78 ± 1.09	**97.12 ± 1.05**
	$r_o = 0.30$	95.61 ± 1.31	86.24 ± 2.73	95.75 ± 1.26	95.61 ± 1.46	95.76 ± 0.84	**96.20 ± 1.12**
Class imbalance	$\rho = 5$	96.34 ± 1.66	95.61 ± 2.01	96.19 ± 1.56	96.19 ± 1.17	96.34 ± 1.38	**97.12 ± 1.08**
	$\rho = 10$	95.17 ± 1.50	95.17 ± 1.36	95.75 ± 1.87	95.46 ± 1.87	96.19 ± 1.41	**96.20 ± 1.42**
Linear attack	$\epsilon = 0.05$	95.32 ± 1.50	94.58 ± 1.94	95.32 ± 1.76	95.61 ± 0.92	95.75 ± 1.41	**96.20 ± 1.28**
	$\epsilon = 0.10$	93.12 ± 1.43	92.53 ± 1.48	93.41 ± 1.85	93.41 ± 1.66	93.56 ± 1.49	**94.12 ± 1.51**

Table 18
Balanced accuracy and minority-class recall on Australian under class imbalance. Each cell reports five-fold mean ± std (%); the corresponding ACC is reported in Table 16. The highest mean in each row is shown in bold.

ρ	Metric	Hinge-SVM	LS-SVM	Pin-SVM	Ramp-SVM	$L_{0/1}$-SVM	L_{ht}-SVM
5	BACC	80.45 ± 1.13	79.07 ± 1.65	79.30 ± 1.58	80.97 ± 3.47	81.98 ± 5.07	**82.74 ± 3.91**
	Min-Recall	66.12 ± 3.77	62.56 ± 3.73	63.55 ± 4.65	67.41 ± 6.68	**72.04 ± 14.82**	69.50 ± 10.62
10	BACC	72.66 ± 3.26	65.41 ± 2.21	59.84 ± 4.57	78.96 ± 5.85	79.93 ± 7.93	**80.81 ± 4.35**
	Min-Recall	48.19 ± 6.46	31.60 ± 4.49	20.21 ± 9.41	68.64 ± 12.87	**73.21 ± 20.53**	68.50 ± 13.87

Table 19
Balanced accuracy and minority-class recall on breast-cancer under class imbalance. Each cell reports five-fold mean ± std (%); the corresponding ACC is reported in Table 17. The highest mean in each row is shown in bold.

ρ	Metric	Hinge-SVM	LS-SVM	Pin-SVM	Ramp-SVM	$L_{0/1}$-SVM	L_{ht}-SVM
5	BACC	95.55 ± 2.23	94.40 ± 2.55	95.34 ± 2.00	95.24 ± 1.24	95.65 ± 1.95	**96.45 ± 1.12**
	Min-Recall	92.90 ± 4.28	90.38 ± 4.48	92.47 ± 3.62	92.05 ± 1.57	93.32 ± 4.24	**94.20 ± 3.44**
10	BACC	93.78 ± 1.86	93.68 ± 1.70	94.62 ± 2.38	94.09 ± 2.45	**95.24 ± 1.73**	95.11 ± 1.47
	Min-Recall	89.13 ± 3.31	88.71 ± 3.11	90.81 ± 4.27	89.54 ± 4.57	**92.06 ± 3.05**	91.50 ± 3.18

(iii) Updating $b^{(v)k+1}$. The $b^{(v)}$-subproblem admits the closed-form solution

$$b^{(v)k+1} = \langle \mathbf{y}, \mathbf{r}^{(v)k} \rangle / m, \tag{58}$$

where $\mathbf{r}^{(v)k} := -M^{(v)}\mathbf{w}^{(v)k+1} + \mathbf{1} - \mathbf{t}^{(v)k+1} - \boldsymbol{\lambda}^{(v)k}/\xi$.

(iv) Updating $\boldsymbol{\lambda}^{(v)k+1}$. The multiplier update follows from (52):

$$\boldsymbol{\lambda}^{(v)k+1}_{F_k^{(v)}} = \boldsymbol{\lambda}^{(v)k}_{F_k^{(v)}} + \tau\xi\boldsymbol{\Lambda}^{(v)k+1}_{F_k^{(v)}}, \quad \boldsymbol{\lambda}^{(v)k+1}_{\overline{F}_k^{(v)}} = \mathbf{0}, \tag{59}$$

where $\boldsymbol{\Lambda}^{(v)k+1} := \mathbf{t}^{(v)k+1} - \mathbf{1} + M^{(v)}\mathbf{w}^{(v)k+1} + b^{(v)k+1}\mathbf{y}$.

For the inner ADMM loop, we terminate when

$$\max\{\epsilon_1^{(v)k}, \epsilon_2^{(v)k}, \epsilon_3^{(v)k}, \epsilon_4^{(v)k}\} < \epsilon,$$

where the error measures are defined analogously to those of Algorithm 1.

5.4.2. *Update view weights* $\boldsymbol{\theta}$

After fixing all classifiers $\{\mathbf{w}^{(v)}, b^{(v)}\}_{v=1}^V$, the $\boldsymbol{\theta}$ optimization becomes a small-scale convex quadratic program:

$$\min_{\boldsymbol{\theta}\in\mathbb{R}^V} \frac{1}{2}\boldsymbol{\theta}^\top\boldsymbol{\pi} + \frac{\eta}{2}\boldsymbol{\theta}^\top\boldsymbol{\rho} + \frac{\alpha}{2}\|\boldsymbol{\theta}\|_2^2, \quad \text{s.t.} \sum_{v=1}^V \theta^{(v)} = 1,\ \theta^{(v)} \geq 0, \tag{60}$$

where $\boldsymbol{\pi} = [\|\mathbf{w}^{(1)}\|^2, \ldots, \|\mathbf{w}^{(V)}\|^2]^\top$ and $\boldsymbol{\rho} = (\rho_1, \ldots, \rho_V)^\top$ with $\rho_v = \sum_{u=1}^V (\mathbf{w}^{(u)})^\top \Sigma^{(u,v)} \mathbf{w}^{(u)}$, where $\Sigma^{(u,v)} := \sum_{j=1}^{K_v} \Sigma_j^{(u,v)}$ and $\Sigma_j^{(u,v)}$ is defined analogously to (47) by computing the covariance of view u features over cluster j from view v. This can be solved efficiently by standard QP solvers.

For the outer loop, we stop when $\|\boldsymbol{\theta}^{(l+1)} - \boldsymbol{\theta}^{(l)}\| < \epsilon$. The complete procedure is summarized in Algorithm 2. Theorem 6 concerns the inner L_{ht}-ADMM. The outer loop follows the block-coordinate alternation used for multi-view classifiers with adaptive view weights by Zhao and

Table 20
Effect of the interior junction δ on accuracy, sparsity, and label-noise robustness, using the same five-fold partitions as the main experiments. ACC entries are means ± standard deviations; NSV entries on clean data are reported in the same format. The noisy ACC uses class-wise label flips at rate $r = 0.10$. In each row, the highest ACC (or the smallest NSV) is shown in bold.

Dataset	Protocol	Interior junction δ				
		1/4	1/3	1/2	2/3	3/4
ACC (%) on clean data						
Australian	Fixed	75.95 ± 13.27	76.38 ± 13.92	82.03 ± 5.40	**86.09 ± 2.35**	85.37 ± 3.04
	Retuned	**86.09 ± 2.79**	85.66 ± 3.19	85.66 ± 2.13	**86.09 ± 2.35**	85.95 ± 3.35
Breast-cancer	Fixed	96.78 ± 0.80	96.78 ± 0.80	96.93 ± 0.86	**97.22 ± 0.86**	96.64 ± 0.85
	Retuned	97.07 ± 1.00	**97.22 ± 1.00**	96.93 ± 0.85	**97.22 ± 0.86**	**97.22 ± 1.19**
NSV on clean data						
Australian	Fixed	101.6 ± 92.2	106.8 ± 57.7	103.2 ± 45.7	**81.4 ± 44.5**	105.4 ± 56.8
	Retuned	151.2 ± 150.6	37.0 ± 16.8	**25.8 ± 3.5**	81.4 ± 44.5	113.4 ± 37.3
Breast-cancer	Fixed	**27.4 ± 2.7**	31.0 ± 3.3	32.0 ± 1.3	33.8 ± 2.7	36.0 ± 1.9
	Retuned	29.6 ± 1.9	**21.6 ± 8.5**	24.4 ± 1.7	33.8 ± 2.7	35.4 ± 2.9
ACC (%) under $r = 0.10$ label noise						
Australian	Fixed	82.76 ± 1.87	83.05 ± 3.85	79.86 ± 10.12	80.44 ± 7.14	**85.37 ± 3.14**
	Retuned	86.23 ± 3.02	86.23 ± 3.02	**86.38 ± 2.96**	86.23 ± 3.02	86.23 ± 3.02
Breast-cancer	Fixed	96.64 ± 0.97	96.78 ± 0.80	**96.93 ± 1.17**	96.20 ± 1.69	**96.93 ± 1.17**
	Retuned	**97.22 ± 0.75**	97.08 ± 0.75	**97.22 ± 1.00**	96.93 ± 0.55	96.93 ± 0.85

Table 21
Details of 10 UCI datasets.

Datasets	#Data (m)	#View 1 $n^{(1)}$	#View 2 $n^{(2)}$
Balance	576	4	4
Bankruptcy	250	6	5
BUPA	345	6	5
German	1000	24	18
Haberman	306	3	3
Heart	270	13	10
Hepatitis	155	19	15
Iris	150	4	2
Vote	435	16	13
Wine	130	13	1

Xu (2024), who show that the objective decreases monotonically when the blocks are minimized in turn. Our scheme has the same structure. For fixed classifiers the objective of (45) is, up to a constant, the strongly convex QP (60), which is solved exactly, so the view-weight step never increases the objective. Each classifier step is warm-started from the previous classifier of the same view and returns a P-stationary point of its subproblem. The objective is bounded below by zero. In the experiments of Section 6.2 and Appendix D, with $\epsilon = 10^{-2}$ and $l_{\max} = 10$, the stopping criterion was met in all reported runs, typically within two to four outer iterations.

Algorithm 2 MvL_{ht}-SVM: alternating ADMM for solving (46).

Input: Multi-view data $\{\mathbf{X}^{(v)}\}_{v=1}^{V}$, labels $\mathbf{y}$, parameters C, η, α, ξ, maximum outer iterations $l_{\max}$, maximum inner iterations K.
Initialize $\theta^{(v)} = 1/V$ for $v \in [V]$, $l = 0$, $\Delta\theta = +\infty$.
while $\Delta\theta \geq \epsilon$ and $l \leq l_{\max}$ **do**
  for $v = 1$ **to** V **do**
    Compute $\Sigma^{(v)}$ by (48).
    Initialize $\mathbf{w}^{(v)^0}, b^{(v)^0}, \mathbf{t}^{(v)^0}, \boldsymbol{\lambda}^{(v)^0}$ (from the previous outer iterate when $l > 0$), $k = 0$.
    while $\max\{\epsilon_1^{(v)^k}, \epsilon_2^{(v)^k}, \epsilon_3^{(v)^k}, \epsilon_4^{(v)^k}\} \geq \epsilon$ and $k \leq K$ **do**
      Compute $F_k^{(v)}$ as in (30).
      Compute $\mathbf{t}^{(v)^{k+1}}$ by (54).
      Compute $\mathbf{w}^{(v)^{k+1}}$ by (56) if $n_v \leq |F_k^{(v)}|$, or by (57) otherwise.
      Compute $b^{(v)^{k+1}}$ by (58).
      Compute $\boldsymbol{\lambda}^{(v)^{k+1}}$ by (59).
      $k \leftarrow k + 1$.
    end while
    return $(\mathbf{w}^{(v)*}, b^{(v)*}) \leftarrow (\mathbf{w}^{(v)^k}, b^{(v)^k})$.
  end for
  Solve QP (60) to obtain θ^{new}.
  $\Delta\theta \leftarrow \|\theta^{\text{new}} - \theta\|$; $\theta \leftarrow \theta^{\text{new}}$; $l \leftarrow l + 1$.
end while
Output: θ^*, $\{\mathbf{w}^{(v)*}, b^{(v)*}\}_{v=1}^{V}$.

5.5. Computational complexity analysis

The computational cost of Algorithm 2 consists of two parts: the inner ADMM loop for each view and the outer loop for updating θ.

For the inner ADMM in view v, let $|F_k^{(v)}|$ denote the working set size at iteration k. The main costs per iteration are as follows. Computing $\mathbf{t}^{(v)^{k+1}}$ via the proximal operator requires $\mathcal{O}(m)$ operations. When $n_v \leq |F_k^{(v)}|$, solving (56) costs $\mathcal{O}(n_v^2|F_k^{(v)}| + n_v^3)$ for forming and inverting the $(n_v \times n_v)$ matrix. When $n_v > |F_k^{(v)}|$, solving (57) costs $\mathcal{O}(n_v|F_k^{(v)}|^2 + |F_k^{(v)}|^3)$ for inverting the $(|F_k^{(v)}| \times |F_k^{(v)}|)$ matrix. Computing $b^{(v)^{k+1}}$ and $\boldsymbol{\lambda}^{(v)^{k+1}}$ requires $\mathcal{O}(m)$ and $\mathcal{O}(|F_k^{(v)}|)$ operations, respectively.

Due to the sparsity of L_{ht} support vectors, $|F_k^{(v)}| \ll m$ typically holds after a few iterations. Suppose the inner loop converges in K iterations on average. The per-view cost is $\mathcal{O}(K(n_v^2|F_k^{(v)}| + n_v^3))$ when $n_v \leq |F_k^{(v)}|$, or $\mathcal{O}(K(n_v|F_k^{(v)}|^2 + |F_k^{(v)}|^3))$ otherwise. In addition, as in the single-view analysis in Section 4.4, updating the full-sample margins, the proximal variables, and the multipliers costs $\mathcal{O}(mn_v)$ per view in each inner iteration. This cost is not reduced by the working set and dominates when $|F_k^{(v)}|$ is small.

For the view weight update, solving the QP (60) involves a $(V \times V)$ quadratic program, which costs $\mathcal{O}(V^3)$. Since V is typically small (e.g., $V \leq 10$), this step is negligible.

Let L be the number of outer iterations. The total complexity of Algorithm 2 is

$$\mathcal{O}\Big(L \cdot \sum_{v=1}^{V} K \cdot \big(mn_v + \max\{n_v^2|F_k^{(v)}|, n_v|F_k^{(v)}|^2\}\big)\Big).$$

The working set strategy reduces the cost of the classifier updates when the support vector ratio is low, while the $\mathcal{O}(mn_v)$ full-sample operations remain, so the total cost per iteration grows linearly with the sample size m.

Table 22
ACC (%) with standard deviations for the UCI multi-view datasets.

Dataset	SVM-2K	MvSVM-2C	MvLSSVC-2C	Wave-MvSVM	MVASY-BX	Mv$SL_{0/1}$-SVM	MvL_{ht}-SVM
Balance	93.75±2.01	94.97±1.77	94.62±1.85	92.19±2.63	95.66±1.45	96.18±2.10	**96.52±2.13**
Bankruptcy	95.20±3.49	98.80±1.60	99.20±1.60	**100.00±0.00**	97.60±2.33	**100.00±0.00**	**100.00±0.00**
BUPA	72.75±4.43	62.61±4.24	73.04±4.45	62.03±3.93	69.86±2.96	**74.78±3.73**	74.20±2.32
German	70.80±0.75	72.20±2.42	**75.00±1.38**	72.80±1.54	70.00±0.00	74.50±1.55	74.50±1.14
Haberman	73.53±0.48	73.53±0.48	73.20±1.26	73.20±0.89	73.53±0.48	74.18±0.81	**74.19±1.09**
Heart	82.96±3.39	85.56±3.19	84.81±3.19	75.19±4.77	85.19±4.22	85.56±5.90	**85.93±3.43**
Hepatitis	92.90±3.16	94.19±3.76	94.84±2.58	96.77±2.04	81.94±3.29	97.42±2.41	**98.06±1.58**
Iris	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
Vote	93.56±4.15	95.63±3.12	95.63±2.56	93.33±2.23	92.18±3.87	**96.09±2.37**	**96.09±2.37**
Wine	91.54±2.88	93.08±2.88	96.92±2.88	90.00±3.08	95.38±2.88	**100.00±0.00**	**100.00±0.00**
Avg. Acc.	86.70	87.06	88.73	85.55	86.13	89.87	**89.95**
Avg. Rank	5.40	4.40	3.90	5.35	5.10	2.10	**1.75**

Table 23
Precision (%) with standard deviations for the UCI multi-view datasets.

Dataset	SVM-2K	MvSVM-2C	MvLSSVC-2C	Wave-MvSVM	MVASY-BX	Mv$SL_{0/1}$-SVM	MvL_{ht}-SVM
Balance	94.22±3.48	95.43±4.28	94.00±3.33	91.49±2.93	95.75±3.78	95.30±2.83	**97.20±2.35**
Bankruptcy	99.00±2.00	98.14±2.28	99.05±1.90	**100.00±0.00**	97.14±2.34	**100.00±0.00**	**100.00±0.00**
BUPA	69.87±7.31	53.54±3.74	70.79±7.07	54.41±4.09	70.07±5.12	70.01±5.51	**70.93±3.12**
German	71.04±0.70	72.65±2.72	77.84±0.73	78.69±0.77	70.00±0.00	**78.76±1.31**	77.48±0.95
Haberman	73.53±0.48	73.53±0.48	73.60±0.80	73.60±0.21	73.53±0.48	**75.12±1.97**	74.02±0.91
Heart	84.07±4.34	**86.57±4.57**	85.58±5.20	80.20±6.13	85.24±3.36	85.73±7.55	85.59±3.25
Hepatitis	85.48±13.31	91.67±10.54	89.64±9.48	95.56±8.89	40.00±48.99	94.29±7.00	**100.00±0.00**
Iris	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
Vote	91.04±6.97	**93.89±5.56**	92.38±4.86	92.79±6.06	89.10±7.78	92.47±4.85	93.01±5.16
Wine	92.03±6.90	91.97±4.60	98.46±3.08	88.83±6.29	98.33±3.33	**100.00±0.00**	**100.00±0.00**
Avg. Prec.	86.03	85.74	88.13	85.56	81.92	89.17	**89.82**
Avg. Rank	5.40	4.30	3.95	4.35	5.20	2.65	**2.15**

Table 24
Recall (%) with standard deviations for the UCI multi-view datasets.

Dataset	SVM-2K	MvSVM-2C	MvLSSVC-2C	Wave-MvSVM	MVASY-BX	Mv$SL_{0/1}$-SVM	MvL_{ht}-SVM
Balance	93.39±3.38	94.77±3.52	95.50±2.81	93.04±2.71	95.84±3.54	**97.22±2.09**	95.84±2.58
Bankruptcy	89.78±7.88	99.05±1.90	99.05±1.90	**100.00±0.00**	97.14±3.81	**100.00±0.00**	**100.00±0.00**
BUPA	63.45±4.14	**92.41±6.69**	62.07±3.78	60.00±10.37	49.66±6.01	71.03±7.10	65.52±3.08
German	98.43±1.23	97.14±2.47	89.86±1.46	83.86±2.62	**100.00±0.00**	87.14±3.50	89.71±3.82
Haberman	**100.00±0.00**	**100.00±0.00**	99.11±1.09	99.11±1.09	**100.00±0.00**	97.33±3.56	**100.00±0.00**
Heart	86.00±5.33	88.00±4.00	88.00±2.67	74.00±6.46	88.67±4.52	**90.00±4.71**	**90.00±5.96**
Hepatitis	84.29±10.61	80.48±16.45	87.14±12.38	90.48±7.82	11.90±16.77	**93.81±7.62**	90.48±7.82
Iris	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
Vote	92.82±4.91	95.22±4.89	97.01±3.81	90.48±6.29	91.62±4.06	**98.20±2.42**	97.61±2.93
Wine	89.85±3.27	93.18±6.28	95.00±6.67	89.70±8.43	91.52±5.28	**100.00±0.00**	**100.00±0.00**
Avg. Rec.	89.80	94.03	91.27	88.07	82.64	**93.47**	92.92
Avg. Rank	4.75	3.85	4.15	5.50	4.40	**2.65**	2.70

6. Numerical experiments

6.1. Single-view experiments

6.1.1. Experimental settings

This subsection benchmarks the proposed L_{ht}-SVM against five representative single-view SVM solvers in terms of classification accuracy and training efficiency. The platform is a Windows 11 workstation (Intel Core i9-13980HX, 32 GB RAM) running Python 3.11.

Datasets. We assemble 15 real-world datasets drawn from the LIBSVM[1] and UCI[2] repositories (Table 2). All multi-class labels are binarised by mapping non-"+1" classes to −1, and each feature is normalised to $[-1, 1]$. For UCI datasets, results are averaged over 5-fold cross-validation; for LIBSVM datasets, the official train/test partitions are adopted directly.

For the synthetic experiments, we sample two-dimensional data from a pair of Gaussian distributions. Specifically, m positive samples are generated from $\mathcal{N}(\theta_1, \Phi_1)$ with $\theta_1 = (0.5, -3)^\top$, and m negative samples from $\mathcal{N}(\theta_2, \Phi_2)$ with $\theta_2 = (-0.5, 3)^\top$; both classes share the covariance matrix $\Phi_1 = \Phi_2 = \left(\begin{smallmatrix} 0.2 & 0 \\ 0 & 3 \end{smallmatrix}\right)$. The pooled dataset is then evenly split into training and testing subsets.

Comparison methods. (1) Hinge-SVM (hinge loss): the standard SVM with the hinge soft-margin loss, implemented by LIBSVM (Cortes & Vapnik, 1995). (2) LS-SVM (least squares loss): SVM with the least squares soft-margin loss (Suykens & Vandewalle, 1999). (3) Pin-SVM (pinball loss): SVM with the pinball soft-margin loss (Huang et al., 2014). (4) Ramp-SVM (ramp loss): SVM with the ramp soft-margin loss, solved by the concave-convex procedure (Collobert et al., 2006). (5) $L_{0/1}$-SVM ($L_{0/1}$ loss): SVM with the $L_{0/1}$ soft-margin loss, solved by ADMM (Wang et al., 2022).

Experimental setup and evaluation metrics. Hyper-parameters of every method are selected by grid search over $\{2^i \mid i = -4, -3, \ldots, 4\}$ for all penalty terms. The proposed L_{ht}-ADMM starts from $\mathbf{w}^0 = \mathbf{1}/100$, $b^0 = 0$,

[1] https://www.csie.ntu.edu.tw/~cjlin/libsvmtools/datasets/
[2] https://archive.ics.uci.edu/ml/index.php

Table 25
F1-score (%) with standard deviations for the UCI multi-view datasets.

Dataset	SVM-2K	MvSVM-2C	MvLSSVC-2C	Wave-MvSVM	MVASY-BX	Mv$SL_{0/1}$-SVM	MvL_{ht}-SVM
Balance	93.72±2.05	94.97±1.75	94.68±1.81	92.25±2.60	95.68±1.44	96.23±2.07	**96.50±2.15**
Bankruptcy	93.97±4.49	98.58±1.90	99.05±1.90	**100.00±0.00**	97.12±2.83	**100.00±0.00**	**100.00±0.00**
BUPA	66.29±4.38	67.56±2.40	66.03±4.75	56.70±6.01	57.91±5.25	**70.23±4.44**	68.10±2.86
German	82.52±0.39	83.05±1.06	**83.42±0.97**	81.17±1.31	82.35±0.00	82.69±1.35	83.09±1.18
Haberman	84.75±0.32	84.75±0.32	84.47±0.74	84.47±0.54	84.75±0.32	84.71±0.25	**85.07±0.60**
Heart	84.85±3.09	87.15±2.70	86.63±2.34	76.76±4.61	86.90±3.82	87.54±4.47	**87.60±3.23**
Hepatitis	83.47±5.46	84.18±11.14	87.31±6.02	92.33±4.15	17.71±23.86	93.79±5.53	**94.83±4.26**
Iris	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**
Vote	91.82±5.22	94.42±3.94	94.54±3.22	91.29±2.98	90.17±4.78	**95.17±2.88**	95.15±2.88
Wine	90.69±2.80	92.37±3.33	96.51±3.37	88.86±4.03	94.69±3.25	**100.00±0.00**	**100.00±0.00**
Avg. F1	87.21	88.70	89.26	86.38	80.73	**91.04**	91.03
Avg. Rank	5.20	3.90	4.05	5.65	5.00	2.45	**1.75**

Table 26
TIME (s) for the UCI multi-view datasets.

Dataset	SVM-2K	MvSVM-2C	MvLSSVC-2C	Wave-MvSVM	MVASY-BX	Mv$SL_{0/1}$-SVM	MvL_{ht}-SVM
Balance	7.220	53.320	**0.540**	15.540	3.110	2.790	2.130
Bankruptcy	0.360	3.650	**0.080**	7.890	2.800	1.090	0.690
BUPA	2.040	7.530	**0.110**	10.060	3.520	2.360	1.080
German	16.700	222.640	**0.300**	37.180	18.180	4.000	3.890
Haberman	0.540	2.100	**0.050**	8.770	2.210	2.180	1.970
Heart	0.470	6.260	**0.220**	6.420	3.820	1.250	1.420
Hepatitis	0.220	1.410	**0.020**	7.030	4.090	0.790	0.500
Iris	0.230	1.910	**0.040**	6.040	1.340	0.480	0.390
Vote	2.450	88.650	**0.160**	11.770	9.540	2.320	1.790
Wine	0.110	2.020	**0.110**	5.710	1.640	0.900	0.960
Avg. Time	3.030	38.950	**0.160**	11.640	5.030	1.820	1.480
Avg. Rank	2.75	6.00	**1.05**	6.70	5.10	3.60	2.80

with tolerance $\epsilon = 10^{-3}$ and iteration budget $K = 10^3$; the P-stationary point condition (Theorem 2) serves as the stopping rule. To guarantee a fair comparison, all baseline solvers are tuned over the same parameter grid. Unless stated otherwise, this configuration is used throughout the experiments; its sensitivity is examined in Section 6.1.5. For UCI datasets, the five folds are stratified and generated with a fixed random seed of 42, while for LIBSVM datasets the official training and test partitions are adopted and the test set is evaluated only once after model selection. The hyper-parameters of all methods are selected on the training data alone, so no test information is involved in model selection. All methods share the same folds, the same normalization fitted on the training data, and the same evaluation code. The default initialization $\mathbf{w}^0 = \mathbf{1}/100$, $b^0 = 0$ is deterministic. The label flips, feature noise, outliers, and downsampling in Section 6.1.6 and the random initializations in Section 6.1.5 use seeds derived from the base seed 42 and the fold index.

Three metrics are reported: (i) ACC, the test-set classification accuracy; (ii) TIME, the mean time in seconds needed to fit the final model with the selected hyper-parameters on each training fold. Hyper-parameter search is not included, so TIME is a fitting time and not the total cost. The search requires 81 candidate fits per fold for L_{ht}-SVM with (C, ξ), and 243 if δ is selected as well; (iii) NSV, the number of support vectors retained. A superior solver should exhibit high ACC together with low TIME and NSV.

6.1.2. Results on real datasets

The average results of ACC, NSV, and TIME of six solvers are recorded in Tables 3–8, where the best results are highlighted in bold.

On the UCI datasets, L_{ht}-SVM ranks first in average ACC; on breast-cancer, for example, it obtains 97.22%, compared with 97.07% for the closest baseline. It also retains the lowest average NSV, and together with $L_{0/1}$-SVM it forms the fastest group in terms of TIME.

The LIBSVM benchmarks contain 1 605–32 561 training samples. Tables 6–8 show that the two bounded sparse models remain competitive in both accuracy and fitting time as the sample size grows, although neither is fastest on every dataset.

6.1.3. Results on synthetic datasets

We examine sample-size scaling on synthetic data with $n = 2$ and $m \in \{5000, 10000, 15000, 20000, 25000, 30000\}$. The means and standard deviations in Tables 9–11 are computed from 50 independent runs. All six methods obtain similar accuracies, with most differences below 0.3 percentage points. The main distinction is computational: LS-SVM and Pin-SVM retain many samples, whereas L_{ht}-SVM and $L_{0/1}$-SVM use much smaller support-vector sets. Their fitting times are also of the same low order on the tested range, although the faster one varies with m; for example, at $m = 30\,000$, $L_{0/1}$-SVM takes 0.012s and L_{ht}-SVM takes 0.031s.

Fig. 3 visualizes the tabulated results: the support-vector ratio of the proposed method stays below 0.8% over the whole tested range, and its fitting time grows slowly with m.

Table 12 examines the working-set strategy itself on the same synthetic protocol. For each sample size, the selected (C, ξ) pair is frozen and the solver is run once with the working-set update of Algorithm 1 (WS) and once with a full update that keeps all samples in every $\mathbf{w}$-subproblem (Full), with the same initialization, tolerance, and iteration budget. Three observations stand out. First, at least one previously excluded sample re-enters the working set in 88.4–99.2% of the iterations, so the working set behaves as a dynamic active-set estimate rather than a one-shot pruning. Second, the accuracy of the working-set variant differs from the full update by at most 0.09 percentage points. This difference is within one standard deviation in every row, while the final working-set solutions retain fewer support vectors. Hence the working-set strategy does not cause observable underfitting in the tested range. Third, the TIME and Iter columns show that the working-set variant needs more iterations than the full update, because the active set is re-estimated at every iteration, but its fitting time is shorter at every sample size (0.002–0.031s against 0.011–0.109s), since each iteration solves the linear system on a small subset only. The absolute times are small and

Table 27
ACC (%) with standard deviations for the STL-10 dataset.

Datasets	SVM-2K	MvSVM-2C	MvLSSVC-2C	Wave-MvSVM	MVASY-BX	Mv$SL_{0/1}$-SVM	MvL_{ht}-SVM
air. vs bird.	84.40±2.50	89.80±1.40	89.30±1.50	89.20±2.29	90.00±2.17	90.80±2.46	**91.60±1.16**
air. vs car.	89.20±0.60	91.20±1.29	92.50±1.38	92.20±1.60	90.40±0.58	94.70±0.93	**95.30±0.68**
air. vs cat.	91.00±1.48	93.60±0.97	**94.50±0.84**	93.50±1.30	93.60±1.36	93.80±1.50	94.10±1.39
air. vs deer.	93.20±1.12	93.70±1.36	**96.10±0.97**	93.70±0.75	95.40±0.97	94.80±2.66	95.70±0.40
air. vs dog.	92.60±0.86	92.40±1.46	**95.40±1.62**	93.30±1.08	93.80±0.98	94.30±0.93	95.20±1.50
air. vs horse.	92.90±1.11	94.20±0.40	96.20±0.93	94.20±2.89	95.00±0.63	95.70±1.17	**96.60±0.73**
air. vs monkey.	93.40±0.58	97.10±1.28	97.10±1.56	95.40±1.28	95.70±0.93	95.50±1.30	**97.60±0.49**
air. vs ship.	**83.70±2.48**	80.30±2.79	80.40±2.67	79.90±3.07	79.50±1.26	78.80±1.29	80.20±3.31
air. vs truck.	89.90±1.53	91.50±1.00	91.70±1.50	84.70±2.16	90.50±0.89	91.10±2.27	**92.40±2.29**
bird. vs car.	94.60±1.59	95.10±1.28	96.20±0.98	94.90±1.02	94.70±1.57	92.90±1.50	**96.60±1.07**
bird. vs cat.	73.30±1.96	74.50±3.79	77.40±2.06	71.80±3.30	72.70±2.16	70.90±3.31	**77.60±2.40**
bird. vs deer.	71.70±2.14	73.00±1.26	76.40±2.67	76.60±0.97	**78.70±2.42**	76.60±3.77	76.00±3.30
bird. vs dog.	70.80±1.44	70.50±3.05	**72.60±1.50**	71.30±2.50	71.30±2.66	71.00±4.83	72.50±3.10
bird. vs horse.	78.10±2.40	84.70±1.50	82.10±1.07	81.50±2.41	80.40±2.27	82.60±1.36	**87.20±1.91**
bird. vs monkey.	71.60±0.73	73.60±3.07	76.20±2.73	75.80±3.33	75.00±3.32	77.90±1.98	**79.20±3.19**
bird. vs ship.	91.60±1.96	94.00±2.47	**95.40±1.96**	91.60±0.73	91.10±1.56	92.80±1.25	93.20±1.50
bird. vs truck.	94.80±1.21	**96.20±0.98**	93.60±1.39	94.50±2.70	94.20±1.21	92.20±1.50	95.30±1.17
car. vs cat.	93.80±1.72	95.40±1.46	95.20±1.33	94.20±1.47	95.20±1.03	94.30±1.50	**95.70±1.63**
car. vs deer.	92.50±1.05	96.00±0.89	96.70±0.60	95.80±1.47	93.40±0.86	96.70±0.87	**96.80±1.29**
car. vs dog.	84.20±2.32	94.60±1.16	93.10±1.53	94.60±1.71	92.80±1.91	**95.10±1.11**	94.80±1.86
car. vs horse.	93.40±1.02	95.40±1.83	96.60±0.66	94.40±1.36	93.50±1.34	93.80±1.21	**97.70±0.98**
car. vs monkey.	96.00±1.22	96.30±1.33	95.70±1.21	95.90±1.16	96.00±0.55	96.30±1.40	**96.70±0.93**
car. vs ship.	**89.10±2.24**	88.20±1.36	86.60±3.12	87.70±2.36	88.30±1.91	86.70±3.23	88.40±1.98
car. vs truck.	85.30±1.54	87.00±2.43	88.30±1.63	84.70±1.60	84.90±2.62	87.80±2.18	**88.60±2.42**
cat. vs deer.	74.70±3.20	74.50±2.76	**77.40±2.13**	72.50±3.36	73.70±3.20	74.60±4.14	76.90±2.20
cat. vs dog.	60.10±4.85	62.60±3.48	68.60±1.28	64.30±1.89	66.00±2.86	60.80±2.77	**68.90±2.33**
cat. vs horse.	79.60±2.06	77.30±3.14	82.90±1.88	79.90±2.87	80.80±1.47	82.60±2.92	**83.30±1.86**
cat. vs monkey.	72.30±3.80	73.00±3.18	76.60±3.51	73.90±3.56	73.10±3.60	76.00±4.00	**78.30±4.63**
cat. vs ship.	92.80±1.44	93.80±2.50	**94.20±1.03**	92.60±1.39	91.90±1.77	91.50±1.38	93.10±3.40
cat. vs truck.	92.90±1.36	94.50±0.89	93.50±1.84	94.30±0.81	92.80±1.50	91.10±2.03	**95.60±1.93**
deer. vs dog.	74.50±1.00	74.70±2.14	76.00±2.90	72.70±2.04	75.70±1.72	75.10±2.13	**76.60±2.27**
deer. vs horse.	79.90±2.60	80.10±2.58	**84.20±2.66**	77.00±1.90	81.70±5.69	82.10±2.13	83.90±1.83
deer. vs monkey.	77.50±2.35	77.60±3.51	82.00±2.12	81.20±2.23	77.20±1.89	80.70±2.16	**82.40±3.04**
deer. vs ship.	94.90±1.39	96.60±1.53	97.20±1.60	94.70±1.36	94.20±1.44	93.70±2.77	**97.70±1.75**
deer. vs truck.	95.80±1.17	96.70±1.29	97.60±0.73	94.60±1.16	94.70±1.44	93.10±1.88	**97.90±0.66**
dog. vs horse.	71.60±2.40	69.50±2.07	73.50±3.11	73.20±2.25	72.30±3.50	72.70±2.84	**73.80±3.52**
dog. vs monkey.	66.20±1.86	67.20±3.75	69.60±3.41	64.80±4.23	66.60±1.91	62.70±1.57	**70.00±2.85**
dog. vs ship.	**96.10±1.50**	94.00±1.14	93.00±1.14	94.00±1.22	92.20±1.08	91.20±0.87	95.00±1.38
dog. vs truck.	93.20±0.68	94.50±1.05	94.60±1.59	94.30±2.14	92.70±1.75	**96.60±1.39**	93.90±1.28
horse. vs monkey.	74.30±3.17	77.70±3.01	**81.90±3.34**	78.50±2.35	76.60±2.96	72.50±2.39	**81.90±3.47**
horse. vs ship.	93.10±1.36	94.10±1.39	**95.70±1.36**	93.90±1.39	92.20±0.68	92.20±0.81	94.60±2.01
horse. vs truck.	91.90±2.24	93.60±0.80	91.50±1.67	92.90±3.01	91.20±1.21	91.10±2.01	**96.10±1.59**
monkey. vs ship.	95.40±1.98	95.60±1.53	**96.90±1.32**	95.40±1.16	95.00±0.84	93.10±1.02	96.40±0.80
monkey. vs truck.	95.30±1.29	95.60±1.24	94.80±1.54	**95.90±0.58**	94.50±1.10	94.30±2.01	94.80±0.68
ship. vs truck.	76.50±2.49	79.80±1.17	77.70±1.78	76.20±3.14	76.40±1.32	77.50±1.58	**80.80±1.54**
Avg. Acc.	85.33	86.70	87.88	86.18	86.17	86.27	**88.60**
Avg. Rank	5.39	3.86	2.66	4.80	4.99	4.62	**1.69**

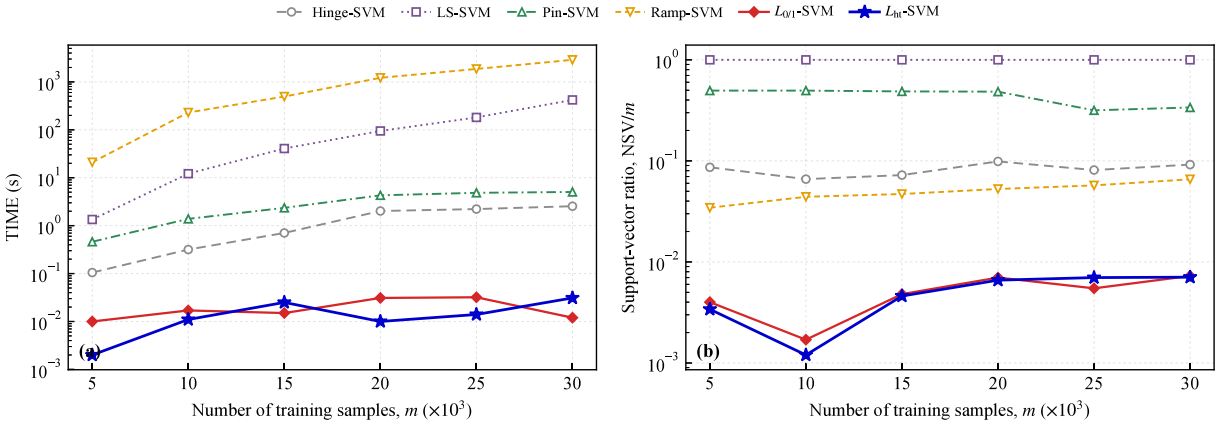


Fig. 3. Sample-size scaling on the synthetic data ($n = 2$), using the results in Tables 10 and 11: (a) TIME and (b) support-vector ratio. Both vertical axes use logarithmic scales.

Table 28
Precision (%) with standard deviations for the STL-10 dataset.

Datasets	SVM-2K	MvSVM-2C	MvLSSVC-2C	Wave-MvSVM	MVASY-BX	Mv$SL_{0/1}$-SVM	MvL_{ht}-SVM
air. vs bird.	87.26±2.74	**92.93±2.20**	90.49±2.89	85.58±2.69	92.28±3.34	91.07±3.42	92.52±2.40
air. vs car.	**98.28±0.98**	91.41±2.07	95.89±0.85	94.14±1.45	96.99±1.15	95.91±1.40	95.58±0.99
air. vs cat.	94.21±1.61	97.68±2.29	97.26±1.36	94.20±2.50	**98.11±2.29**	96.85±2.52	94.63±2.20
air. vs deer.	96.18±1.33	95.69±2.12	98.12±0.79	95.11±1.63	**98.93±0.66**	96.85±2.24	94.94±1.50
air. vs dog.	94.97±1.24	95.33±2.08	96.38±2.49	92.00±2.67	**97.44±1.44**	95.69±1.33	96.37±2.10
air. vs horse.	96.82±2.03	97.26±1.33	97.95±1.55	94.77±2.85	**98.73±1.00**	95.97±0.66	95.34±1.09
air. vs monkey.	98.02±0.80	98.15±0.42	97.98±1.81	96.34±1.36	**98.51±0.52**	95.25±1.44	98.39±0.77
air. vs ship.	86.08±2.50	**88.73±3.26**	86.74±3.18	81.84±2.70	84.70±3.88	78.97±1.31	84.35±1.59
air. vs truck.	92.50±2.90	91.94±3.14	**93.09±2.96**	89.31±3.73	92.28±3.25	90.76±4.31	90.23±3.15
bird. vs car.	93.73±1.30	94.36±2.07	95.00±1.92	**97.50±1.68**	92.46±2.20	90.58±2.02	95.71±1.53
bird. vs cat.	83.80±3.60	79.70±5.06	81.68±3.61	74.33±2.45	80.79±1.92	71.97±4.55	**84.19±3.29**
bird. vs deer.	**95.04±1.64**	90.24±2.65	86.71±2.98	88.08±3.35	89.20±3.75	76.96±5.53	81.32±8.18
bird. vs dog.	**90.43±3.88**	82.33±3.90	78.16±1.56	74.20±1.41	78.05±4.37	76.06±8.21	82.18±2.56
bird. vs horse.	89.76±1.20	84.53±2.87	88.90±0.93	86.85±1.59	**91.96±1.43**	85.39±5.70	89.99±2.34
bird. vs monkey.	**92.41±2.57**	85.73±4.47	78.15±3.18	74.60±5.06	81.84±4.11	80.15±3.02	84.37±3.58
bird. vs ship.	89.25±1.79	93.23±3.24	93.69±2.27	**94.07±0.48**	86.91±3.04	91.73±2.30	92.07±1.90
bird. vs truck.	93.46±1.72	95.49±1.24	91.80±1.60	**97.22±2.20**	92.20±1.50	88.99±2.14	92.98±0.98
car. vs cat.	96.19±0.56	97.15±2.20	97.69±1.53	95.30±1.47	**98.93±1.14**	95.40±2.83	94.25±3.83
car. vs deer.	97.33±0.52	98.34±1.06	**98.96±0.66**	98.12±1.21	96.81±1.28	95.63±2.67	94.54±2.19
car. vs dog.	76.23±2.60	**97.68±1.17**	96.77±0.55	95.21±2.55	97.58±1.24	96.92±1.60	93.96±2.39
car. vs horse.	96.99±1.19	95.99±2.45	98.16±1.34	94.04±0.92	96.81±1.42	95.43±1.06	**99.18±0.76**
car. vs monkey.	98.35±1.36	97.73±1.21	98.74±1.01	95.46±1.42	**99.57±0.52**	95.47±3.28	96.81±1.40
car. vs ship.	84.39±2.38	87.64±3.35	85.61±8.44	**88.04±3.41**	84.46±2.72	87.85±6.87	87.34±3.07
car. vs truck.	**90.04±2.63**	88.94±4.47	89.43±2.84	84.81±2.08	89.38±2.51	89.46±3.17	89.97±3.32
cat. vs deer.	79.34±4.50	77.22±3.26	**80.95±2.92**	73.70±3.49	**80.95±2.32**	79.95±5.22	79.82±3.26
cat. vs dog.	58.53±3.78	62.14±2.57	67.81±0.49	66.19±2.43	63.17±2.18	62.34±3.43	**68.31±1.69**
cat. vs horse.	76.44±2.29	77.22±2.78	83.17±2.52	82.11±3.05	80.37±3.24	82.02±3.29	**83.52±1.79**
cat. vs monkey.	72.90±3.30	73.66±2.02	77.52±2.90	74.62±2.27	75.07±2.73	76.77±3.32	**79.27±4.53**
cat. vs ship.	90.57±1.88	91.75±3.56	92.05±1.48	91.75±3.08	87.13±2.21	**92.32±2.28**	90.68±5.53
cat. vs truck.	91.41±2.49	93.23±1.25	91.84±2.97	**97.30±2.31**	89.24±1.92	93.25±4.08	95.81±2.43
deer. vs dog.	71.01±0.73	**73.81±2.14**	72.83±3.13	71.39±1.68	72.98±2.20	71.11±2.47	73.55±2.37
deer. vs horse.	75.57±2.86	78.72±3.58	82.12±3.26	78.43±3.52	81.18±8.82	81.56±3.22	**82.56±3.04**
deer. vs monkey.	74.35±2.19	74.86±3.97	**78.59±2.10**	77.90±2.40	74.88±2.06	78.12±2.48	78.30±3.29
deer. vs ship.	92.46±1.47	94.50±2.11	95.78±2.40	95.53±1.48	90.82±2.10	96.63±3.53	**97.79±1.61**
deer. vs truck.	93.92±2.07	95.72±1.77	96.31±0.93	95.17±1.61	91.58±2.93	95.21±3.14	**97.81±0.95**
dog. vs horse.	71.56±2.57	68.93±1.97	73.34±3.36	73.05±3.01	**75.90±5.30**	72.62±3.14	74.10±4.03
dog. vs monkey.	68.02±2.13	68.18±3.73	71.05±4.91	61.59±3.26	67.61±2.05	65.67±5.54	**73.24±5.07**
dog. vs ship.	**96.02±1.75**	93.02±2.07	91.24±1.98	94.58±1.73	87.99±1.75	94.65±1.69	93.46±1.58
dog. vs truck.	91.71±1.05	93.94±2.17	92.29±2.43	95.66±1.70	88.81±2.50	**95.89±1.87**	93.13±1.20
horse. vs monkey.	81.91±3.16	79.53±3.91	**84.41±4.12**	78.48±4.00	75.64±3.41	78.44±4.60	84.31±3.41
horse. vs ship.	89.78±2.43	93.05±2.30	93.75±2.28	93.75±2.75	87.85±1.25	91.28±1.58	**98.11±1.67**
horse. vs truck.	89.22±2.81	92.50±2.24	88.15±2.65	93.25±3.84	86.82±1.77	93.56±1.89	**94.14±2.61**
monkey. vs ship.	93.87±2.69	93.90±2.31	95.22±2.14	93.84±1.60	91.52±1.14	95.13±2.51	**95.67±0.98**
monkey. vs truck.	92.86±2.11	95.09±1.81	92.85±2.92	**96.77±1.27**	90.87±1.86	94.16±4.29	95.22±1.76
ship. vs truck.	**86.23±3.87**	80.45±1.53	79.83±3.15	77.78±1.78	79.02±1.88	79.50±1.43	82.93±3.16
Avg. Prec.	87.76	88.21	88.77	87.20	87.43	87.23	**89.18**
Avg. Rank	4.44	3.86	3.18	4.76	4.26	4.42	**3.09**

do not increase monotonically with m, because the selected (C, ξ) and the iteration count differ between sample sizes. The table compares the two variants under the same settings and is not a scaling curve.

6.1.4. Performance analysis of L_{ht}-ADMM

We illustrate the optimization trajectory of L_{ht}-ADMM on w5a ($m = 9\,888$) in Fig. 4. Panel (a) shows that the stationarity residual falls below the prescribed 10^{-3} tolerance within about twenty iterations. Panel (b) shows that the number of retained support vectors decreases from about 800 to 115, while panels (c) and (d) record the accompanying stabilization of the training accuracy and the cumulative training time.

6.1.5. Optimization stability and sensitivity

Because the objective is non-convex, the solution returned by L_{ht}-ADMM may depend on initialization and algorithmic controls. For each of four UCI datasets (Australian, breast-cancer, wine-red, and wine-white), we start from the (C, ξ) pair selected under the default configuration and vary one factor at a time. The value of C remains fixed throughout, while ξ remains at its selected value except in the penalty-sensitivity experiment. We test four initializations, nine values of ξ, and three stopping tolerances, keeping the five-fold evaluation protocol unchanged.

Table 13 summarizes the results. On breast-cancer and wine-white, the accuracy is insensitive to all tested controls: the ranges are 96.93–97.22% and 74.58–75.38% across initializations, 96.64–97.22% and 74.09–75.38% across ξ, and 96.49–97.22% and 73.95–75.38% across stopping tolerances. The support-vector counts vary more, ranging over all tested controls from 22.9 to 30.2 on breast-cancer and from 108.3 to 208.7 on wine-white. The stability is, however, data-dependent. On Australian, zero initialization terminates at a degenerate solution with 44.49% accuracy in which every training sample remains a support vector, and the intermediate penalty $\xi = 1$ yields 72.61% with a fold-wise standard deviation of 27.40, whereas the default configuration reaches 86.23%. A simple safeguard is to run several initializations and keep the solution with the smallest training objective. In the tested runs, this rule avoids the degenerate outcome without using test information, although the selected run can still trail the best single initialization by up to 0.74 percentage points of test accuracy. These observations are consistent with the conditional nature of Theorem 6: different settings can terminate at different stationary points, which is why the default configuration is reported explicitly. In practice, we recommend $\mathbf{w}^0 = \mathbf{1}/100$

Table 29
Recall (%) with standard deviations for the STL-10 dataset.

Datasets	SVM-2K	MvSVM-2C	MvLSSVC-2C	Wave-MvSVM	MVASY-BX	Mv$SL_{0/1}$-SVM	MvL_{ht}-SVM
air. vs bird.	80.60±3.72	86.20±1.33	88.00±2.68	**94.40±3.01**	87.40±1.20	90.60±2.65	90.60±0.80
air. vs car.	79.80±0.75	91.00±1.10	88.80±2.14	90.00±2.10	83.40±1.02	93.40±0.80	**95.00±0.63**
air. vs cat.	87.40±2.58	89.40±1.85	91.60±1.36	92.80±1.94	89.00±3.03	90.60±1.20	**93.60±3.01**
air. vs deer.	90.00±2.28	91.60±2.73	94.00±1.41	92.20±2.32	91.80±1.94	92.60±3.72	**96.60±1.85**
air. vs dog.	90.00±2.19	89.20±1.60	94.40±1.36	**95.00±2.10**	90.00±2.19	92.80±1.17	94.00±2.76
air. vs horse.	88.80±2.93	91.00±1.79	94.40±1.02	93.60±4.08	91.20±2.04	95.40±1.85	**98.00±0.63**
air. vs monkey.	88.60±1.36	96.00±2.28	96.20±1.60	94.40±1.74	92.80±1.72	95.80±2.14	**96.80±1.33**
air. vs ship.	**80.40±3.20**	69.40±3.38	71.80±3.31	76.80±4.45	72.40±4.03	78.60±4.18	74.20±7.73
air. vs truck.	87.00±3.46	91.20±2.99	90.20±1.17	79.00±1.79	88.60±1.96	91.80±0.98	**95.20±1.72**
bird. vs car.	95.60±2.65	96.00±2.28	**97.60±1.85**	92.20±1.94	97.40±1.96	95.80±1.17	**97.60±1.36**
bird. vs cat.	58.00±4.43	66.20±6.24	**71.00±5.18**	66.40±5.68	59.60±4.76	69.60±8.09	68.20±5.42
bird. vs deer.	45.80±4.45	51.80±4.71	62.40±4.80	61.80±3.76	65.60±6.12	**77.00±6.07**	69.80±9.06
bird. vs dog.	46.80±4.31	52.20±4.96	62.80±3.97	**65.20±5.56**	59.80±6.40	63.20±9.66	57.60±8.11
bird. vs horse.	63.40±4.72	**85.20±4.17**	73.40±3.01	74.20±4.02	66.60±4.13	79.80±8.26	83.80±3.82
bird. vs monkey.	47.20±2.93	56.80±6.05	72.80±3.49	**79.20±2.40**	64.40±6.05	74.40±3.67	71.80±5.31
bird. vs ship.	94.60±2.65	95.00±3.16	**97.40±2.65**	88.80±1.47	97.00±2.00	94.20±2.99	94.60±2.58
bird. vs truck.	96.40±2.42	97.00±1.67	95.80±2.71	91.60±3.50	96.60±1.96	96.40±2.73	**98.00±1.41**
car. vs cat.	91.20±3.25	93.60±2.24	92.60±1.85	93.00±2.19	91.40±1.85	93.20±1.94	**97.60±1.62**
car. vs deer.	87.40±2.15	93.60±2.06	94.40±1.02	93.40±2.58	89.80±2.32	98.00±1.41	**99.40±0.80**
car. vs dog.	**99.60±0.49**	91.40±2.65	89.20±3.66	94.00±2.10	87.80±3.87	93.20±2.32	95.80±1.60
car. vs horse.	89.60±2.06	94.80±1.60	95.00±0.63	94.80±2.04	90.00±2.97	92.00±1.55	**96.20±1.72**
car. vs monkey.	93.60±2.50	94.80±1.60	92.60±2.50	96.40±1.36	92.40±1.02	**97.40±2.50**	96.60±1.02
car. vs ship.	**96.00±2.19**	89.20±1.72	90.00±6.23	87.40±1.85	94.00±0.63	86.80±10.17	90.00±3.16
car. vs truck.	79.60±4.72	84.80±1.72	**87.00±2.00**	84.60±1.85	79.20±3.66	85.80±1.47	**87.00±2.10**
cat. vs deer.	67.00±3.63	69.60±4.59	71.80±3.31	70.00±5.02	62.00±7.24	65.80±4.66	**72.20±2.71**
cat. vs dog.	69.40±8.50	64.00±7.69	70.80±4.07	58.80±4.02	**77.00±8.22**	55.00±2.28	70.40±4.76
cat. vs horse.	**85.80±4.92**	77.60±6.56	82.60±2.58	76.60±5.75	82.00±5.10	83.60±3.01	83.00±3.35
cat. vs monkey.	70.80±5.53	71.40±5.68	74.80±4.87	72.20±6.94	70.00±12.99	74.40±5.82	**76.60±5.39**
cat. vs ship.	95.60±1.96	96.40±1.85	96.80±1.83	93.80±1.83	**98.40±2.06**	90.60±1.85	96.60±2.42
cat. vs truck.	94.80±1.47	96.00±1.79	95.60±1.20	91.20±1.17	**97.40±2.58**	89.00±5.80	95.40±1.62
deer. vs dog.	82.80±2.04	76.60±2.58	83.20±2.64	75.80±4.45	81.80±2.64	**84.80±2.79**	83.20±3.37
deer. vs horse.	**88.60±3.07**	82.80±3.87	87.60±3.20	74.80±2.04	84.80±2.56	83.20±2.56	86.20±3.06
deer. vs monkey.	84.00±2.45	83.40±2.33	88.00±2.10	87.20±1.60	82.00±3.85	85.40±2.06	**89.80±2.32**
deer. vs ship.	97.80±2.04	**99.00±1.26**	98.80±0.75	93.80±1.94	98.40±0.80	90.80±6.08	97.60±2.06
deer. vs truck.	98.00±0.89	97.80±1.17	**99.00±0.63**	94.00±1.67	98.60±1.36	91.00±4.94	98.00±0.63
dog. vs horse.	71.80±3.82	71.00±2.45	**74.00±4.29**	73.80±2.93	66.00±5.33	73.00±3.41	73.40±4.84
dog. vs monkey.	61.20±2.86	64.40±4.88	66.80±3.76	**78.80±6.76**	64.60±10.37	57.60±12.24	63.80±5.11
dog. vs ship.	96.20±1.47	95.20±1.17	95.20±1.72	93.40±2.24	**97.80±0.75**	87.40±2.42	96.80±1.94
dog. vs truck.	95.00±0.89	95.20±0.40	97.40±0.49	92.80±2.93	**97.80±0.75**	97.40±1.36	94.80±1.94
horse. vs monkey.	62.40±6.62	74.80±2.40	78.40±3.93	**79.00±2.37**	78.80±5.56	62.60±4.03	78.40±4.50
horse. vs ship.	97.40±1.85	95.40±1.85	**98.00±1.67**	94.20±2.48	**98.00±1.79**	93.40±3.01	91.00±4.15
horse. vs truck.	95.40±2.42	95.00±1.67	96.00±1.10	92.60±3.14	97.20±0.75	88.40±5.39	**98.40±0.80**
monkey. vs ship.	97.20±1.33	97.60±1.02	98.80±0.75	97.20±0.75	**99.20±0.75**	91.00±3.35	97.20±0.75
monkey. vs truck.	98.20±0.40	96.20±1.47	97.20±0.40	95.00±1.67	**99.00±0.63**	94.80±3.25	94.40±1.85
ship. vs truck.	63.20±3.97	**78.80±2.71**	74.40±2.73	73.20±5.81	72.00±3.03	74.20±4.40	77.80±2.23
Avg. Rec.	82.89	84.79	86.86	85.10	84.91	85.15	**87.84**
Avg. Rank	5.09	4.38	2.96	4.46	4.22	4.19	**2.71**

and $b^0 = 0$ as the default initialization, which avoided the degenerate solution in all tested runs. The multi-start safeguard is needed only when the returned solution shows signs of degeneracy, such as a training balanced accuracy close to 0.5, a single predicted class, or an empty support vector set. In this case the solver is rerun from the zero, random, and hinge warm-started initializations, and the solution with the smallest training objective is kept. Since these signs are computed on the training data, the safeguard does not affect the evaluation protocol.

The sensitivity is data-dependent, so universal fixed values are not recommended. To limit the search cost, we use a staged validation procedure. We first select C and ξ for the single-view model. Their selected values provide starting points for the multi-view model, after which η and α are tuned on smaller grids. Table 13 indicates that the default ξ is a useful starting point on the four tested datasets, but the Australian results also show that it is not generally safe to exclude ξ from validation. This staged procedure reduces the simultaneous four-parameter search without assuming that any parameter is insensitive.

6.1.6. Robustness under label and practical corruptions

We first retain the class-wise random-label experiment from the original evaluation. On breast-cancer and Australian, a fraction $r \in \{0.00, 0.02, 0.04, 0.06, 0.08, 0.10\}$ of the labels in each training class is flipped, while the test labels remain clean. Results are averaged over five-fold cross-validation and reported in Tables 14–15.

As the noise rate increases, the accuracy of every method decreases. L_{ht}-SVM remains among the best-performing methods and attains the highest or tied-highest accuracy at most, though not all, noise levels. In terms of NSV, L_{ht}-SVM and $L_{0/1}$-SVM retain substantially fewer support vectors than LS-SVM and Pin-SVM, and they also form the fastest group in these experiments. Note that the smallest support-vector count does not automatically imply the highest ACC or the lowest TIME: fitting time also depends on the numerical path and the iteration count, while accuracy depends on whether the retained samples preserve useful boundary information after corruption.

Additional stress protocols. To examine robustness beyond random label flips, we use the same two datasets and six classifiers under five further conditions. Boundary-label noise flips the class-wise fraction $r_b \in \{0.1, 0.2, 0.3\}$ closest to a clean linear decision boundary. Gaussian feature corruption adds zero-mean noise scaled by the training-feature standard deviation with $\gamma \in \{0.2, 0.4\}$. Sparse outlier contamination selects $r_o \in \{0.1, 0.3\}$ of the training samples and multiplies one randomly

Table 30
F1-score (%) with standard deviations for the STL-10 dataset.

Datasets	SVM-2K	MvSVM-2C	MvLSSVC-2C	Wave-MvSVM	MVASY-BX	Mv$SL_{0/1}$-SVM	MvL_{ht}-SVM
air. vs bird.	83.76±2.76	89.43±1.38	89.16±1.48	89.74±2.18	89.76±2.10	90.79±2.40	**91.53±1.06**
air. vs car.	88.08±0.66	91.19±1.23	92.20±1.49	92.02±1.68	89.68±0.63	94.63±0.92	**95.29±0.67**
air. vs cat.	90.65±1.63	93.32±0.99	**94.34±0.87**	93.46±1.27	93.28±1.48	93.61±1.50	94.06±1.45
air. vs deer.	92.96±1.23	93.56±1.45	**96.01±1.01**	93.60±0.81	95.22±1.06	94.66±2.76	95.74±0.42
air. vs dog.	92.39±0.96	92.15±1.48	**95.36±1.60**	93.42±0.99	93.55±1.07	94.21±0.93	95.13±1.52
air. vs horse.	92.58±1.27	94.00±0.47	96.13±0.92	94.14±3.00	94.79±0.73	95.68±1.20	**96.65±0.72**
air. vs monkey.	93.06±0.66	97.05±1.33	97.08±1.57	95.35±1.32	95.56±0.99	95.51±1.33	**97.58±0.51**
air. vs ship.	**83.13±2.66**	77.87±3.27	78.54±3.00	79.21±3.47	77.90±1.54	78.71±1.79	78.73±4.43
air. vs truck.	89.58±1.65	91.47±0.97	91.59±1.41	83.80±2.06	90.33±0.70	91.21±2.01	**92.63±2.16**
bird. vs car.	94.64±1.64	95.14±1.29	96.25±0.97	94.75±1.06	94.84±1.50	93.11±1.42	**96.64±1.06**
bird. vs cat.	68.40±2.99	72.10±4.54	**75.79±2.66**	70.08±4.20	68.49±3.27	70.34±4.04	75.18±3.34
bird. vs deer.	61.68±4.10	65.61±2.99	72.47±3.76	72.48±1.70	75.34±3.79	**76.67±3.51**	74.23±3.80
bird. vs dog.	61.46±3.11	63.79±4.43	**69.56±2.41**	69.32±3.60	67.40±4.08	68.24±6.17	67.37±5.40
bird. vs horse.	74.23±3.55	84.75±1.68	80.36±1.56	79.99±2.87	77.19±3.18	81.94±2.35	**86.72±2.15**
bird. vs monkey.	62.38±1.99	68.12±4.66	75.35±2.98	76.68±2.27	71.91±4.42	77.08±2.15	**77.47±3.73**
bird. vs ship.	91.84±1.95	94.06±2.45	**95.49±1.94**	91.35±0.81	91.62±1.31	92.89±1.28	93.29±1.52
bird. vs truck.	94.87±1.21	**96.23±0.98**	93.73±1.41	94.31±2.82	94.33±1.18	92.51±1.48	95.42±1.14
car. vs cat.	93.61±1.90	95.31±1.48	95.07±1.39	94.12±1.50	95.00±1.11	94.25±1.46	**95.82±1.43**
car. vs deer.	92.08±1.19	95.89±0.96	96.62±0.62	95.68±1.55	93.14±0.97	96.76±0.78	**96.89±1.22**
car. vs dog.	86.34±1.75	94.40±1.28	92.78±1.77	94.57±1.69	92.38±2.18	**95.00±1.18**	94.86±1.82
car. vs horse.	93.13±1.12	95.38±1.83	96.55±0.65	94.41±1.38	93.24±1.50	93.68±1.24	**97.66±1.00**
car. vs monkey.	95.89±1.31	96.24±1.36	95.55±1.32	95.92±1.15	95.85±0.58	96.35±1.36	**96.70±0.92**
car. vs ship.	**89.81±2.07**	88.34±1.02	87.16±2.13	87.69±2.19	88.96±1.65	86.56±3.87	88.58±1.90
car. vs truck.	84.35±2.12	86.75±2.23	88.16±1.56	84.69±1.54	83.96±2.94	87.57±2.08	**88.43±2.32**
cat. vs deer.	72.59±3.46	73.14±3.26	**76.04±2.36**	71.74±3.84	69.98±5.14	72.13±4.54	75.76±2.21
cat. vs dog.	63.33±5.12	62.92±4.82	69.22±2.07	62.17±2.36	69.20±3.65	58.40±2.45	**69.29±2.95**
cat. vs horse.	80.75±2.22	77.26±3.62	82.85±1.85	79.13±3.57	80.99±1.65	82.78±2.81	**83.23±2.03**
cat. vs monkey.	71.80±4.39	72.46±3.88	76.12±3.90	73.29±4.62	71.58±6.97	75.53±4.46	**77.89±4.83**
cat. vs ship.	93.00±1.40	93.99±2.35	**94.35±1.01**	92.71±1.21	92.40±1.62	91.43±1.35	93.42±3.00
cat. vs truck.	93.05±1.25	94.58±0.90	93.66±1.72	94.13±0.78	93.11±1.45	90.85±2.29	**95.60±1.92**
deer. vs dog.	76.44±1.10	75.17±2.12	77.63±2.44	73.47±2.49	77.10±1.54	77.31±1.76	**78.04±2.17**
deer. vs horse.	81.52±2.21	80.62±2.49	**84.73±2.50**	76.50±1.57	82.55±3.86	82.31±1.85	84.26±1.67
deer. vs monkey.	78.87±2.18	78.87±3.02	83.02±1.96	82.28±1.96	78.22±2.02	81.58±1.92	**83.64±2.66**
deer. vs ship.	95.04±1.37	96.69±1.47	97.26±1.54	94.65±1.40	94.45±1.34	93.44±3.02	**97.69±1.76**
deer. vs truck.	95.90±1.10	96.74±1.26	97.63±0.72	94.57±1.16	94.92±1.28	92.91±2.11	**97.90±0.66**
dog. vs horse.	71.63±2.61	69.94±2.12	73.61±3.19	73.36±2.04	70.41±3.76	72.78±2.83	**73.67±3.66**
dog. vs monkey.	64.40±2.18	66.21±4.19	68.74±3.17	**69.05±4.06**	65.50±4.88	60.05±5.06	67.97±3.07
dog. vs ship.	**96.11±1.49**	94.08±1.07	93.15±1.09	93.96±1.27	92.62±0.95	90.84±1.00	95.09±1.37
dog. vs truck.	93.32±0.65	94.55±0.96	94.76±1.48	94.20±2.22	93.07±1.58	**96.63±1.37**	93.95±1.30
horse. vs monkey.	70.66±4.62	77.06±2.83	**81.24±3.44**	78.64±1.85	77.05±3.14	69.45±2.55	81.21±3.71
horse. vs ship.	93.40±1.21	94.18±1.34	**95.80±1.31**	93.92±1.37	92.63±0.65	92.28±0.95	94.36±2.21
horse. vs truck.	92.18±2.14	93.70±0.72	91.88±1.53	92.89±3.00	91.71±1.05	90.77±2.41	**96.21±1.52**
monkey. vs ship.	95.50±1.92	95.70±1.45	**96.97±1.28**	95.49±1.11	95.20±0.79	92.94±1.16	96.43±0.79
monkey. vs truck.	95.45±1.20	95.63±1.22	94.95±1.40	**95.86±0.61**	94.75±0.98	94.36±1.90	94.78±0.68
ship. vs truck.	72.85±3.16	79.58±1.38	76.94±1.68	75.36±3.86	75.29±1.63	76.67±2.31	**80.22±1.35**
Avg. F1	84.42	86.12	87.60	85.96	85.70	85.94	**88.29**
Avg. Rank	5.41	4.01	2.71	4.47	5.13	4.47	**1.80**

selected feature by 10. Severe imbalance retains all majority-class samples and downsamples the minority class to majority-to-minority ratios $\rho \in \{5, 10\}$. Finally, for each trained linear classifier, a margin-reducing white-box ℓ_∞ perturbation with $\epsilon \in \{0.05, 0.10\}$ is applied to the test features and clipped to the observed feature range. All other corruptions are applied only to the training folds; evaluation uses clean test data.

Tables 16 and 17 report all tested levels. Each entry is the five-fold mean ACC ± standard deviation in percent.

The stress tests extend the evaluation to targeted mislabeling, feature corruption, 30% outlier contamination, 10:1 imbalance, and test-time linear perturbations. Across all these settings, L_{ht}-SVM attains the highest mean ACC in both tables, although several margins are small relative to the fold-wise variation: under 10:1 imbalance on breast-cancer, $L_{0/1}$-SVM and L_{ht}-SVM obtain 96.19 ± 1.41% and 96.20 ± 1.42%, respectively, and under 10% sparse outliers on Australian, four methods lie between 86.81% and 86.96%. Overall, the bounded loss keeps the proposed model competitive in ACC under every tested corruption. The class imbalance setting is examined further below.

Since overall accuracy may hide poor minority-class performance under strong imbalance, Tables 18 and 19 further report the balanced accuracy (BACC) and the recall of the downsampled minority class (Min-Recall) for the class imbalance rows of Tables 16–17, using the same five-fold protocol. On BACC, L_{ht}-SVM remains the best or among the best methods, as it preserves a high majority-class recall while remaining competitive on the minority class. On Min-Recall, however, its advantage is not uniform. $L_{0/1}$-SVM attains the highest minority recall on Australian at both ratios and on breast-cancer at $\rho = 10$, and Ramp-SVM is also competitive on Australian at $\rho = 10$. Therefore, when minority-class detection is the primary concern, the truncated zero-one and ramp losses can be comparable to or better than L_{ht}, and the three metrics should be read together in imbalanced settings.

6.1.7. *Sensitivity to the interior junction δ*

We evaluate $\delta \in \{1/4, 1/3, 1/2, 2/3, 3/4\}$ on Australian and Breast-cancer using the same five-fold partitions as in the main experiments. To separate sensitivity to δ from compensation through hyperparameter selection, the fixed protocol retains the values of C and ξ selected for the clean $\delta = 2/3$ model, whereas the retuned protocol reselects C and ξ for each junction with an identical search budget. Every junction is optimized by the same solver with the δ-specific constants of Section 3 under the same initialization, tolerance, and iteration budget. Because the junction controls where the quadratic branch saturates, it

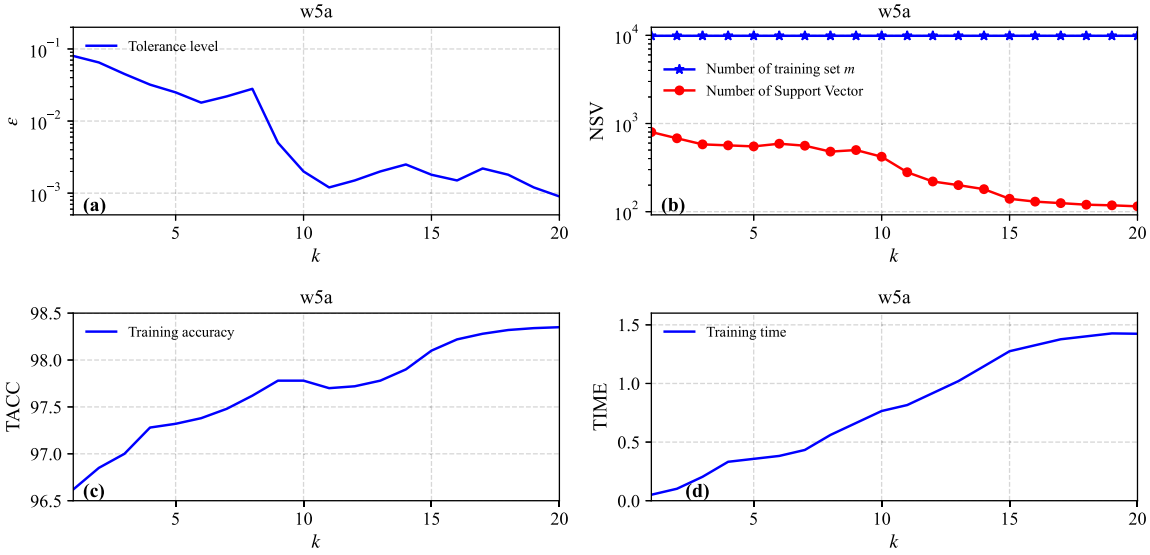


Fig. 4. Optimization process of L_{ht}-ADMM on w5a: (a) stationarity residual, (b) training-set size and number of support vectors, (c) training accuracy, and (d) cumulative fitting time.

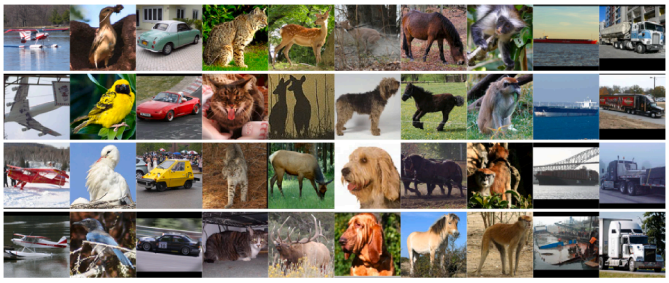


Fig. 5. STL-10 dataset.

can in principle also reduce the influence of hard informative samples rather than only that of outliers. In addition to clean accuracy, Table 20 therefore reports the associated support-vector counts and the accuracy when 10% of the training labels in each class are flipped, following the protocol of Section 6.1.6.

Three observations follow from Table 20. First, with frozen hyperparameters the junction matters: $\delta = 2/3$ gives the highest clean-data accuracy on both datasets, but under 10% label noise the ranking changes, and on Australian the largest junction $3/4$ becomes the most accurate (85.37% versus 80.44% for $2/3$). Second, sparsity depends on the junction in a data-dependent way: on breast-cancer under the fixed protocol, `NSV` grows monotonically from 27.4 at $\delta = 1/4$ to 36.0 at $\delta = 3/4$, consistent with a later saturation retaining more active samples. On Australian under the retuned protocol, however, the smallest `NSV` occurs at $\delta = 1/2$ (25.8) rather than at the reference junction $\delta = 2/3$ (81.4), because retuning selects different (C, ξ) pairs across junctions and the reference setting is not the sparsest one. Third, once C and ξ are reselected per junction, the accuracy differences shrink to within the fold-wise standard deviations on both clean and noisy data, so hyperparameter selection compensates for much of the junction effect. The sparser Australian solution at $\delta = 1/2$ (25.8 against 81.4 support vectors) corresponds to a different (C, ξ) pair selected by accuracy alone, and its accuracy of 85.66% is within one standard deviation of the 86.09% obtained at $\delta = 2/3$. The sparsity depends on (δ, C, ξ) together and not on the junction alone. Since the bounded branch may in principle also down-weight hard informative samples, we inspect the class imbalance results in Tables 16–19. The accuracy and balanced accuracy of L_{ht}-SVM remain competitive under 5:1 and 10:1 downsampling, whereas its minority-class recall on Australian is lower than that of $L_{0/1}$-SVM, so the bounded branch does lose part of the minority class in some cases. The results therefore support $\delta = 2/3$ as a reference value rather than a universal choice. In principle, any $\delta \in (0, 1)$ defines a valid member of the loss family, and Table 20 covers the five values $\{1/4, 1/3, 1/2, 2/3, 3/4\}$. Under the fixed protocol, however, the two smallest junctions give clearly lower and less stable accuracy (e.g., 75.95% and 76.38% on Australian), so the useful range in practice lies in the upper part of the grid. For a new dataset, especially when heavy noise or imbalance is expected, we therefore recommend selecting δ from $\{1/2, 2/3, 3/4\}$ by cross-validation on the training folds, which at most triples the search cost.

6.2. Multi-view experiments

6.2.1. Experimental settings

This subsection compares MvL_{ht}-SVM with six multi-view SVM baselines on two families of datasets.

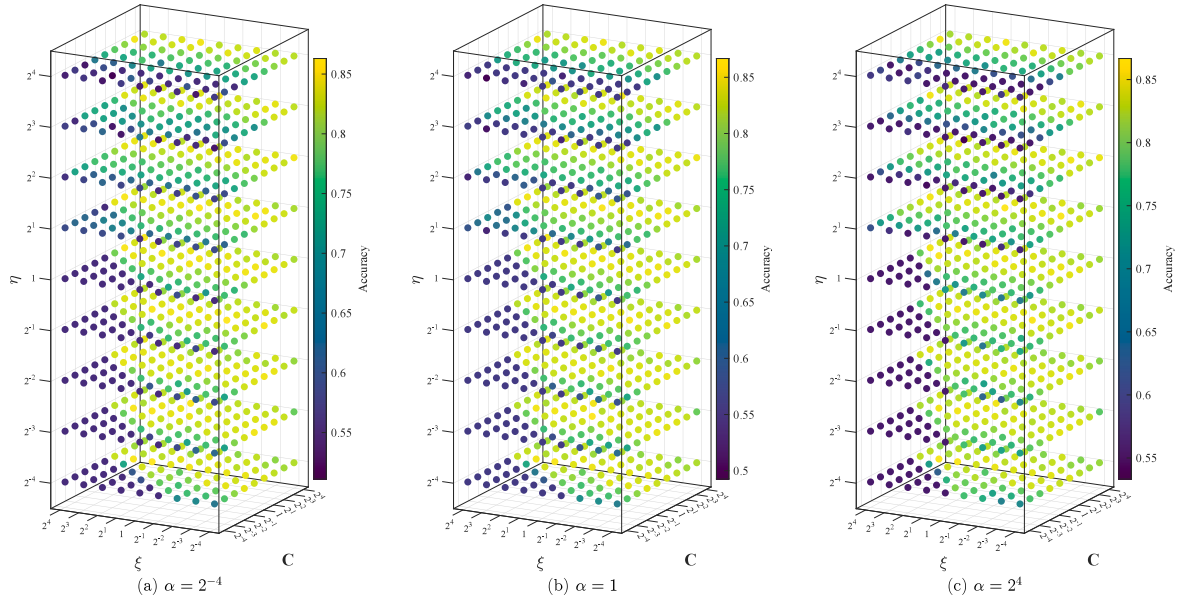
Datasets. We use 10 UCI datasets and 45 binary tasks derived from STL-10.[3]

(1) UCI datasets: Table 21 summarises the 10 UCI datasets employed. Because these datasets are originally single-view, we adopt an artificial two-view setting: view 1 consists of the raw features and view 2 comprises the leading principal components that retain 95% of the variance. The PCA mapping and the scaling of both views are fitted on the training portion of each fold and then applied to the held-out fold. These datasets are therefore not naturally multi-view; complementary experiments on a naturally multi-view dataset are reported in Appendix D.

(2) STL-10: STL-10 contains 96×96-pixel images from ten object categories (see Fig. 5). We extract HOG and LBP descriptors as two views and randomly sample 500 images per class. Following the treatment of CIFAR-10 by Chen et al. (2025a) and of AwA by Quadir et al. (2025), we construct all $\binom{10}{2} = 45$ one-vs-one binary tasks to test generalisation across diverse class pairs.

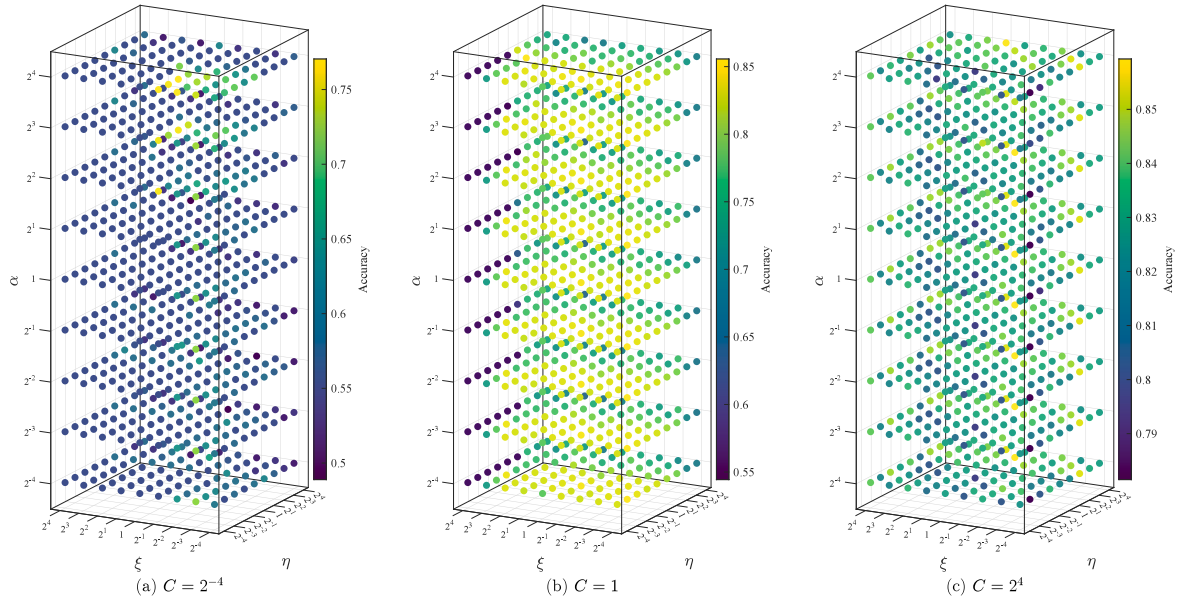
Benchmark methods. Six multi-view SVM baselines are considered: (1) SVM-2K (hinge loss): enforces inter-view consistency via KCCA (Farquhar et al., 2005). (2) MvSVM-2C (hinge loss): jointly captures consensus and complementarity through adaptive view weighting (Xie & Sun,

[3] https://cs.stanford.edu/~acoates/stl10/



(a) $\alpha = 2^{-4}$ (b) $\alpha = 1$ (c) $\alpha = 2^{4}$

Fig. 6. Parameter sensitivity on heart: fix α at 2^{-4}, 1, 2^{4}.



(a) $C = 2^{-4}$ (b) $C = 1$ (c) $C = 2^{4}$

Fig. 7. Parameter sensitivity on heart: fix C at 2^{-4}, 1, 2^{4}.

2020). (3) MvLSSVC-2C (least squares loss): couples multiple LS-SVM classifiers with a view-interaction term (Zhang et al., 2025). (4) Wave-MvSVM (wave loss): applies a bounded wave loss for robustness under the 2C framework (Quadir et al., 2025). (5) MVASY-BX (bounded LINEX loss): employs a bounded asymmetric loss to resist outliers (Tang et al., 2023). (6) Mv$SL_{0/1}$-SVM ($L_{0/1}$ loss): adopts the 0/1 soft-margin loss in multi-view classification (Chen et al., 2025a).

Evaluation metrics. We measure accuracy (ACC), precision, recall, and F1-score, all estimated via 5-fold cross-validation. Hyper-parameters are tuned by grid search over $\{2^i \mid i = -4, -2, 0, 2, 4\}$. On the larger STL-10 tasks the search grid is narrowed to $\{2^i \mid i = -2, 0, 2\}$ to keep the total cost manageable. The fold construction, the random seed, and the model selection procedure follow the single-view protocol in Section 6.1.1. TIME is defined as in Section 6.1.1 and does not include hyper-parameter search. For MvL_{ht}-SVM with (C, ξ, η, α) the search requires $5^4 = 625$ candidate fits per fold on the UCI datasets and $3^4 = 81$ on STL-10.

6.2.2. Results on UCI multi-view datasets

Tables 22–26 show that MvL_{ht}-SVM obtains the best or tied-best accuracy on 8 of the 10 UCI datasets and the best reported average rank for accuracy, precision, and F1-score. For recall, however, Mv$SL_{0/1}$-SVM has both the higher average value and the better average rank. MvLSSVC-2C is the fastest solver on these small-to-medium datasets, while MvL_{ht}-SVM ranks second in average fitting time. Because the two views of each UCI dataset are constructed from the same underlying features and are therefore strongly correlated, these datasets mainly assess the robustness of the fusion procedure; complementary evidence on naturally multi-view data is provided in Appendix D. Formal significance analysis is conducted on the 45 STL-10 tasks in Section 6.3.

6.2.3. Results on STL-10 datasets

Tables 27–30 report the results on all 45 STL-10 binary tasks. MvL_{ht}-SVM has the highest average accuracy (88.60%), followed by MvLSSVC-2C (87.88%). It also has the highest reported averages for precision

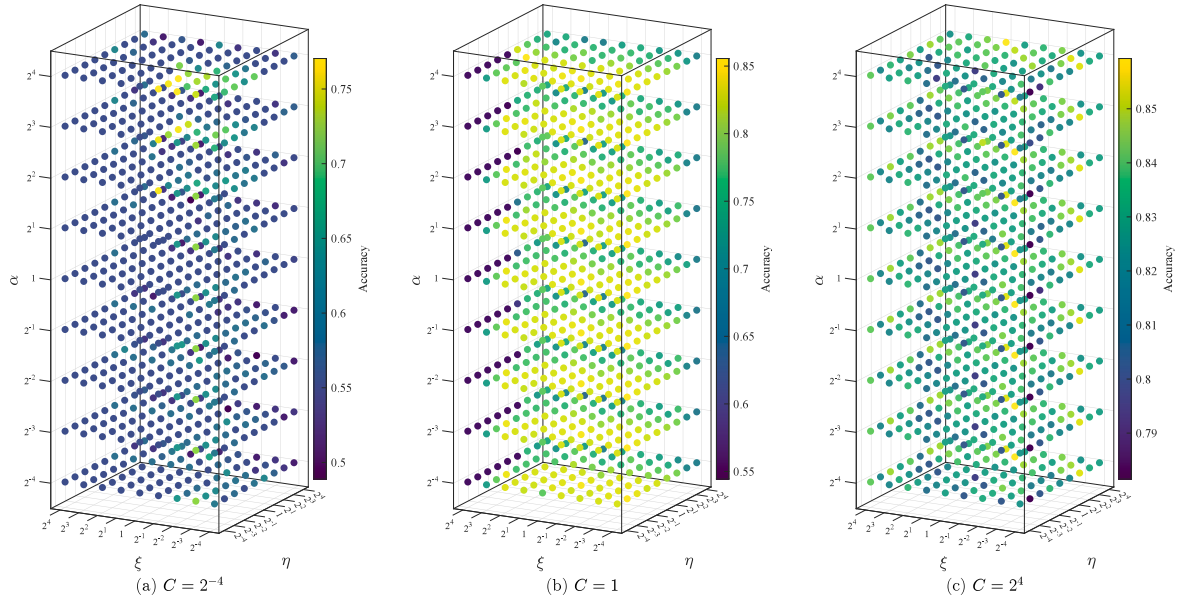


Fig. 8. Parameter sensitivity on heart: fix η at 2^{-4}, 1, 2^{4}.

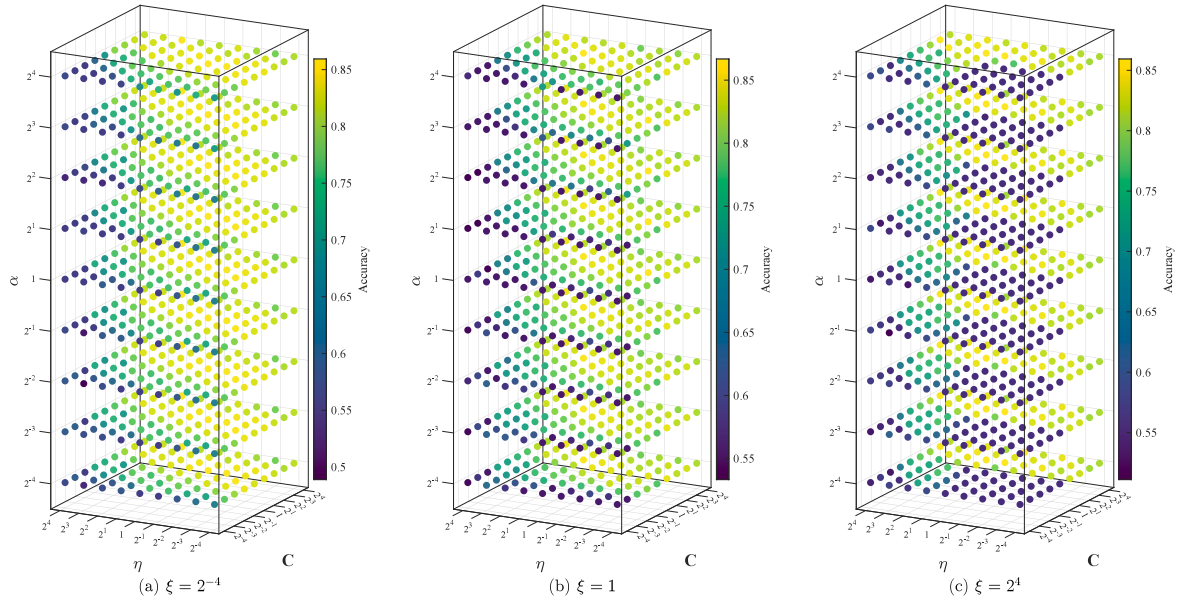


Fig. 9. Parameter sensitivity on heart: fix ξ at 2^{-4}, 1, 2^{4}.

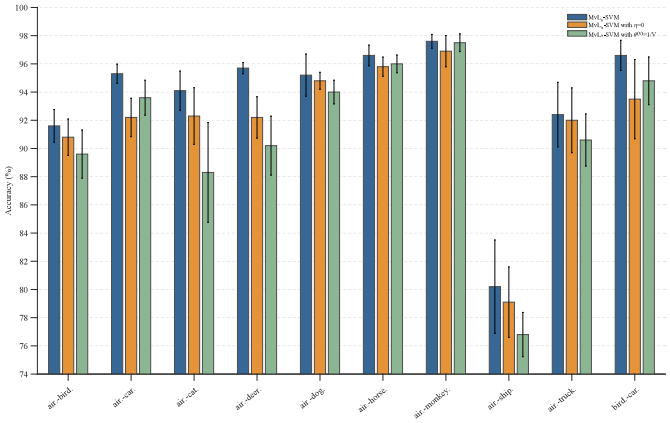


Fig. 10. Ablation experiments on the first 10 STL-10 datasets.

(89.18%), recall (87.84%), and F1-score (88.29%). The margins vary substantially across class pairs: the proposed model leads on several difficult pairs, including cat vs. dog and dog vs. monkey, but it is not uniformly best. The corresponding Friedman–Iman–Davenport tests, which reject equality of the seven methods on all four metrics, are reported in Section 6.3.

6.2.4. *Parameter sensitivity analysis*

Here, we use the UCI datasets to examine the parameter sensitivity of MvL_{ht}-SVM in order to analyze the impact of parameters. The accuracy of MvL_{ht}-SVM on the Heart dataset is shown in Figs. 6–9, where the color of the dot indicates the accuracy for the relevant parameter combination. We fix each parameter in turn and explore combinations of the remaining three. These graphs show that the accuracy of MvL_{ht}-SVM can be sensitive to all four parameters. The number of SVs is mostly controlled by C and α. Thus, no universal fixed values are assumed; the

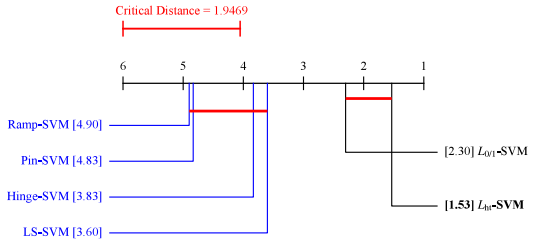


Fig. 11. Critical-difference diagram for single-view ACC on 15 datasets. Lower ranks indicate better performance. Red horizontal segments mark groups not separated by the all-pairs Nemenyi procedure at $\alpha = 0.05$; CD = 1.9469.

Table 31
Wilcoxon signed-rank comparisons between L_{ht}-SVM and the single-view baselines. The paired difference is $\Delta\text{ACC} = \text{ACC}_{L_{ht}} - \text{ACC}_{\text{baseline}}$ in percentage points; $n_{\neq 0}$ is the number of nonzero differences.

Baseline	$n_{\neq 0}$	Median ΔACC	R^+/R^-	T	Holm p
Hinge-SVM	14	0.20	103/2	2	0.0010
LS-SVM	13	0.09	91/0	0	0.0010
Pin-SVM	13	0.42	91/0	0	0.0010
Ramp-SVM	14	0.48	105/0	0	0.0006
$L_{0/1}$-SVM	13	0.07	75/16	16	0.0386

Table 32
Friedman–Iman–Davenport results on the 45 STL-10 binary tasks. The last column gives the average rank of MvL_{ht}-SVM (lower is better).

Metric	χ_F^2	F_F	p-value	Proposed rank
ACC	107.0714	28.9154	$< 10^{-15}$	1.69
Precision	24.4833	4.3878	3.03×10^{-4}	3.09
Recall	42.1690	8.1439	4.34×10^{-8}	2.71
F1-score	98.4786	25.2625	$< 10^{-15}$	1.80

staged validation procedure described in Section 6.1.5 reduces the simultaneous search while retaining data-dependent tuning.

6.2.5. Multi-view mechanism ablation

We compare three variants on the first ten STL-10 tasks: (1) the complete MvL_{ht}-SVM; (2) a no-structure variant with $\eta = 0$, which retains adaptive view weighting but removes both within-view and cross-view covariance regularization; and (3) an equal-weight variant with $\theta^{(v)} = 1/V$, which retains structural regularization but removes adaptive weighting. Apart from the stated restriction, each variant follows the same training and parameter-selection protocol.

Fig. 10 reports the corresponding accuracy, with the task index on the horizontal axis. The complete model attains the highest accuracy on all ten tasks, and the equal-weight restriction is generally more competitive than the $\eta = 0$ restriction, indicating that the structural term contributes information beyond weighted score aggregation.

Because HOG and LBP are two descriptors of the same image, Appendix D complements this ablation on the naturally multi-view WebKB data: the fused model is compared with the individual Fulltext and Inlinks views, the weighting and structural components are removed separately, one training view is corrupted at a time, and the learned weights and cross-view support-set overlap are reported.

6.3. Statistical tests

This subsection tests whether the performance differences observed above are statistically significant. The statistical unit is a dataset or a binary classification task rather than an individual cross-validation fold. All tests are based on the mean values reported in Tables 3, 6, and 27–30. Within each block, rank 1 is assigned to the best result and average ranks are used for ties.

6.3.1. Single-view statistical tests

Following the single-view evaluation protocol, the ACC values from the five UCI and ten LIBSVM datasets form $N = 15$ blocks for $k = 6$ methods. Let R_j be the average rank of method j. The Friedman statistic and its Iman–Davenport correction are

$$\chi_F^2 = \frac{12N}{k(k+1)}\left[\sum_{j=1}^{k} R_j^2 - \frac{k(k+1)^2}{4}\right], \tag{61}$$

$$F_F = \frac{(N-1)\chi_F^2}{N(k-1) - \chi_F^2}, \tag{62}$$

where F_F follows an F distribution with $k-1$ and $(k-1)(N-1)$ degrees of freedom under the null hypothesis of equal performance. We obtain $\chi_F^2 = 39.2857$ and $F_F = 15.4000$. Because $F_F > F_{0.05}(5, 70) = 2.3456$ ($p = 3.41 \times 10^{-10}$), the null hypothesis is rejected. Fig. 11 shows the resulting mean ranks and the Nemenyi critical distance. The proposed method has the lowest mean rank (1.53), but the all-pairs Nemenyi procedure does not separate it from $L_{0/1}$-SVM.

We next use a two-sided Wilcoxon signed-rank test to compare L_{ht}-SVM with each baseline on the same 15 datasets. Zero differences are omitted, tied absolute differences receive average ranks, and exact sign-permutation p-values are adjusted by Holm's procedure over the five planned comparisons. The results are reported in Table 31. Here, R^+ and R^- are the sums of ranks associated with positive and negative paired differences, respectively. All five adjusted p-values are below 0.05, so the improvements are consistent across datasets, although the median gain over $L_{0/1}$-SVM is only 0.07 percentage points and hence small in magnitude. These tests confirm that the improvements are consistent rather than large, and the main practical advantage of L_{ht}-SVM lies in its sparser solutions and shorter fitting time at comparable accuracy.

The two procedures answer different questions: the Nemenyi test adjusts for all 15 method pairs simultaneously and is consequently more conservative, whereas the Holm family contains only the five planned comparisons with the proposed method. This explains why the Wilcoxon tests separate L_{ht}-SVM from $L_{0/1}$-SVM while the Nemenyi procedure does not.

6.3.2. Multi-view Friedman test on STL-10

The multi-view tests use the 45 STL-10 image tasks represented by the HOG and LBP views; the artificially paired two-view UCI datasets are excluded. Thus $N = 45$ and $k = 7$. The null hypothesis states that the seven algorithms perform equally; the alternative is that at least one differs. Applying Eqs. (61) and (62) separately to ACC, Precision, Recall, and F1-score gives Table 32. The 45 tasks are formed by pairing the ten STL-10 classes, so each class appears in nine tasks and the tasks share images. The tasks are not independent. Following Chen et al. (2025a), Quadir et al. (2025), and Tang et al. (2019), who test across binary tasks built from the classes of one dataset, we apply the Friedman test across the 45 STL-10 tasks. The p-values below measure how consistently the methods are ranked across these related tasks and should not be read as evidence from 45 independent benchmarks. WebKB, the only naturally multi-view dataset in this study, provides a single task and is reported descriptively in Appendix D.

At $\alpha = 0.05$, the critical value is $F_{0.05}(6, 264) = 2.1330$. Every observed F_F exceeds this value, so the null hypothesis is rejected for all four metrics, with the qualification above. MvL_{ht}-SVM has the lowest average rank in each case, although its precision rank (3.09) is close to that of MvLSSVC-2C (3.18).

7. Conclusion and future work

This paper introduced a generalized family of hybrid truncated losses and developed single- and multi-view SVM classifiers based on its

$\delta = 2/3$ specialization. For L_{ht}-SVM, we established necessary and sufficient P-stationary conditions for local optimality, derived the closed-form proximal operator, and designed an ADMM solver whose working set reduces the linear-system cost when the active set is small; whenever the generated sequence converges, its limit is a P-stationary point and hence a local minimizer.

Experiments on synthetic, UCI, LIBSVM, and STL-10 data show that L_{ht}-SVM matches the strongest bounded-loss baseline in accuracy (a median gain of 0.07 percentage points over $L_{0/1}$-SVM) with markedly sparser solutions and shorter fitting time. The contribution lies in the loss construction, the support-vector characterization, and the resulting sparsity and fitting time, not in accuracy. MvL_{ht}-SVM attains the best average ranks on the 45 related STL-10 tasks. The single-view improvements, although small in median magnitude, are consistent across datasets, the bounded loss remains competitive under label noise, feature corruption, outliers, imbalance, and test-time perturbations, and $\delta = 2/3$ is a useful reference value. On imbalanced data, however, the minority-class recall of L_{ht}-SVM is not always the highest, and $L_{0/1}$-SVM can be a better choice when the minority class is the main concern. Moreover, the reported advantages in fitting time are verified only with respect to the sample size, and the multi-view mechanisms are examined only on one naturally multi-view dataset. The multi-view significance tests use related tasks from a single dataset, and evaluation on further naturally multi-view benchmarks is left for future work. The main caution is that some initializations lead to degenerate solutions; a multi-start safeguard reduced this risk in the tested runs but provides no global guarantee.

Several questions remain open: establishing convergence of the whole sequence together with its rate, and of the outer alternation in MvL_{ht}-SVM; validating scalability on genuinely high-dimensional sparse data; extending the loss to kernel and multi-class formulations, where the proximal update carries over but the support-vector characterization and the storage of the kernel matrix require new treatment; and using the bounded loss as a robust classification component on top of learned representations.

CRediT authorship contribution statement

Yuliang Yang: Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization; **Chen Chen:** Validation, Writing – review & editing; **Yuxiang Liu:** Formal analysis, Writing – review & editing; **Huiru Wang:** Conceptualization, Methodology, Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

Appendix A. Proof of Theorem 1

Proof. Assume $(\mathbf{w}^*; b^*; \mathbf{q}^*)$ to be a local optimal solution of (7). Following variational analysis (Rockafellar & Wets, 2009), there exists a multiplier $\boldsymbol{\psi}^* \in \mathbb{R}^m$ satisfying

$$\begin{cases} \mathbf{w}^* + N^\top \boldsymbol{\psi}^* = \mathbf{0}, \\ \langle \mathbf{y}, \boldsymbol{\psi}^* \rangle = 0, \\ \mathbf{q}^* + N\mathbf{w}^* + b^*\mathbf{y} = \mathbf{1}, \\ \boldsymbol{\psi}^* + C\partial L_{\mathrm{ht}}(\mathbf{q}^*) \ni \mathbf{0}. \end{cases} \tag{A.1}$$

Combining (16) with (A.1), it suffices to verify that whenever $0 < \nu \le \nu^*$ and $(\boldsymbol{\psi}^*; \mathbf{q}^*)$ satisfies $\mathbf{0} \in \boldsymbol{\psi}^* + C\partial L_{\mathrm{ht}}(\mathbf{q}^*)$, the inclusion $\mathbf{q}^* \in \mathrm{prox}_{\nu C L_{\mathrm{ht}}}(\mathbf{q}^* - \nu\boldsymbol{\psi}^*)$ holds. Invoking Lemma 1 alongside (A.1), we deduce

$$\psi_k^* \in \begin{cases} 0, & q_k^* \ge 1, \\ -18C(1-q_k^*)/5, & q_k^* \in (2/3, 1), \\ -6C/5, & q_k^* \in (0, 2/3], \\ [-6C/5, 0], & q_k^* = 0, \\ 0, & q_k^* < 0. \end{cases} \tag{A.2}$$

Utilizing Lemma 2, we establish (16) by examining three regimes: $\nu C \in (0, 5/18)$, $\nu C \in [5/18, 25/18)$, and $\nu C \ge 25/18$.

Case (i): Consider $\nu C \in (0, 5/18)$. Set $\mathbf{p}^* := \mathbf{q}^* - \nu\boldsymbol{\psi}^* \in \mathbb{R}^m$.

(i) For $k \in \mathcal{A}^*$, one has $q_k^* > 1$ with $\psi_k^* = 0$ via (A.2), leading to

$$p_k^* := q_k^* - \nu\psi_k^* = q_k^* > 1. \tag{A.3}$$

(ii) For $k \in \mathcal{B}^*$, one has $q_k^* \in (2/3, 1]$ with $\psi_k^* = -18C(1-q_k^*)/5$ via (A.2). Consequently,

$$p_k^* = q_k^* - \nu\psi_k^* = q_k^* + 18\nu C(1-q_k^*)/5. \tag{A.4}$$

Given $\nu \le \nu_1^* = 5/(18C)$, it follows that $18\nu C \le 5$; combining with (A.4) yields $p_k^* \in (2/3 + 6\nu C/5, 1]$.

(iii) For $k \in \mathcal{C}^*$, one has $q_k^* \in (0, 2/3]$ with $\psi_k^* = -6C/5$ via (A.2). Hence,

$$p_k^* = q_k^* - \nu\psi_k^* = q_k^* + 6\nu C/5. \tag{A.5}$$

Together with (A.5), this implies $p_k^* \in (6\nu C/5, 2/3 + 6\nu C/5]$.

(iv) For $k \in \mathcal{D}^*$, one has $q_k^* = 0$ with $\psi_k^* \in [-6C/5, 0]$ via (A.2), so that

$$p_k^* = q_k^* - \nu\psi_k^* = -\nu\psi_k^* \in [0, 6\nu C/5]. \tag{A.6}$$

(v) For $k \in \mathcal{E}^*$, one has $q_k^* < 0$ with $\psi_k^* = 0$, giving

$$p_k^* = q_k^* - \nu\psi_k^* = q_k^* < 0. \tag{A.7}$$

Summarizing (i)–(v), we arrive at (11), finishing Case (i).

Case (ii): Let $\nu C \in [5/18, 25/18)$, $\mathbf{p}^* := \mathbf{q}^* - \nu\boldsymbol{\psi}^*$, $\widetilde{\beta} := 6\nu C/5$, and $\widehat{\beta} := 5/6 + 3\nu C/5$. We partition $[m]$ into $\mathcal{R}^* \cup \mathcal{Q}^* \cup \mathcal{P}^* \cup \mathcal{D}^* \cup \mathcal{E}^*$ and verify five sub-cases.

(i) For $k \in \mathcal{R}^*$ (i.e., $q_k^* > 5/6$), the restriction $\nu \le \nu_2^* = 5(6q_k^* - 5)/(18C)$ yields $3\nu C/5 \le (6q_k^* - 5)/6$, hence $q_k^* \ge 5/6 + 3\nu C/5 = \widehat{\beta}$. Noting $\widehat{\beta} \ge 1$ when $\nu C \ge 5/18$, we deduce $q_k^* \ge 1$ and $\psi_k^* = 0$ by (A.2). Therefore $p_k^* = q_k^* > \widehat{\beta}$, and (12) returns $\mathrm{prox}(p_k^*) = p_k^* = q_k^*$.

(ii) For $k \in \mathcal{Q}^*$ (i.e., $q_k^* = 5/6$), observe that $5/6 \in [2/3, 1]$, so $k \in \mathcal{B}^*$. However, $\mathcal{B}^* \ne \emptyset$ would force $\nu \le \nu_1^* = 5/(18C)$, i.e., $\nu C \le 5/18$, contradicting the assumption $\nu C \ge 5/18$. Consequently, $\mathcal{Q}^*$ is empty in this regime.

(iii) For $k \in \mathcal{P}^*$ (i.e., $q_k^* \in (0, 5/6)$), since $\mathcal{B}^* = \emptyset$ (as shown above), all such q_k^* satisfy $q_k^* \le 2/3$, giving $\psi_k^* = -6C/5$ via (A.2) and $p_k^* = q_k^* + \widetilde{\beta}$. The restriction $\nu \le \nu_3^* = 5(5 - 6q_k^*)/(18C)$ ensures $q_k^* \le 5/6 - 3\nu C/5$, from which $p_k^* \le \widehat{\beta}$. Combined with $p_k^* > \widetilde{\beta}$, we obtain $p_k^* \in (\widetilde{\beta}, \widehat{\beta}]$, so (12) delivers $\mathrm{prox}(p_k^*) = p_k^* - \widetilde{\beta} = q_k^*$.

(iv) For $k \in \mathcal{D}^*$ (i.e., $q_k^* = 0$), (A.2) provides $\psi_k^* \in [-6C/5, 0]$, leading to $p_k^* = -\nu\psi_k^* \in [0, \widetilde{\beta}]$. Then (12) returns $\mathrm{prox}(p_k^*) = 0 = q_k^*$.

(v) For $k \in \mathcal{E}^*$ (i.e., $q_k^* < 0$), we have $\psi_k^* = 0$ and $p_k^* = q_k^* < 0$, so (12) gives $\mathrm{prox}(p_k^*) = p_k^* = q_k^*$.

Collecting (i)–(v) establishes (12) for Case (ii).

Case (iii): Let $\nu C \geq 25/18$, $\mathbf{p}^* := \mathbf{q}^* - \nu\boldsymbol{\psi}^*$, and $\gamma := \sqrt{2\nu C}$. As in Case (ii), $\mathcal{B}^* = \emptyset$ follows from $\nu \leq \nu_1^*$. Furthermore, $\mathcal{P}^* \neq \emptyset$ would imply $\nu \leq \nu_3^* \leq 5/(18C)$ and thus $\nu C \leq 5/18 < 25/18$, a contradiction, so $\mathcal{P}^* = \emptyset$. Because $\mathcal{Q}^* \subset \mathcal{B}^*$, we also have $\mathcal{Q}^* = \emptyset$. Hence every $q_k^* > 0$ belongs to $\mathcal{R}^*$. Three sub-cases remain.

(i) For $k \in \mathcal{R}^*$ (i.e., $q_k^* > 0$), the bound $\nu \leq \nu_5^*$ provides $q_k^* \geq \gamma$. Since $\gamma \geq \sqrt{2 \cdot 25/18} = 5/3 > 1$, (A.2) yields $\psi_k^* = 0$ and $p_k^* = q_k^* \geq \gamma$. When $p_k^* > \gamma$, (13) gives $\mathrm{prox}(p_k^*) = p_k^* = q_k^*$; when $p_k^* = \gamma$, we have $q_k^* = \gamma \in \mathrm{prox}(p_k^*) = \{\gamma, 0\}$.

(ii) For $k \in \mathcal{D}^*$ (i.e., $q_k^* = 0$), (A.2) supplies $\psi_k^* \in [-6C/5, 0]$ and $p_k^* = -\nu\psi_k^* \in [0, 6\nu C/5]$. Provided $\mathcal{D}^* \neq \emptyset$, the bound $\nu \leq \nu_4^* = 25/(18C)$ ensures $6\nu C/5 \leq 5/3 \leq \gamma$, placing $p_k^* \in [0, \gamma]$. By (13), $\mathrm{prox}(p_k^*)$ includes $0 = q_k^*$.

(iii) For $k \in \mathcal{E}^*$ (i.e., $q_k^* < 0$), $\psi_k^* = 0$ and $p_k^* = q_k^* < 0$. By (13), $\mathrm{prox}(p_k^*) = p_k^* = q_k^*$.

Collecting (i)–(iii) establishes (13) and completes the proof. □

Appendix B. Proof of Theorem 2

Proof. Let $\Sigma^* := (\mathbf{w}^*; b^*; \mathbf{q}^*)$ and define $\Pi := \{\Sigma = (\mathbf{w}; b; \mathbf{q}) : \mathbf{q} + N\mathbf{w} + b\mathbf{y} = \mathbf{1}\}$. When $\Sigma \in \Pi \cap U(\Sigma^*, \widetilde{\varrho})$ for some $\widetilde{\varrho} > 0$, the equality constraint $\mathbf{q} + N\mathbf{w} + b\mathbf{y} = \mathbf{1}$ together with (16) leads to

$$-N(\mathbf{w} - \mathbf{w}^*) = (b - b^*)\mathbf{y} + \mathbf{q} - \mathbf{q}^*. \tag{B.1}$$

Since $\|\mathbf{w}\|^2$ is convex, we can lead to

$$\begin{aligned}\|\mathbf{w}\|^2 - \|\mathbf{w}^*\|^2 &\geq 2\langle \mathbf{w} - \mathbf{w}^*, \mathbf{w}^*\rangle \\ &\overset{(16)}{=} -2\langle N(\mathbf{w} - \mathbf{w}^*), \boldsymbol{\psi}^*\rangle \\ &\overset{(B.1)}{=} 2\langle \boldsymbol{\psi}^*, \mathbf{q} - \mathbf{q}^*\rangle + 2(b - b^*)\langle \boldsymbol{\psi}^*, \mathbf{y}\rangle \\ &\overset{(16)}{=} 2\langle \boldsymbol{\psi}^*, \mathbf{q} - \mathbf{q}^*\rangle. \end{aligned} \tag{B.2}$$

According to (B.1) and (B.2), we now show that Σ^* is a local minimizer of (7). For any $\Sigma \in \Pi \cap U(\Sigma^*, \varrho)$ with $\varrho > 0$, the target inequality is:

$$\frac{1}{2}\|\mathbf{w}\|^2 + CL_{\mathrm{ht}}(\mathbf{q}) \geq \frac{1}{2}\|\mathbf{w}^*\|^2 + CL_{\mathrm{ht}}(\mathbf{q}^*). \tag{B.3}$$

We verify (B.3) by analyzing three regimes of νC: $(0, 5/18)$, $[5/18, 25/18)$, and $[25/18, +\infty)$.

Case (i): Suppose $\nu C \in (0, 5/18)$. Define $\mathbf{p}^* := \mathbf{q}^* - \nu\boldsymbol{\psi}^*$ and partition $[m]$ into five subsets:

$$\begin{aligned}&\mathbb{H}_1^* := \{i \in [m] : p_i^* < 0\}, \quad \mathbb{H}_2^* := \{i \in [m] : p_i^* \in [0, \widetilde{\beta}]\}, \\ &\mathbb{H}_3^* := \{i \in [m] : p_i^* \in (\widetilde{\beta}, \widehat{\beta})\}, \\ &\mathbb{H}_4^* := \{i \in [m] : p_i^* \in [\widehat{\beta}, 1)\}, \quad \mathbb{H}_5^* := \{i \in [m] : p_i^* \geq 1\},\end{aligned}$$

where $\widetilde{\beta} := \frac{6\nu C}{5}$ and $\widehat{\beta} := \frac{2}{3} + \frac{6\nu C}{5}$. From the proximal characterization $\mathbf{q}^* \in \mathrm{prox}_{\nu C L_{\mathrm{ht}}}(\mathbf{p}^*)$ combined with (11) and (A.2), we deduce

$$\begin{aligned}&\psi_{\mathbb{H}_1^*}^* = \mathbf{0}_{\mathbb{H}_1^*}, \quad \mathbf{q}_{\mathbb{H}_2^*}^* = \mathbf{0}_{\mathbb{H}_2^*}, \quad \psi_{\mathbb{H}_3^*}^* = -\frac{6C}{5}\mathbf{1}_{\mathbb{H}_3^*}, \\ &\psi_{\mathbb{H}_4^*}^* = -\frac{18C(1 - \mathbf{q}^*)_{\mathbb{H}_4^*}}{5}, \quad \psi_{\mathbb{H}_5^*}^* = \mathbf{0}_{\mathbb{H}_5^*}.\end{aligned} \tag{B.4}$$

Together with (A.2), this implies

$$\begin{aligned}&\psi_i^* = 0, q_i^* < 0, \ i \in \mathbb{H}_1^*, \\ &\psi_i^* \in [-\frac{6C}{5}, 0], q_i^* = 0, \ i \in \mathbb{H}_2^*, \\ &\psi_i^* = -\frac{6C}{5}, q_i^* \in (0, \frac{2}{3}), \ i \in \mathbb{H}_3^*, \\ &\psi_i^* = -\frac{18C(1 - q_i^*)}{5}, q_i^* \in [\frac{2}{3}, 1), \ i \in \mathbb{H}_4^*, \\ &\psi_i^* = 0, q_i^* \geq 1, \ i \in \mathbb{H}_5^*.\end{aligned} \tag{B.5}$$

To confirm (B.3), introduce $\overline{\mathbb{H}}^* := \mathbb{H}_1^* \cup \mathbb{H}_5^*$ and $\mathbb{H}^* := \mathbb{H}_2^* \cup \mathbb{H}_3^* \cup \mathbb{H}_4^*$. Noting that $\psi_i^* = 0$ for $i \in \overline{\mathbb{H}}^*$, it suffices by (B.2) to prove:

$$CL_{\mathrm{ht}}(\mathbf{q}_{\mathbb{H}^*}) - CL_{\mathrm{ht}}(\mathbf{q}_{\mathbb{H}^*}^*) + \langle \boldsymbol{\psi}_{\mathbb{H}^*}^*, \mathbf{q}_{\mathbb{H}^*} - \mathbf{q}_{\mathbb{H}^*}^*\rangle \geq 0, \tag{B.6}$$

$$CL_{\mathrm{ht}}(\mathbf{q}_{\overline{\mathbb{H}}^*}) - CL_{\mathrm{ht}}(\mathbf{q}_{\overline{\mathbb{H}}^*}^*) \geq 0. \tag{B.7}$$

Verification of (B.7): For $i \in \mathbb{H}_1^*$, we have $q_i^* < 0$. There exists $\varrho_1 > 0$ such that $q_i < 0$ for all $\Sigma \in \Pi \cap U(\Sigma^*, \varrho_1)$, yielding $\ell_{\mathrm{ht}}(q_i) = \ell_{\mathrm{ht}}(q_i^*) = 0$. For $i \in \mathbb{H}_5^*$, we have $q_i^* \geq 1$. There exists $\varrho_5 > 0$ such that $q_i \geq 1$ for all $\Sigma \in \Pi \cap U(\Sigma^*, \varrho_5)$, giving $\ell_{\mathrm{ht}}(q_i) = \ell_{\mathrm{ht}}(q_i^*) = 1$. In both cases, $\ell_{\mathrm{ht}}(q_i) - \ell_{\mathrm{ht}}(q_i^*) = 0$, confirming (B.7).

Verification of (B.6): We examine the three subsets in turn.

(i) For $i \in \mathbb{H}_4^*$, we have $q_i^* \in [2/3, 1)$ and $\psi_i^* = -18C(1 - q_i^*)/5 = -C\nabla\ell_{\mathrm{ht}}(q_i^*)$, since $\ell_{\mathrm{ht}}(s) = (-9s^2 + 18s - 4)/5$ yields $\nabla\ell_{\mathrm{ht}}(s) = 18(1 - s)/5$ on $(2/3, 1)$. There exists $\varrho_4 > 0$ ensuring $q_i \in (2/3, 1)$ for $\Sigma \in \Pi \cap U(\Sigma^*, \varrho_4)$. The proximal characterization $q_i^* = \mathrm{prox}_{\nu C\ell_{\mathrm{ht}}}(p_i^*)$ gives

$$\ell_{\mathrm{ht}}(q_i) + \frac{(q_i - p_i^*)^2}{2\nu C} \geq \ell_{\mathrm{ht}}(q_i^*) + \frac{(q_i^* - p_i^*)^2}{2\nu C}.$$

Expanding $(q_i^* - p_i^*)^2 - (q_i - p_i^*)^2 = 2(q_i^* - q_i)(p_i^* - q_i^*) + O(\|q_i - q_i^*\|^2) = 2\nu\psi_i^*(q_i - q_i^*) + O(\|q_i - q_i^*\|^2)$ and rearranging, we arrive at $C[\ell_{\mathrm{ht}}(q_i) - \ell_{\mathrm{ht}}(q_i^*)] + \psi_i^*(q_i - q_i^*) \geq 0$.

(ii) For $i \in \mathbb{H}_3^*$, we have $q_i^* \in (0, 2/3)$ and $\psi_i^* = -6C/5$. On $(0, 2/3]$, $\ell_{\mathrm{ht}}(s) = 6s/5$ is linear with derivative $6/5$, so $\psi_i^* = -C\nabla\ell_{\mathrm{ht}}(q_i^*)$. For Σ near Σ^*, q_i remains in $(0, 2/3)$, and linearity yields $C[\ell_{\mathrm{ht}}(q_i) - \ell_{\mathrm{ht}}(q_i^*)] + \psi_i^*(q_i - q_i^*) = \frac{6C}{5}(q_i - q_i^*) - \frac{6C}{5}(q_i - q_i^*) = 0$.

(iii) For $i \in \mathbb{H}_2^*$, we have $q_i^* = 0$ and $\psi_i^* \in [-6C/5, 0]$. At $s = 0$, the subdifferential is $\partial\ell_{\mathrm{ht}}(0) = [0, 6/5]$ (noting $\ell_{\mathrm{ht}}(0) = 0$ and $\ell'_{\mathrm{ht}}(0^+) = 6/5$). Since $-\psi_i^*/C \in [0, 6/5]$, the subdifferential inequality $\ell_{\mathrm{ht}}(q_i) \geq (-\psi_i^*/C)q_i$ implies $C\ell_{\mathrm{ht}}(q_i) + \psi_i^* q_i \geq 0$. With $q_i^* = 0$ and $\ell_{\mathrm{ht}}(0) = 0$, this becomes $C[\ell_{\mathrm{ht}}(q_i) - \ell_{\mathrm{ht}}(q_i^*)] + \psi_i^*(q_i - q_i^*) \geq 0$.

Aggregating (i)–(iii), all terms are non-negative, confirming (B.6). This finishes Case (i).

Case (ii): Suppose $\nu C \in [5/18, 25/18)$. Define $\mathbf{p}^* := \mathbf{q}^* - \nu\boldsymbol{\psi}^*$, $\widetilde{\beta} := 6\nu C/5$, and $\widehat{\beta} := 5/6 + 3\nu C/5$. From (12), the proximal condition partitions $[m]$ into $\mathbb{C}_1^*$ ($q_i^* < 0$, $\psi_i^* = 0$), $\mathbb{C}_2^*$ ($q_i^* = 0$, $\psi_i^* \in [-6C/5, 0]$), $\mathbb{C}_3^*$ ($q_i^* \in (0, 2/3]$, $\psi_i^* = -6C/5$), and $\mathbb{C}_4^*$ ($q_i^* \geq 1$, $\psi_i^* = 0$).

Let $\overline{\mathbb{C}}^* := \mathbb{C}_1^* \cup \mathbb{C}_4^*$ and $\mathbb{C}^* := \mathbb{C}_2^* \cup \mathbb{C}_3^*$.

For $\overline{\mathbb{C}}^*$: When $i \in \mathbb{C}_1^*$, $q_i^* < 0$ implies $\ell_{\mathrm{ht}}(q_i) = \ell_{\mathrm{ht}}(q_i^*) = 0$ locally. When $i \in \mathbb{C}_4^*$, $q_i^* \geq 1$ implies $\ell_{\mathrm{ht}}(q_i) = \ell_{\mathrm{ht}}(q_i^*) = 1$ locally. Either way, $\ell_{\mathrm{ht}}(q_i) - \ell_{\mathrm{ht}}(q_i^*) = 0$.

For $\mathbb{C}^*$: (i) For $i \in \mathbb{C}_3^*$, linearity of $\ell_{\mathrm{ht}}(s) = 6s/5$ on $(0, 2/3]$ with $\psi_i^* = -C\nabla\ell_{\mathrm{ht}}(q_i^*)$ yields $C[\ell_{\mathrm{ht}}(q_i) - \ell_{\mathrm{ht}}(q_i^*)] + \psi_i^*(q_i - q_i^*) = 0$. (ii) For $i \in \mathbb{C}_2^*$, using $-\psi_i^*/C = \partial\ell_{\mathrm{ht}}(0)$ and $q_i^* = 0$, we get $C[\ell_{\mathrm{ht}}(q_i) - \ell_{\mathrm{ht}}(q_i^*)] + \psi_i^*(q_i - q_i^*) \geq 0$.

Both contributions are non-negative, completing Case (ii).

Case (iii): Suppose $\nu C \geq 25/18$. Define $\mathbf{p}^* := \mathbf{q}^* - \nu\boldsymbol{\psi}^*$ and $\gamma := \sqrt{2\nu C}$. By (13), the index set splits into $\mathbb{E}_1^*$ ($q_i^* < 0$, $\psi_i^* = 0$), $\mathbb{E}_2^*$ ($q_i^* = 0$, $\psi_i^* \in [-6C/5, 0]$), and $\mathbb{E}_3^*$ ($q_i^* \geq \gamma$, $\psi_i^* = 0$).

Let $\overline{\mathbb{E}}^* := \mathbb{E}_1^* \cup \mathbb{E}_3^*$ and $\underline{\mathbb{E}}^* := \mathbb{E}_2^*$.

For $\overline{\mathbb{E}}^*$: When $i \in \mathbb{E}_1^*$, $q_i^* < 0$ yields $\ell_{\mathrm{ht}}(q_i) = \ell_{\mathrm{ht}}(q_i^*) = 0$ locally. When $i \in \mathbb{E}_3^*$, $q_i^* \geq \gamma \geq 5/3 > 1$ yields $\ell_{\mathrm{ht}}(q_i) = \ell_{\mathrm{ht}}(q_i^*) = 1$ locally. Thus $\ell_{\mathrm{ht}}(q_i) - \ell_{\mathrm{ht}}(q_i^*) = 0$.

For $\underline{\mathbb{E}}^*$: With $q_i^* = 0$, $\psi_i^* \in [-6C/5, 0]$, and $-\psi_i^*/C \in \partial\ell_{\mathrm{ht}}(0)$, the subdifferential property gives $C[\ell_{\mathrm{ht}}(q_i) - \ell_{\mathrm{ht}}(q_i^*)] + \psi_i^*(q_i - q_i^*) \geq 0$.

Combining all cases, $(\mathbf{w}^*; b^*; \mathbf{q}^*)$ is a local minimizer of (7) in $\Pi \cap U(\Sigma^*, \eta^*)$ for sufficiently small $\eta^* > 0$. □

Appendix C. Proof of Theorem 3

Proof. Suppose $(\mathbf{w}^*; b^*; \mathbf{q}^*)$ with $\boldsymbol{\psi}^*$ is a P-stationary point of (7). From Definition 3, the first equation $\mathbf{w}^* + N^\top\boldsymbol{\psi}^* = \mathbf{0}$ gives

$$\mathbf{w}^* = -N^\top\boldsymbol{\psi}^* = -\sum_{i=1}^m \psi_i^* y_i \mathbf{x}_i. \tag{C.1}$$

Set $\mathbf{p}^* := \mathbf{q}^* - \nu\boldsymbol{\psi}^* \in \mathbb{R}^m$. For $\nu C \in (0, 5/18)$, invoking Lemma 2 and the proximal inclusion $\mathbf{q}^* \in \mathrm{prox}_{\nu C L_{\mathrm{ht}}}(\mathbf{p}^*)$, we partition the index set $[m]$ into five disjoint subsets $\mathbb{G}_1^*, \mathbb{G}_2^*, \mathbb{G}_3^*, \mathbb{G}_4^*, \mathbb{G}_5^*$ as defined in Theorem 3. By (11), the proximal mapping yields:

$$q_i^* = \begin{cases} p_i^*, & i \in \mathbb{G}_1^*, \\ 0, & i \in \mathbb{G}_2^*, \\ p_i^* - 6\nu C/5, & i \in \mathbb{G}_3^*, \\ (5p_i^* - 18\nu C)/(5 - 18\nu C), & i \in \mathbb{G}_4^*, \\ p_i^*, & i \in \mathbb{G}_5^*. \end{cases}$$

Substituting $p_i^* = q_i^* - \nu\psi_i^*$ into the above expressions, we obtain:

(i) For $i \in \mathbb{G}_1^*$, we have $q_i^* = p_i^* = q_i^* - \nu\psi_i^*$, which implies $\psi_i^* = 0$.

(ii) For $i \in \mathbb{G}_2^*$, from (A.2), when $q_i^* = 0$, we have $\psi_i^* \in [-6C/5, 0]$, thus $\psi_i^* \neq 0$ in general.

(iii) For $i \in \mathbb{G}_3^*$, we have $q_i^* = p_i^* - 6\nu C/5 = q_i^* - \nu\psi_i^* - 6\nu C/5$, which gives $\psi_i^* = -6C/5 \neq 0$.

(iv) For $i \in \mathbb{G}_4^*$, solving $(5 - 18\nu C)q_i^* = 5p_i^* - 18\nu C$ with $p_i^* = q_i^* - \nu\psi_i^*$ yields $\psi_i^* = -18C(1 - q_i^*)/5$. Since $q_i^* \in (2/3, 1)$ for $i \in \mathbb{G}_4^*$, we have $\psi_i^* \neq 0$.

(v) For $i \in \mathbb{G}_5^*$, we have $q_i^* = p_i^* = q_i^* - \nu\psi_i^*$, which implies $\psi_i^* = 0$.

Summarizing the above analysis, we have $\psi_i^* = 0$ for $i \in \mathbb{G}_1^* \cup \mathbb{G}_5^* = \overline{\mathbb{G}}^*$ and $\psi_i^* \neq 0$ for $i \in \mathbb{G}_2^* \cup \mathbb{G}_3^* \cup \mathbb{G}_4^* = \mathbb{G}^*$. Substituting into (C.1) yields (22).

For the support vector conditions (23), from the feasibility constraint $\mathbf{q}^* + N\mathbf{w}^* + b^*\mathbf{y} = \mathbf{1}$, we have $q_i^* = 1 - y_i(\langle \mathbf{w}^*, \mathbf{x}_i \rangle + b^*)$. Thus:

(i) For $i \in \mathbb{G}_2^*$, we have $q_i^* = 0$, hence $y_i(\langle \mathbf{w}^*, \mathbf{x}_i \rangle + b^*) = 1$.

(ii) For $i \in \mathbb{G}_3^*$, we have $q_i^* \in (0, 2/3)$, hence $y_i(\langle \mathbf{w}^*, \mathbf{x}_i \rangle + b^*) = 1 - q_i^* \in (1/3, 1)$.

(iii) For $i \in \mathbb{G}_4^*$, we have $q_i^* \in [2/3, 1)$, hence $y_i(\langle \mathbf{w}^*, \mathbf{x}_i \rangle + b^*) = 1 - q_i^* \in (0, 1/3]$.

This completes the proof. □

Appendix D. Natural multi-view mechanism analysis on WebKB

We use the WebKB course-versus-non-course task as a natural multi-view diagnostic. The two views are Fulltext and Inlinks. TF–IDF vocabularies, PCA mappings, and max-absolute scaling are fitted within each training fold, and the same five outer folds are used for every method. Since WebKB provides a single classification task, it is used here to examine the behavior of the multi-view mechanisms and is not included in the cross-task significance tests.

The fused model improves the mean ACC (Table D.1) from 96.48% and 93.72% for the two individual views to 97.62%, and raises F1 from 91.63% and 84.09% to 94.24%. Its margin over the strongest multi-view baseline is modest but consistent across metrics.

On clean WebKB (Table D.2), removing adaptive weights (Equal-weight) reduces ACC/F1 to 95.72/89.27%, while removing structural regularization (No-structure) gives 96.95/93.42%; removing both gives 95.91/89.69%. Hence each component contributes to the full model, with the adaptive weights playing the larger role on this dataset.

Table D.3 reports the learned view weights. On clean data, the stronger Fulltext view receives the larger mean weight (0.56 versus 0.44). The mean weight of the corrupted view is lower than in the clean case in both tested settings: corrupting Fulltext lowers its weight from 0.56 to 0.53, while corrupting Inlinks lowers its weight from 0.44 to 0.36 and shifts the weight of Fulltext up to 0.64. These descriptive shifts are consistent with Fulltext being the stronger view on clean data, where the single-view baselines in Table D.1 reach 96.48% and 93.72%, respectively. The clean-data scores do not establish which view remains more informative after corruption.

Since each view selects its own working set, we further examine whether independent selection harms cross-view consistency. For the four folds with complete per-view support records, the Full model has support-set Jaccard overlap 0.283 ± 0.391, working-set overlap 0.282 ± 0.391, and view-prediction disagreement 0.100 ± 0.038. Under the Equal-weight and Neither variants, the support-set overlaps drop to $0.109 \pm$

Table D.1
Five-fold WebKB results (mean±standard deviation). Fulltext-only and Inlinks-only are single-view diagnostics and are not included in the multi-view method ranking.

Method	ACC (%)	Precision (%)	Recall (%)	F1 (%)	TIME (s)
SVM-2K	96.48 ± 1.79	89.34 ± 7.49	96.09 ± 2.38	92.41 ± 3.59	14.709 ± 4.936
MvSVM-2C	97.24 ± 0.62	98.18 ± 2.91	89.13 ± 2.17	93.40 ± 1.45	238.500 ± 115.730
MvLSSVC-2C	97.05 ± 2.29	91.94 ± 9.14	96.09 ± 4.18	93.65 ± 4.45	0.332 ± 0.037
Wave-MvSVM	46.43 ± 1.81	27.66 ± 0.65	89.57 ± 1.82	42.27 ± 0.75	25.566 ± 0.158
MVASY-BX	94.48 ± 1.53	**100.00 ± 0.00**	74.78 ± 6.98	85.42 ± 4.61	16.954 ± 0.077
Mv$SL_{0/1}$-SVM	97.34 ± 0.43	96.59 ± 4.23	91.30 ± 3.07	93.76 ± 0.81	9.884 ± 0.402
MvL_{ht}-SVM	**97.62 ± 1.01**	99.51 ± 1.09	89.57 ± 4.18	**94.24 ± 2.52**	7.488 ± 13.888
Fulltext-only	96.48 ± 1.20	95.60 ± 4.79	88.26 ± 5.00	91.63 ± 2.87	0.006 ± 0.001
Inlinks-only	93.72 ± 1.77	94.04 ± 3.95	76.09 ± 5.10	84.09 ± 4.56	0.006 ± 0.001

Table D.2
WebKB mechanism diagnostics (five-fold mean±standard deviation). View corruption adds Gaussian noise with amplitude 0.4 to one transformed training view; the test views remain clean.

Block	Variant	Condition	ACC (%)	F1 (%)
Clean ablation	Full	Clean	97.62 ± 1.01	94.24 ± 2.52
	Equal-weight	Clean	95.72 ± 2.36	89.27 ± 6.90
	No-structure	Clean	96.95 ± 2.37	93.42 ± 4.68
	Neither	Clean	95.91 ± 2.85	89.69 ± 7.97
	No-working-set	Clean	87.81 ± 21.69	82.44 ± 26.10
	Own-structure-only	Clean	97.05 ± 2.34	93.63 ± 4.58
View corruption	Full	Fulltext corrupted	96.38 ± 1.49	91.50 ± 3.52
	Equal-weight	Fulltext corrupted	95.62 ± 2.68	88.71 ± 7.99
	No-structure	Fulltext corrupted	94.87 ± 2.84	88.82 ± 5.05
	Full	Inlinks corrupted	96.48 ± 2.67	92.45 ± 4.83
	Equal-weight	Inlinks corrupted	95.15 ± 2.19	89.01 ± 4.01
	No-structure	Inlinks corrupted	96.10 ± 2.77	91.54 ± 5.32

Table D.3
Learned view weights of MvL_{hl}-SVM on WebKB (mean±standard deviation) on clean data and when one training view is corrupted.

Condition	Fulltext weight	Inlinks weight
Clean	0.56 ± 0.16	0.44 ± 0.16
Fulltext corrupted	0.53 ± 0.13	0.47 ± 0.13
Inlinks corrupted	0.64 ± 0.16	0.36 ± 0.16

0.052 and 0.105 ± 0.061, and the disagreement roughly doubles to 0.197 ± 0.165 and 0.202 ± 0.158. In these four folds, different active sets coexist with low mean prediction disagreement, and the Full variant has lower mean disagreement than the two ablations. These results are descriptive evidence rather than a general guarantee of cross-view consistency.

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